# Cash Transfers, Health, and the Transition to Adulthood

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**Abstract**

Can a schooling-conditioned cash transfer improve the later employment of adolescents with disabilities? Finland offers a €700 monthly benefit to encourage continued education among people who might otherwise begin claiming disability benefits. Using administrative data, I estimate the program's long-run effects with an instrumental-variables design based on quasi-random assignment to examiners of different leniency. Among marginal applicants—those whose approval depends on the assigned examiner—approval to the benefit makes them 12 percentage points less likely to be working or studying after payments end. The benefit does not extend their education. Instead, it weakens their early attachment to the labor market. I use comparisons of observably similar accepted and rejected applicants to assess the effects on the wider population of applicants. This analysis suggests that the program helps adolescents with the poorest labor market prospects, moving them from disability benefits into education and improving later employment. Accounting for what recipients would do in the absence of the transfer could improve program targeting.

**Keywords**: disability benefits, conditional cash transfers, labor supply, mental health

**JEL**: I38, J13, J14, J22

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# 1 Introduction

Across high-income countries, disability affects a substantial share of adolescents[1]. These adolescents face barriers to entering the labor market. For example, in the UK, 29.6% of disabled 16–24-year-olds are not in employment, education, or training (NEET), compared with just 8.7% of their non-disabled peers. Large education and employment gaps remain even in countries such as Finland, where public health insurance is universal and tuition-free education widely available. Despite this extensive public support, many young individuals with disabilities drop out of education early and instead enter disability-benefit programs. To counter this, Finland offers these adolescents a cash transfer conditional on continued education or another rehabilitative activity[2]. By offering an alternative to disability benefits, the policy provides incentives to invest in human capital aimed at improving their labor-market outcomes in the long run.

The cash transfer is generous, $\sim$ 700 euros/month. It targets adolescents (ages 16–20) who have a documented health problem that threatens their education or employment prospects. The most common disabilities among the recipients are mental health and neurodevelopmental problems, ranging from intellectual disability and autism to depression, and thus covering applicants with varying labor-market prospects.[3] While many countries offer income support to people with health conditions who participate in rehabilitation, the Finnish program is distinctive in targeting adolescents with health problems before labor-market entry and treating education as the primary form of rehabilitation. Many programs share the aim of improving the prospects of disadvantaged youth through income support (Aizer et al., 2022), education, employment and training programs (Schochet et al., 2008; Heller, 2014; Kluve et al., 2019) as well as mentoring and psychotherapy programs (Oreopoulos et al., 2017; Heller et al., 2017; Resnjanskij et al., 2024).

Theoretically, the effects of this kind of policy are ambiguous. Conditional cash transfers are a popular and intuitive policy tool (e.g. Parker and Todd, 2017) since instead of just providing cash, they can be conditioned on a desirable behaviour like attendance in education. In this setting a policy that provides cash with education conditionality could improve outcomes if the recipients would have otherwise received passive disability benefits (Deshpande, 2016; Levere, 2021). But correct targeting of the benefits is also important (Diamond and Sheshinski, 1995; Haller et al., 2024): if the recipients would have attended school anyway, the transfer may generate little additional human capital while displacing early work experience and job connections.

In this paper, I study how the program affects applicants' education, labor-market outcomes, and health into their thirties. I use two complementary empirical strategies to both estimate these effects and to understand how they vary across applicants who would likely follow different paths without the program. To implement these strategies, I link comprehensive administrative data on labor-market outcomes and healthcare use to novel

[1] In the EU, 8.1% of 16–29-year-olds report health-related limitations in usual activities (Eurostat, 2026).

[2] In practice, around 85% of recipients attend secondary education.

[3] The most common diagnoses were developmental mental health disorders (learning disorders, autism), intellectual disability, ADHD/conduct disorders, mood disorders, and anxiety disorders. Applicants can also have physical or sensory conditions, including arthritis, visual or hearing impairments, epilepsy, cerebral palsy, and Down syndrome.

records on applications, diagnoses, and examiners for around 30,000 applications between 2004 and 2018.

My main identification strategy uses the fact that the applications for the benefit are allocated to examiners quasi-randomly from a running queue. Examiner leniency varies substantially: a one-standard-deviation increase in examiner leniency increases the probability of approval by 4.9 percentage points. This instrument is uncorrelated with the applicant's background characteristics and passes a variety of other validity checks. The main caveat is that the effects estimated are for marginal applicants. As I show, this is a group with better labor-market prospects than the average applicant to the benefit.

I find economically meaningful adverse labor-market effects among marginal applicants. Benefit approval reduces employment by 11 percentage points and the probability of being in education or employment by 12 percentage points after the benefit ends. These estimates correspond to 28% and 21%, respectively, of the corresponding means without approval. Approval for the benefit does not increase the probability of continuing studies beyond secondary education, and I find no statistically detectable effects on serious adverse health events, criminal convictions, or financial distress. For marginal applicants, the IV estimates therefore provide no evidence that approval improves long-run labor-market attachment or produces offsetting gains in the other measured domains.

The transfer changes the type of human capital the adolescents acquire. For marginal applicants, approval reduces labor supply and cumulative earnings by 36% during the benefit period without materially changing time in education. This pattern is consistent with most marginal applicants having attended school without the transfer: approval appears to displace work rather than induce additional education. The relevant no-program alternative therefore differs from that of the program's intended beneficiaries, who would otherwise receive disability benefits. Because responses to cash transfers may vary across recipients (e.g., Løken et al., 2012), the IV estimates for marginal applicants need not generalize to this population.

I complement the IV design with a selection-on-observables approach to study the effects on applicants with low predicted labor-market prospects. I combine detailed application records with pretreatment information on applicants' backgrounds and use double/debiased machine learning (DML) to adjust for selection into approval (Chernozhukov et al., 2018). To assess the credibility of the selection-on-observables approach, I benchmark its estimates against the IV estimates in the complier-weighted sample. The selection-on-observables estimates closely match the IV estimates, remain stable across alternative covariate sets, and pass standard balance and placebo tests.

I find that effects vary sharply with predicted labor-market prospects. Among applicants with the lowest prospects, approval raises employment by 4.7 percentage points from a mean of 16% and lowers NEET status. This group mainly has disabilities like intellectual disability and chromosomal disorders, which are clearly diagnosable and often lead to early entry to the disability insurance program. During the benefit period, their time on disability insurance falls by more than two years, and their time in education rises, which may help explain their higher employment after the benefit ends.

What explains these heterogeneous effects? I show that a decomposition based on a simple human-capital framework—in which education and work experience or work attach-

ment in adolescence raise adult employment—matches both the sign and order of magnitude of the estimates. I find little evidence for alternative channels involving substance use (e.g. Evans and Moore, 2011; Chorniy et al., 2025) or spillovers to related health-based benefits (Dahl et al., 2014).

To evaluate the welfare effects of the policy, I calculate the Marginal Value of Public Funds (MVPF) for the policy, inclusive of all benefits, healthcare expenses, and costs of crime. The MVPF for the marginal applicants is 0.8, indicating that the fiscal costs exceed the recipients' willingness to pay for the benefit. In contrast, providing the benefit to the population with low labor-market prospects appears to pay for itself, although there is considerable uncertainty around the point estimates.

**Related literature.** This paper studies a cash transfer for adolescents with health-based screening and schooling conditionality. Some previous papers study programs that share some of these features, but the combination of monetary support with human capital investment targeted to this large and vulnerable group has not been studied before in economics.

This paper contributes to the literature on disability benefits for adolescents by studying a transfer that incentivizes human capital accumulation. Previous work has studied the effects of Supplemental Security Income (SSI), a non-temporary program in the US (Deshpande, 2016; Deshpande and Mueller-Smith, 2022; Levere, 2021)[4]. In line with theoretical models that abstract from human capital accumulation (Diamond and Sheshinski, 1995), this evidence shows that disability programs without incentives for human capital investment can have negative effects on labor supply but also considerable insurance value. My contribution is to extend this literature into a different case: I study a non-means-tested temporary program that also incentivizes investment in human capital, all features that leave less room for the negative labor-market consequences and a possibility for positive ones[5]. Yet these features do not resolve the trade-offs, as I observe that while helping some, this kind of benefit can still inhibit the labor-market success of many.

Second, I extend the literature on schooling-conditioned cash transfers by studying a large transfer for adolescents targeted on health rather than, as more commonly, family income. I show that effects depend on what applicants would otherwise have done. Prior studies find that financial support tied to education increases participation or enrollment (Dearden et al., 2009; Dynarski, 2003; Denning et al., 2019; Barr et al., 2021; Britton et al., 2025), yet may not increase or even reduce subsequent earnings (Britton et al., 2025; Barr et al., 2021). My results provide evidence consistent with one proposed explanation: these transfers can displace early work experience and job connections, since the youth employment during education might be consumption driven (similarly to Humlum and

[4]There is also a large literature on the effects of (mostly permanent) adult health-based/disability transfers (Bound, 1991; Maestas et al., 2013; French and Song, 2014; Kostøl and Mogstad, 2015; Autor et al., 2019; Black et al., 2024; Huynh and Chan, 2024; Haller et al., 2024; Silver and Zhang, 2022; Gelber et al., 2023). Compared to the program studied here, the target populations and incentives imposed are very different in these programs.

[5]From a social welfare perspective, the transfer is intended to solve the problem of privately optimal but publicly too low levels of investment in human capital in the presence of government disability insurance programs for adults.

Vejlin (2013)).[6]

Third, I extend the literature on active labor-market programs (ALMPs) to education-based support delivered during adolescence, at the school-to-work transition (Card et al., 2018; Kluve et al., 2019, for reviews). Borgan Reiling et al. (2024) compare regular education with subsidized employment among work-impaired young adults and find that education's medium-run employment advantage disappears in the long run. I identify a different margin: within a single program for adolescents, approval improves later employment when it redirects applicants from disability insurance to education, but reduces employment when it displaces early work among those who would have attended school anyway. This heterogeneity is consistent with evidence that activation can offset the negative employment effects of cash support (Filippucci, 2025), that placement-oriented approaches can outperform conventional vocational rehabilitation (Markussen and Røed, 2014; Fogelgren et al., 2023), and that formal education can yield substantial returns when disability would otherwise push workers out of employment (Humlum et al., 2025). Relatedly, Reiterä (2021) estimates that vocational-rehabilitation approval increases subsequent benefit receipt among young applicants.[7] My mechanism evidence also connects to research on labor-market entry conditions (Raaum and Røed, 2006; Nordström Skans, 2011; Oreopoulos et al., 2012) and on early work experience and job connections acquired during school (Le Barbanchon et al., 2020; Hensvik et al., 2023; Bruhn et al., 2025).

Beyond employment and earnings, I also provide evidence on broader effects of disability benefits among adolescents in a non-US setting. I find no evidence of previously documented reductions in crime (Deshpande and Mueller-Smith, 2022) or financial distress (Deshpande et al., 2021), or of large effects on health (previous evidence varies; see (Black et al., 2024; Huynh and Chan, 2024; Silver and Zhang, 2022; Gelber et al., 2023)).

Overall, this paper finds evidence that the effects of a schooling-conditioned cash transfer can be very heterogeneous with respect to the counterfactual activity. This suggests that when designing disability, education, and rehabilitation benefits, close attention should be paid to the target population and their incentives, not only the average effects.

Sections 2 and 3 describe the benefit, examiner assignment, and administrative data. Section 4 presents the identification strategies. Section 5 reports the effects during the benefit period and in adulthood and examines how they vary with labor-market prospects. Section 6 discusses mechanisms and welfare implications. Section 7 concludes.

# 2 Institutional background

## 2.1 Youth Rehabilitation Allowance

The Youth Rehabilitation Allowance (YRA) is a cash benefit for 16–20-year-olds whose disability threatens their ability to study or work.

Policymakers introduced YRA as a temporary alternative to Disability Insurance (DI),

[6] In contrast, education-linked transfers and subsidies in developing countries have improved adult outcomes (Parker and Vogl, 2023; Duflo et al., 2024). In these settings, the counterfactual is more often leaving school or not enrolling.

[7] Reiterä (2021) instruments approval with examiner leniency; this design inspired the identification strategy in this paper.

which can lead to a permanent exit from the labor market. The aim was to improve recipients' long-run outcomes by making the temporary benefit conditional on participation in education or rehabilitation. If most applicants would otherwise receive DI, YRA adds little to public spending in the short run. It may even pay for itself if recipients claim fewer benefits as adults. Additional motivations for the benefit include the possibility that disabled individuals may face additional costs for rehabilitative treatments.

YRA may affect recipients through more than the cash itself. The program can change their incentives to attend school or rehabilitation and may lead to closer monitoring of school attendance. The application process can also attach a medical label, increase applicants' knowledge of the benefit system, and put them in contact with school and social service workers. In the rest of the paper, I refer to the treatment as a cash transfer or benefit, but I do not attempt to separate the effect of cash from these other channels.

Over time, both annual applications and the stock of YRA recipients have increased considerably (Fig. 1). The stock of YRA recipients has overtaken the pre-YRA stock of DI recipients (while DI recipiency has also continued to increase Fig. A.11b). Most of this growth comes from applications based on mental health disorders, which now account for approximately 89 % of applicants and 91 % of accepted applicants. This pattern mirrors a broader rise in mental health diagnoses among adolescents. Finnish birth cohorts contain about 60,000 people, so first-time applicants each year correspond to roughly 2–6 % of a cohort. Figure 2 shows the distribution of applicants' main diagnoses.

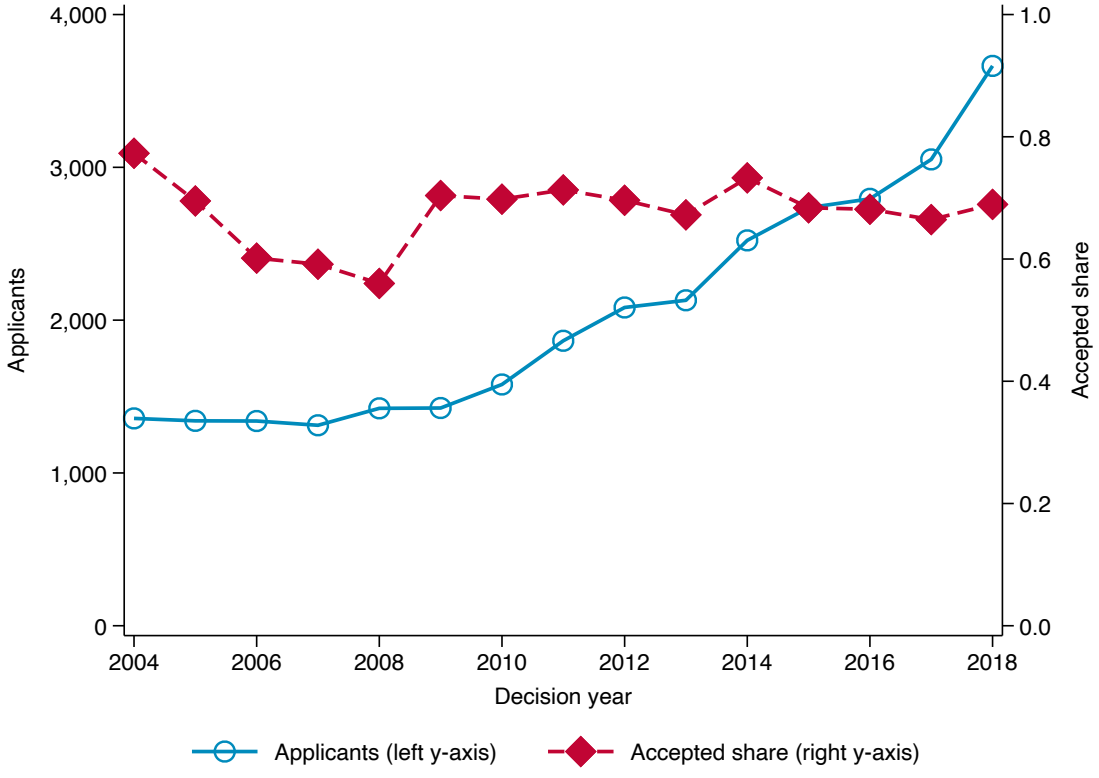



Figure 1: Number of applicants to the Youth Rehabilitation Allowance and share of accepted applicants, years 2004–2018.

In addition to a diagnosis, applicants must submit a rehabilitation plan. This plan

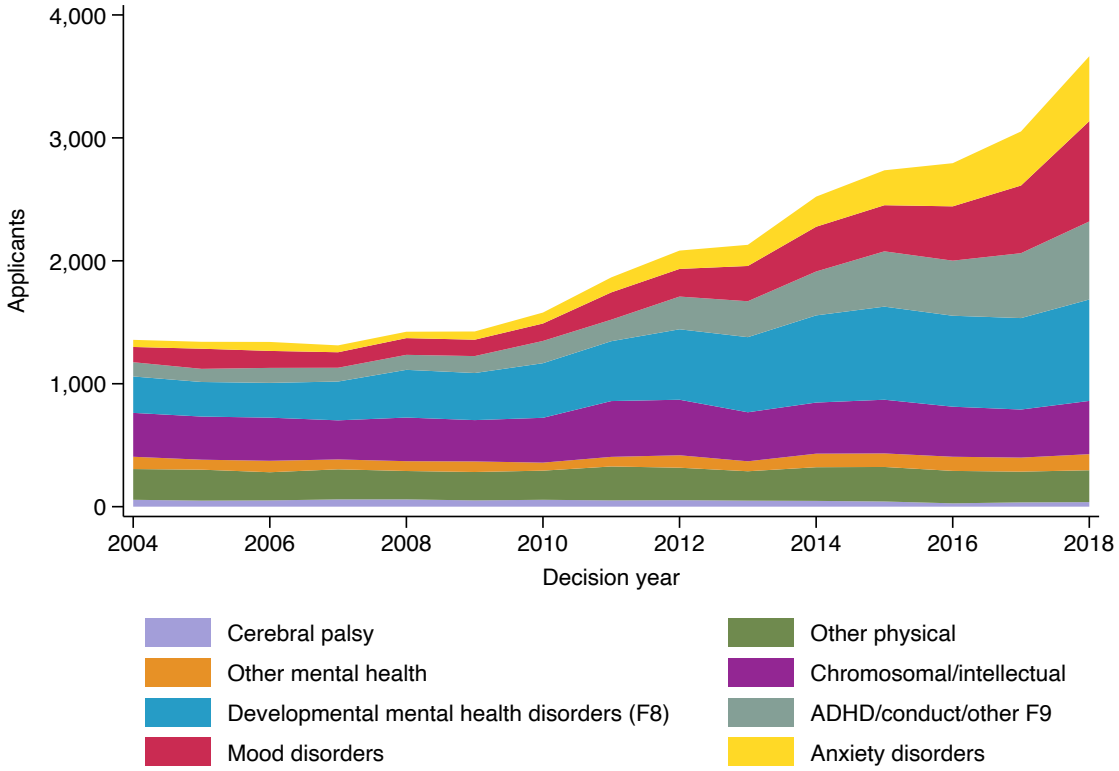



Figure 2: Main diagnosis indicated in the applications across the years 2004–2018. The sample includes all applicants, not only accepted ones.

is prepared together with medical personnel, social workers, or school-based support staff and specifies the accommodations needed to support school attendance or describes other rehabilitative activities the individual plans to undertake.

Figure 3 summarizes the application process. Applicants submit their materials to KELA. Each application must document the diagnosis and the required accommodations. Examiners assess the applicant's circumstances, the severity of the disability, and the adequacy of the proposed rehabilitation plan.

Applications are processed within insurance districts, whose boundaries follow regional borders, although their exact configuration changed during the study period. Within each district and language group, applications enter a common queue and are assigned to the next available examiner. Applicants cannot choose their examiner. The existence of this system is confirmed by written and verbal communication from KELA, and no written or verbal sources, despite specifically asking about this, have revealed other forms of sorting the applications. As assignment varies by language [8] and district, I account for this variation using language and area-year fixed effects in the empirical analysis (see the methods section).

There are no income or asset tests, either for the individual or the family, and earned income does not reduce the benefit amount.

The benefit is temporary. Once an application is approved, the benefit is paid monthly

[8] Finland has a large Swedish-speaking minority that uses Swedish in administrative matters.

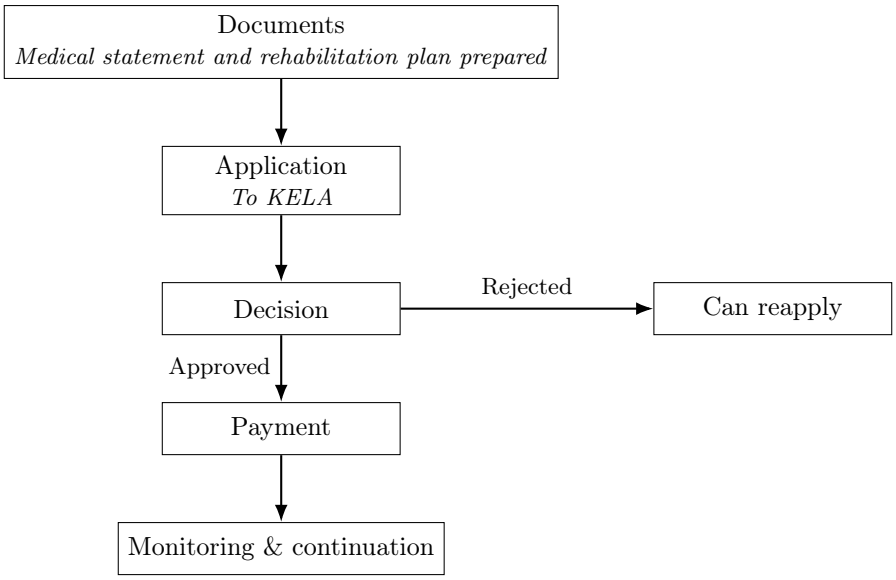



Figure 3: Application process for the Youth Rehabilitation Allowance.

until one of the following occurs: (i) the recipient graduates from secondary education or completes the approved rehabilitative activity; (ii) the recipient turns 20; or (iii) participation in education or rehabilitation is deemed insufficient. Attendance is monitored, and KELA may discontinue the benefit when participation is insufficient. Grades are not assessed, and graduation is not a condition of receiving the benefit. The monthly benefit was about €700 until 2017 and €900 thereafter [9]. Accepted applicants receive €25,400 on average over the full benefit spell. YRA is taxable. In practice, recipients with no other income pay little tax.

YRA recipients can meet the rehabilitation requirement through several types of activity, although the data do not record which one they undertake. For benefit spells lasting multiple years, the main rehabilitative activity can be assumed to be secondary education. Recipients may also attend other forms of education that do not lead to a secondary education diploma. These studies are not recorded very well in the administrative data, and some individuals attending them have their main activity recorded as *unknown*.

Among YRA recipients, 59 % attend vocational secondary education, 10 % attend general upper-secondary school, 14 % are completing basic education or attending other non-degree programs, and 1 % attend higher education. The remaining 16 % are neither classified primarily as students nor enrolled in secondary education, although some may participate in education not captured by the registers or in other rehabilitation. These shares are close to those reported by Koskenvuo et al. (2014).

## 2.2 Schooling system in Finland

Finland provides tuition-free comprehensive education from ages 7 to 16. Almost all comprehensive schools are public, and students receive meals and learning materials free of charge. During the study period, compulsory education ended when a student completed comprehensive school or, at the latest, ten years after it began.

[9] In 2023 euros.

Students who enter secondary education choose between an academic and a vocational track. Admission is based primarily on teacher-assigned ninth-grade marks. In the academic track, students follow a standardized curriculum covering mathematics, literature, languages, history, and the natural sciences. Vocational education is organized into occupation-specific tracks that prepare students for specific professions, such as car mechanic, plumber, or practical nurse. While the academic track is primarily intended to prepare students for university studies, both tracks qualify students for higher education. Secondary education typically lasts three years and is tuition-free, as is higher education.

Secondary schools offer a range of support measures for students with learning difficulties and disabilities. Special schools and programs provide more individualized instruction and allow for slower progression, with graduation often occurring after four to five years instead of the standard three. These schools and programs enroll only a small share of students; most special-needs education takes place in regular schools. All special educational support is offered free of charge.

### 2.3 Healthcare

Finland has a universal public health insurance system that covers the entire population. Healthcare services, particularly for children and adolescents, are predominantly provided by the public sector. The public sector provides almost all treatment for serious conditions and emergencies. Private healthcare is also available, but it is accessed mainly through employer-provided coverage and typically focuses on preventive care or mild conditions. Private healthcare therefore plays a limited role for the population studied in this paper, and the public healthcare registers capture most healthcare my sample receives.

Adults make modest out-of-pocket payments for public healthcare, subject to an annual cap of about €600. A similar annual cap applies to out-of-pocket expenditures on prescription drugs reimbursed by KELA, which include the vast majority of commonly prescribed medications.

Health insurance coverage does not depend on YRA receipt. Copayments in the public system are generally low, although patients often pay a substantial share of psychotherapy costs. Unlike Supplemental Security Income in the United States, YRA does not confer health insurance eligibility. Any estimated health effects therefore do not operate through a statutory change in insurance coverage.

### 2.4 Other benefits

Those not receiving YRA can receive multiple alternative benefits. Disability Insurance/Pension (DI) [10] is similar to YRA in that it is based on medical screening, but with a separate application process. Individuals applying need both a diagnosis from a medical doctor and acceptance from an insurance institution. For young individuals with little or no work history, this institution is usually KELA.

For the first few years, DI is awarded as a temporary DI, but if the inability to work persists, it can be made permanent. An individual in DI can have small earnings without

[10]The Finnish program is formally a disability pension. I use Disability Insurance (DI) to match the terminology in the international literature.

the benefit being reduced. Eligibility for DI starts at age 16, and the amount during this period was mostly equal to YRA or somewhat higher.

Applicants may also qualify for unemployment benefits, student aid, housing allowance, or social assistance. Many of these programs are means-tested on personal or household income and provide little support to young people who live with their parents. For instance, student aid for individuals attending secondary education and living with parents is limited only to the families with little or no earnings. Applicants denied YRA therefore tend to receive no other benefits or smaller transfers, usually without a school-attendance requirement.

# 3 Data

I combine KELA records on YRA applications from 2004–2018 with Finnish administrative registers that follow applicants annually and, where relevant, link them to their parents. Outcome coverage varies across registers, with the longest series running through 2023. Statistics Finland stores the linked data on secure servers, where I conduct all analyses. Information on how to apply for access is available from the author. All euro amounts in the paper, including earnings, benefits, and healthcare costs, are adjusted for inflation to 2023 prices using the consumer price index.

## 3.1 Youth Rehabilitation Allowance Application data

KELA records all YRA applications and benefit payments.[11]

The application records identify the applicant and examiner and report the decision (approved, rejected, or discontinued), decision date, diagnosis code, and municipality of residence. They cover 2004–2018. Applicant identifiers link to the other registers, but examiner identifiers do not.

Separate payment records describe the payment dates and amounts and whether KELA paid the benefit to the applicant or a parent or guardian. About 98 % of beneficiaries receive payments directly.

I construct the analysis sample as follows. I use individuals' first application, no matter if they were eventually accepted later. I exclude applicants with missing diagnosis codes—all of whom were rejected—and those older than 20 at their first application. The final sample contains 30,623 applicants. The sample is unbalanced: follow-up differs, individuals leave the panel after death, and some registers do not cover the latest years. Sample sizes therefore vary across outcomes and follow-up periods.

## 3.2 Labor market and education

Statistics Finland's FOLK modules provide annual labor-market and demographic outcomes. The FOLK PERUS module (Statistics Finland, 2025d) includes yearly information on earnings, main activity, disability insurance receipt, place of residence, and other outcomes for the full Finnish population, including applicants and their parents. These data are mainly available for the years 1987–2023. I classify a person as not in employment,

[11] KELA is the Social Insurance Institution of Finland.

education, or training (NEET) if their main activity during the year is neither employment nor study and they are not completing military service.[12] I use wage income as a measure of earnings.

The FOLK LAPS module includes links between biological parents and children (Statistics Finland, 2025c). Using these links, I am able to match the applicants to their parents. The parents not found in this data are presumably dead, unknown, or have migrated from the country. For these individuals, I use an indicator for missing parents in analyses where this is necessary.

I use two datasets on education. First, I use the TYHR (Statistics Finland, 2025b) dataset to describe the educational history of the individuals. These data are available for the years 1998–2023 and include the GPA of the students graduating from the basic education and their applications to the secondary education. I use whether a student appears in these data as an indicator of graduation and GPA as a measure of academic performance. Since the data do not cover the whole study period, a student not found in the data can also be one who graduated outside the years covered. The EDUC register (Statistics Finland, 2025a) records enrollment in secondary and higher education.

### 3.3 Healthcare data

I use three Finnish healthcare registers to describe the health and healthcare use of the population. The National Institute for Health and Welfare maintains HILMO, which covers inpatient and specialist outpatient care from 1995 to 2023, and AvoHILMO, which covers public primary care from 2011 to 2023. The registers report visit dates, diagnoses or reasons for care, emergency status, and procedures. Because the public sector provides most serious and emergency care, these registers capture most such events in the sample.

The third register covers reimbursed prescription drugs. KELA reimburses most prescription medicines, usually at the pharmacy, so the register captures most prescribed drug use in Finland.

I use the registers to measure both serious health events and broader healthcare use. Measures of serious health events include hospital days, emergency visits, and inpatient visits with a mental-health diagnosis. Because these events generally require care regardless of cost, they should primarily reflect underlying health. Broader utilization measures include total healthcare spending, prescription purchases, and specialist and primary-care visits. These outcomes may respond to YRA through changes in health or care-seeking. I return to this distinction when interpreting the estimates in the results section.

I define substance-use problems using public healthcare diagnoses for substance use and addiction (ICD-10 codes F10–F19). Where both registers are available, the outcome uses specialist and primary-care records.

HILMO reports provider costs from 2012 onward, although costs remain unavailable for some administrative units. I cap visits at one per person, day, and diagnosis to avoid counting transfers between units as separate visits. Among records with observed costs, I calculate average costs by main ICD-10 diagnosis and length of stay and use these averages to impute costs for all visits. I validate the imputation among visits with observed costs.

[12] Finland requires men to complete 6–12 months of military or civilian service, although people with health problems may be exempt.

The prescription register reports the public reimbursement for each purchase. The primary-care data begin in 2011 but contain no costs, and no official method is available to estimate them from the information recorded. I therefore exclude primary care from the cost calculations.

### 3.4 Other data

To assess participation in criminal activities, I use both criminal convictions and police reports. The conviction data cover the years 1995–2023 and the police report data the years 2006–2023. Both data sources include dates of the crimes and the type of crime committed. I construct measures of drug-related crime and violent crime from the crime codes.

I use data from the National Enforcement Authority (in Finnish, Ulosottolaitos) to measure financial distress. The data include all enforced debts from the years 2008–2023. A debt or a bill ends up in enforcement after the creditor files for enforcement, usually with a delay of multiple months from the due date. As argued in Maczulskij et al. (2025), ending up in debt enforcement is seldom desirable for the indebted: wages, benefits, and property can be garnished, and individuals are subjected to severe monitoring and other procedures. Unlike, for instance, seeking bankruptcy in the US, it is not a strategic choice, which makes it a good measure of financial distress.

### 3.5 Descriptive results

Table 1 presents summary statistics for the applicants, separately for those approved and those rejected.

Applicants have poor average labor-market prospects. Figure 4 shows that, at age 25, about 30 % are employed and 15 % are students; the remainder are unemployed or receive DI.

In my main heterogeneity analysis I use an index of labor-market prospects. To estimate the index, I use pretreatment covariates and application details to predict the probability of DI receipt two years after application. To limit overfitting, I construct out-of-fold predictions using two-fold cross-fitting and train the prediction model only on rejected applicants. I randomly split rejected applicants into two folds. I estimate a classification forest on one fold and predict two-year DI receipt in the held-out fold, reverse the training and prediction samples, and combine the resulting out-of-fold predictions. I then use the fitted models to construct predictions for approved applicants. This avoids incorporating the effects of treatment into the index.

The results that use this index are similar if a different time horizon or different outcome for the prediction is used. In practice, the index mostly reflects the diagnosis in the application: individuals with intellectual disability and chromosomal disorders have high values. As can be seen in Fig. 5, the labor-market-prospects index is strongly correlated with the share that is outside of the labor force later in adulthood, showing that it captures labor market prospects.

I classify applicants in the top quintile of the index as having low labor-market prospects and the remaining applicants as having higher prospects within the applicant sample. A

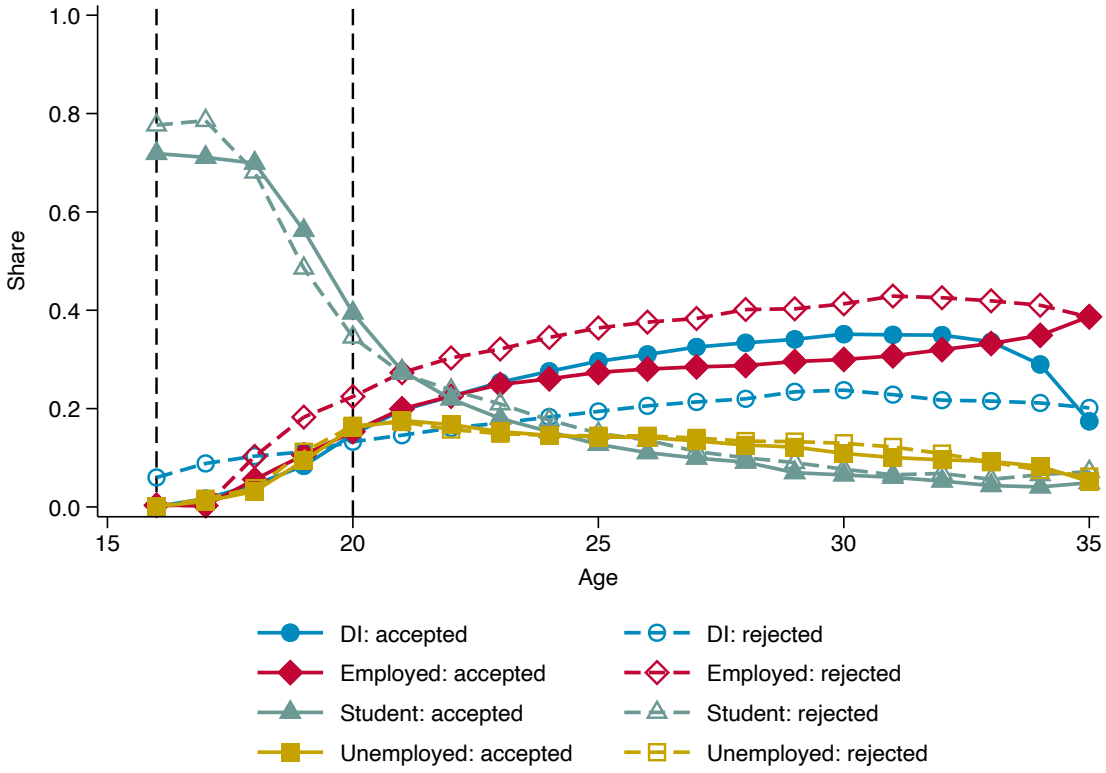



Figure 4: Shares in each main activity by age and YRA approval status. The figure excludes military service and uses an unbalanced panel. Vertical lines mark the main YRA eligibility period.

| | (1) Gen. pop. | (2) Approved | (3) Rejected |
|---|---|---|---|
| Student | 0.976 | 0.997 | 0.996 |
| Employed | 0.024 | 0.003 | 0.004 |
| Distinct admissions | 1.02 | 8.34 | 6.50 |
| Healthcare costs | 559 | 5,779 | 4,094 |
| Anxiety diagnosis | 0.053 | 0.091 | 0.079 |
| Depression diagnosis | 0.052 | 0.098 | 0.082 |
| Single parent HH | 0.211 | 0.290 | 0.305 |
| Urban | 0.807 | 0.781 | 0.804 |
| Mother age | 44.04 | 44.19 | 44.05 |
| Father age | 46.45 | 47.03 | 46.69 |
| Mother DI | 0.027 | 0.085 | 0.075 |
| Father DI | 0.040 | 0.081 | 0.079 |
| Mother married | 0.697 | 0.558 | 0.562 |
| Father married | 0.691 | 0.550 | 0.553 |
| Mother employed | 0.814 | 0.699 | 0.721 |
| Father employed | 0.831 | 0.727 | 0.740 |
| Mother taxable income | 4,467 | 91 | 117 |
| Father taxable income | 6,174 | 123 | 173 |
| Accepted (1st appl.) | 0 | 1 | 0 |
| Ever in YRA | 0 | 1 | 0.196 |
| Total YRA benefits | 0 | 22,967 | 4,397 |
| Observations | 1,476,350 | 20,868 | 9,755 |

Table 1: Summary statistics for the applicants, by approval status.
Notes: All variables are measured in the year before the first application. Euro amounts are in 2023 prices. Gen. pop. is the sample of all other individuals aged 15.

descriptive table for this group is shown in Table A.6. Both groups nevertheless have worse health and socioeconomic backgrounds than the general population.

# 4 Methods

## 4.1 Target populations and parameters

To estimate the effects of YRA, we can start by considering the simple comparison of approved and rejected applicants

$$Y_{it} = \beta_0 + \beta_1 Approved_i + \beta_2 X_i + \epsilon_{it} \tag{1}$$

where $i$ indexes applicants, $t$ indexes time, $Approved_i$ indicates first-application approval, and $X_i$ contains covariates observed before the decision. This strategy relies on the assumption of conditional ignorability: conditional on $X_i$, approval is independent of the potential outcomes,

$$\big(Y_{it}(1), Y_{it}(0)\big) \perp Approved_i \mid X_i.$$

Approved and rejected applicants differ on many observed background variables, even within the applicant sample (Table 4). This suggests the presence of unobserved factors correlated with approval. My main design exploits the quasi-random assignment of applications to examiners who differ in their approval rates (Section 4.2). I also report estimates

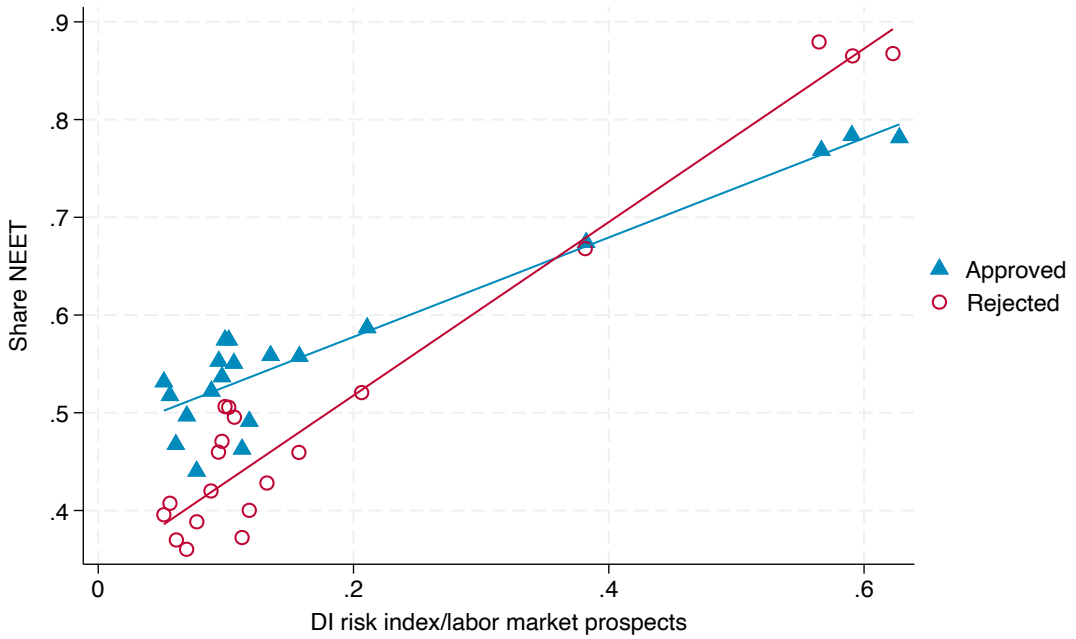



Figure 5: Adult NEET (not in education, employment or training) rates by predicted labor-market prospects and YRA approval status. The figure plots mean NEET rates in 20 equal-sized bins separately for approved and rejected applicants. Higher values of the index indicate worse labor-market prospects.

that adjust for selection on observed covariates.

Before describing the estimators, I define the parameters they estimate. Treatment is approval of the first application. Let $Y_{it}(1)$ and $Y_{it}(0)$ denote the potential outcomes of the applicant $i$ at time $t$ if the first application is approved or rejected, so that the observed outcome is $Y_{it} = Approved_i\, Y_{it}(1) + (1 - Approved_i)\, Y_{it}(0)$. The treatment effect for an individual applicant is $Y_{it}(1) - Y_{it}(0)$.

Treatment effects likely differ across applicants. Applicants vary substantially in their labor market prospects. For those with relatively strong prospects, approval may mainly reduce work while they receive YRA. For those with poor prospects, the alternative to YRA may be Disability Insurance rather than employment. Different averages therefore answer different questions.

The average treatment effect (ATE) over all applicants, $ATE = E[Y_{it}(1) - Y_{it}(0)]$, is the relevant parameter for large changes to the program, such as approving nearly all or hardly any applications. Examiner IV instead estimates a weighted average of the effects on compliers: applicants whose approval depends on their assigned examiner. This local average treatment effect (LATE) is directly relevant for changes to the approval threshold.

The LATE does not describe all applicants. Compliers are concentrated in the middle of the distribution of labor market prospects (Section 4.2.3). The effects on applicants with the poorest prospects are also of interest. These effects may differ from those on marginal applicants and therefore require going beyond the IV.

Examiner IV is the primary design. I use selection on observables for populations and outcomes the IV does not cover well: the full applicant population, applicants with the

poorest labor market prospects, and outcomes for which the IV estimates are imprecise. Where the strategies estimate the same parameter for the same population, I compare their estimates. Similar estimates support the assumptions of the selection-on-observables design. Table 2 summarizes the designs.

Table 2: Empirical designs.

| Design | Identifying assumption | Estimand | Population | Used for |
|---|---|---|---|---|
| Examiner IV | Relevance, conditional random assignment, exclusion, average monotonicity, and extensive-margin compliers only (A1–A4) | LATE | Compliers (about 23% of applicants), concentrated in the middle of the distribution of labor market prospects | The main short-run and long-run estimates |
| Selection on observables (DDML) | Conditional ignorability given the covariates in Section 4.3, and overlap | ATE, also by subgroup | All applicants, and subgroups defined by predicted DI risk; reweighted to the compliers for comparison with the IV | Effects for the full applicant population, for the applicants with the lowest labor market prospects, and for outcomes where the IV is imprecise |

## 4.2 Examiner IV

An application arrives at the insurance district covering the applicant's municipality and is handed to the next examiner in the queue. Within region, year, and language, applicants receive different examiners because of their position in the queue. Examiners differ in their approval rates, and this variation provides the instrument for first-application approval.[13]

The first stage relates approval to examiner leniency:

$$Approved_i = \alpha z_{ij} + \mu_c + \eta_i. \tag{2}$$

Here, $z_{ij}$ measures the examiner's tendency to approve applications, and $\mu_c$ denotes the time and place fixed effects needed to make examiner assignment as good as random.

The second-stage equation is

$$Y_{it} = \beta_0 + \beta_1 \widehat{Approved}_i + \mu_c + \epsilon_{it}. \tag{3}$$

In most specifications, I include only the fully non-parametric fixed effects. Because I include the necessary controls in fully saturated form, the specification satisfies the *rich covariates* condition of Blandhol et al. (2026). The 2SLS estimand is therefore weakly causal rather than a combination of treatment effects that can carry negative weights.

I do not cluster standard errors at the judge level in this regression since the level of randomization is effectively the applicant, in line with the previous literature (Chyn et al., 2024). The endogenous variable is approval of the first application.

[13]The design has been used in many papers on the economics of crime (Kling, 2006), foster care (Doyle Jr, 2007; Bald et al., 2022; Helénsdotter, 2026), and disability insurance (Maestas et al., 2013; French and Song, 2014; Kostøl and Mogstad, 2015; Autor et al., 2019), among others.

I use first-application approval as the main treatment because it is most likely to satisfy the exclusion restriction. Around 20% of rejected applicants are approved on a later application, and the instrument also raises the probability of ever being approved (Table 3). Using *ever approved* as the endogenous variable changes the scale of the estimates but otherwise yields similar results (Fig. A.8).

I construct residualized leave-one-out examiner leniency, following Agan et al. (2023). Leaving out an applicant's own case breaks the finite-sample correlation between the instrument and that applicant's decision. Assignment takes place among examiners handling applications in the same region, year, and language group. I therefore residualize approval decisions using region × year × language fixed effects. I call these the *necessary controls* because random assignment holds only conditional on them.

In the Appendix, I check alternative controls and residualization choices. The estimates are similar when, for example, I add the month of decision to the equations (Fig. A.13).

The instrument is

$$z_{ij} = \frac{1}{N_j - 1} \times \left( \sum_{k=1}^{N_j} (Approved^*_{jk}) - Approved^*_i \right). \tag{4}$$

Here, $N_j$ is the number of cases handled by examiner $j$, and $Approved^*$ is the residual from regressing approval decisions on region × year × language indicators. Subtracting $Approved^*_i$ removes the application's own contribution and gives the leave-one-out mean. Each application therefore receives a different value of the instrument. I calculate the main instruments once with the full application sample and use these throughout.

The sample contains 379 examiners, who handle 79 cases on average. Appendix Fig. A.14 shows that the estimates are similar under alternative minimum-caseload restrictions.

### 4.2.1 Empirical Bayes shrinkage

Examiner IV studies usually use either residualized leave-one-out examiner leniency or jackknife IV. Both estimate each examiner's leniency from that examiner's own caseload. These caseloads are small: the average examiner in my data handles 79 applications, and many handle far fewer. The resulting leave-one-out mean is noisy. This noise does not threaten validity, but it weakens the first stage and widens the second-stage confidence intervals.

To reduce this noise, I use an empirical Bayes approach. The observed leave-one-out mean $\hat{z}_j$ is the sum of the examiner's true leniency and estimation error. The procedure shrinks noisier estimates further toward the average:

$$z^{EB}_{ij} = \bar{z} + \lambda_j (z_{ij} - \bar{z}), \qquad \lambda_j = \frac{\sigma^2_z}{\sigma^2_z + s^2_j}, \qquad s^2_j = \frac{\hat{\sigma}^2_j}{N_j - 1} \tag{5}$$

where $\bar{z}$ is mean leniency across examiners and $s^2_j$ is the sampling variance of the leave-one-out mean. Because this mean averages $N_j - 1$ decisions, its sampling variance is the same for every application that examiner $j$ handles. I recover $\sigma^2_z$, the variance of true

leniency across examiners, as the variance of examiners' full-caseload approval rates minus their average sampling variance, $\hat{\sigma}_j^2/(N_j - 1)$.

Examiners with few cases have a large $s_j^2$, so their estimates are shrunk more strongly toward the average. Estimates for examiners with large caseloads remain close to their leave-one-out means. The procedure does not add identifying assumptions beyond A1–A4.

Empirical Bayes shrinkage is common in settings such as teacher value added (Chetty et al., 2014), but less common in examiner IV designs. Humphries et al. (2025) use it as a check against measurement error in the instrument. Frandsen et al. (2026) develop the approach concurrently with my work. They establish its theoretical properties, derive optimal shrinkage formulas, and provide simulation and empirical evidence. My choices are close to theirs, and their paper contains the proofs.

In my application, shrinkage leaves the point estimates similar but reduces standard errors for the main outcomes by around 25% relative to the standard leave-one-out mean instruments (Fig. A.13). This is a combination of being able to include the examiners with small amounts of cases and improving the first stage in general.

Appendix Section A.5.5 provides additional details and simulation evidence on empirical Bayes IV. Unless stated otherwise, my IV results use shrunk instruments because they provide more precise estimates. Section 5.1.4 shows that the point estimates are similar under empirical Bayes and the standard leave-one-out procedure.

### 4.2.2 Identifying assumptions

I use four assumptions to interpret the IV estimate as a LATE for applicants whose approval responds to examiner leniency:

**A1. Relevance.** The instrument must have a strong effect on the endogenous variable, that is, approval of the application.

**A2. Independence.** The instrument is as good as randomly assigned, conditional on the necessary controls.

**A3. Exclusion.** The instrument only affects outcomes through treatment.

**A4. Average monotonicity.** More lenient examiners are more likely to approve an application on average, though this need not hold for every applicant.

Under these assumptions, the IV estimator identifies the average effect of having the first application approved for the compliers (LATE).

#### A1 Instrument Relevance

Examiner leniency strongly predicts approval. The histogram in Figure 6 displays the distribution of the instrument, with fractions in each bin on the right y-axis. The line displays a linear local polynomial (with bandwidth 0.05) of the instrument $z_{ij}$ on first-application approval. A one SD increase in the instrument raises the approval probability by 4.9 pp, or 7.2% of the 68% mean.

Table 3 shows that the first stage is strong: the coefficient on examiner leniency is 0.803, and the first-stage F-statistic in the main specification is 344. F-statistics in leniency designs are not comparable to those for a single, one-dimensional instrument, so the $F = 10$ rule of thumb is not the right benchmark (Hull, 2017; Chyn et al., 2024). Following

Andrews et al. (2019), I also report Anderson–Rubin confidence intervals for the main results (Table 6). These intervals are valid regardless of instrument strength and are close to the conventional intervals.

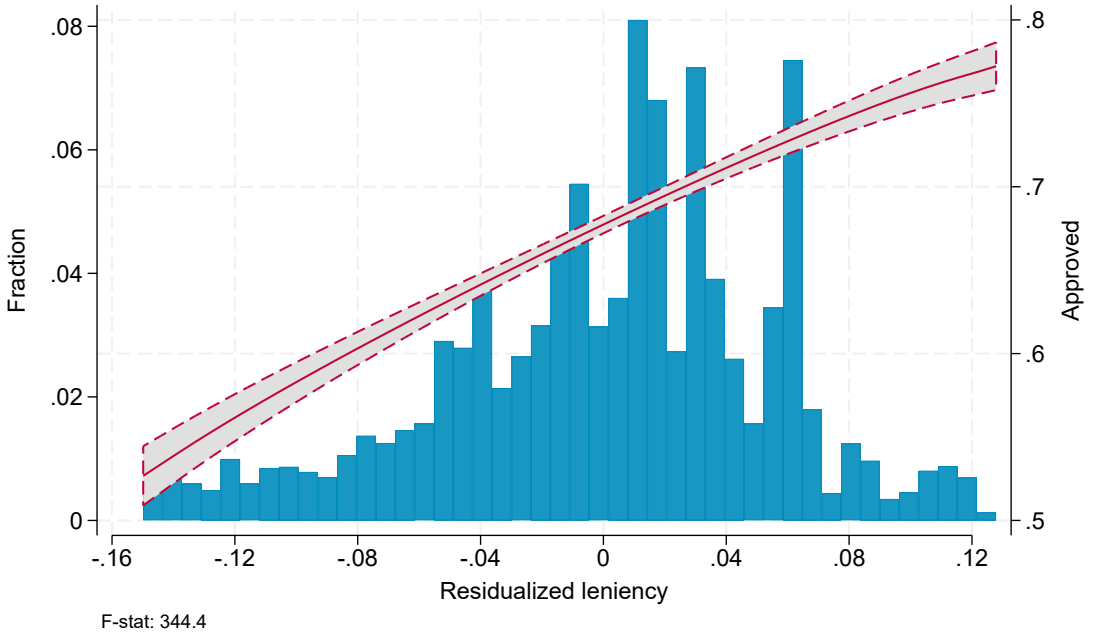



Figure 6: Application approval by examiner leniency. The histogram shows the distribution of $z_{ij}$ in the estimation sample (right y-axis). The line shows a linear local polynomial of first-application approval on residualized examiner leniency (left y-axis).

| | (1) First application approv. | (2) Ever in YRA | (3) Total YRA Benefits |
|---|---|---|---|
| EB IV | 0.803*** | 0.547*** | 13792.6*** |
| | (0.0433) | (0.0395) | (1330.2) |
| N | 30623 | 30623 | 30623 |
| F-Statistic | 344.4 | 191.7 | 107.5 |

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

Table 3: First-stage estimates for alternative treatment measures. "Ever YRA" is the probability of receiving YRA at any point in the observation period. "Total YRA benefits" is total YRA receipt over the observation period. All regressions use the same instrument; it is not estimated again for each outcome. All specifications include the necessary region × year × language fixed effects used to construct the instrument.

### A2 Instrument exogeneity

Within region, year, and language, examiner assignment must be independent of applicants' potential outcomes. Applications are assigned from a running queue. I residualize approval decisions using region × year × language fixed effects and include the same fixed effects in the regressions. Written and oral communication from KELA confirms the queue system, and I found no evidence of additional sorting.

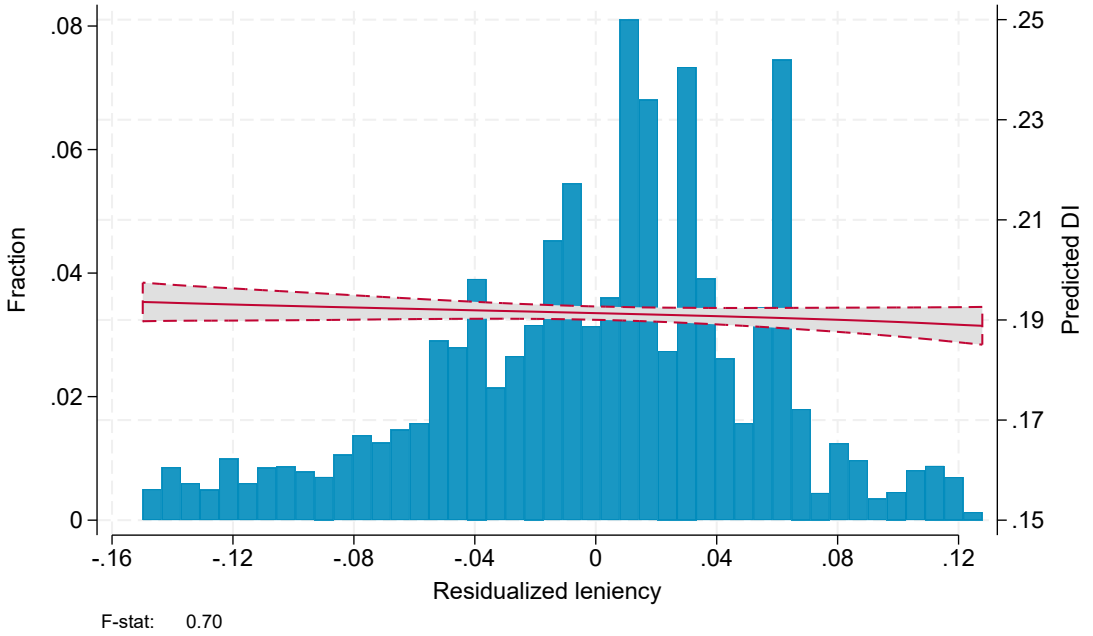



Figure 7: Predicted DI risk by examiner leniency. The histogram shows the distribution of $z_{ij}$ in the estimation sample (right y-axis). The line shows a linear local polynomial of predicted DI risk on residualized examiner leniency (left y-axis) with 95% confidence intervals. Pretreatment variables, mainly health and family background, predict DI receipt two years after application.

To my knowledge, Reiterä (2021) first use this design in the YRA setting to study active rehabilitation policies. They also argue that exogeneity is plausible after including similar controls, but provide little institutional or empirical evidence on possible threats to random assignment.

Random assignment implies that examiner leniency should be unrelated to applicants' pretreatment characteristics. Figure 7 checks this implication using predicted DI risk. The index combines pretreatment measures of socioeconomic status and healthcare use that predict later outcomes.[14]

The curve is flat: examiner leniency is not meaningfully correlated with predicted DI risk. Scaling this association by the correlation between the risk index and the key outcomes implies a bias of around 0.5 pp in post-benefit employment. This is less than 5% of the IV estimate.[15] The small association supports the assumption that examiners are randomly assigned.

Table 4 tests joint balance across observable characteristics. Approval is strongly related to applicants' backgrounds, especially their health measures. In contrast, the coefficients relating these characteristics to examiner leniency are small: two of 30 are significant at the 10% level, and I cannot reject that they are jointly zero ($p = 0.62$).

Two further checks support the assignment mechanism. For health outcomes observed from birth onward, I estimate placebo effects before applicants are eligible for YRA and present these together with the estimates for the periods after application. Random assignment implies no relationship in either levels or trends before application, and the estimates

[14] Any imbalance matters for bias only if the unbalanced characteristic also predicts the outcome.
[15] The implied upper confidence limit for the bias is around 1.1 pp.

are consistent with zero. Table A.10 also shows that the first-stage coefficient changes little when I add controls beyond the necessary fixed effects.

These checks cannot rule out unobserved sorting. If sorting occurs, it would most likely operate through information available at the application stage. In robustness checks, I add decision month, diagnosis category, parental socioeconomic background, and ninth-grade GPA. The estimates change little. For unobserved sorting to explain the results, it would have to be large relative to sorting on these observed characteristics.

#### A3 Exclusion restriction

Random assignment identifies the reduced-form effect of being assigned a stricter examiner. Identifying the effect of first-application approval also requires examiners to affect outcomes only through that decision. This exclusion restriction can fail in settings such as criminal justice, where judges often make decisions about several treatments (Humphries et al., 2025).

The exclusion restriction is plausible here because examiners only approve or reject applications based on the submitted information. The decision is communicated through KELA's centralized system. Examiners neither advise applicants about other benefits nor meet them in person.

#### A4 Monotonicity

Strict monotonicity requires a more lenient examiner to approve every application that a stricter examiner would approve (Imbens and Angrist, 1994). This is demanding in examiner IV designs. It fails, for example, if some examiners are stricter toward mental health conditions but more lenient toward other conditions. Frandsen et al. (2023) and Chyn et al. (2024) call this *strict monotonicity*. Under strict monotonicity, researchers can also estimate marginal treatment effects (Mogstad et al., 2018).

My main results rely on the weaker *average monotonicity* assumption. Applicants may respond differently to examiners, but more lenient examiners must increase approval on average. The IV estimates can then be interpreted as "convex averages of the treatment effects among compliers" (Frandsen et al., 2023). Strict monotonicity would also be needed to estimate marginal treatment effects.

I assess average monotonicity in three ways, described in Appendix Section A.2.1. The first stage is positive and significant in every subsample (Maestas et al., 2013) and in each split by illness category and gender (Bhuller et al., 2020). The binary lenient-against-strict instrument proposed by Sigstad (2024) also gives estimates similar to the main results (Tables A.8 and A.9).

### 4.2.3 Compliers

If assumptions A1–A4 hold, the IV estimates the average treatment effect for compliers. These applicants are approved by some examiners but rejected by others. Compliers likely fall in the middle of the distribution of labor market prospects: applicants with the poorest prospects may be approved by every examiner, while those with the strongest prospects

| | (1) Approved | (2) EB IV |
|---|---|---|
| ***Applicant*** | | |
| Month of dec. | -0.00433*** | -0.0000106 |
| | (0.000766) | (0.000104) |
| Urban | -0.0114* | -0.00130 |
| | (0.00657) | (0.000888) |
| Male | 0.0240*** | 0.000786 |
| | (0.00551) | (0.000745) |
| Student | -0.0205** | -0.00133 |
| | (0.00840) | (0.00114) |
| GPA (9th grade) | -0.0219*** | 0.000104 |
| | (0.00228) | (0.000308) |
| Age | -0.0470*** | -0.000636 |
| | (0.00337) | (0.000455) |
| ***Parents*** | | |
| Father age | 0.000884 | 0.0000557 |
| | (0.000554) | (0.0000750) |
| Mother age | -0.0000544 | -0.00000947 |
| | (0.000620) | (0.0000838) |
| Father wage inc. | 0.000117 | 0.0000113 |
| | (0.000138) | (0.0000187) |
| Mother wage inc. | -0.000235 | -0.0000463* |
| | (0.000203) | (0.0000274) |
| Father employed | -0.00189 | -0.00154 |
| | (0.00716) | (0.000968) |
| Mother employed | -0.0104 | 0.000648 |
| | (0.00739) | (0.001000) |
| Mother gen. HS | -0.00281 | -0.000608 |
| | (0.00629) | (0.000850) |
| Father gen. HS | 0.000239 | 0.000878 |
| | (0.00769) | (0.00104) |
| Mother DI | 0.000363 | 0.00147 |
| | (0.00984) | (0.00133) |
| Father DI | -0.00886 | -0.00240* |
| | (0.0102) | (0.00138) |
| Mother married | 0.00461 | -0.000312 |
| | (0.00540) | (0.000730) |
| ***Diagnosis in application*** | | |
| Mood/Anx/Dev. dis. | -0.0841*** | 0.0000300 |
| | (0.00774) | (0.00105) |
| ADHD/conduct | -0.117*** | -0.00000438 |
| | (0.0101) | (0.00137) |
| Other MH | -0.0368*** | 0.00119 |
| | (0.0141) | (0.00190) |
| Other Physical | -0.146*** | -0.0000604 |
| | (0.00948) | (0.00128) |
| ***Healthcare use*** | | |
| Hosp. visits | 0.00129*** | 0.0000173 |
| | (0.000199) | (0.0000268) |
| Any Depression vis. | 0.0104 | -0.000875 |
| | (0.00853) | (0.00115) |
| Any anxiety vis. | 0.0138 | -0.00108 |
| | (0.00843) | (0.00114) |
| Any intell. dis. visit | 0.0722*** | 0.000783 |
| | (0.00777) | (0.00105) |
| Any ADHD/conduct visit | 0.0455*** | -0.000937 |
| | (0.00791) | (0.00107) |
| Any nervous system vis. | 0.0293*** | -0.000322 |
| | (0.00956) | (0.00129) |
| N of diff. prescr. | -0.00114** | -0.00000734 |
| | (0.000487) | (0.0000658) |
| N of antidepr. prescr. | 0.00348 | -0.000327 |
| | (0.00798) | (0.00108) |
| N of Diabetes prescr. | -0.0134 | 0.000166 |
| | (0.0124) | (0.00168) |
| Observations | 30623 | 30623 |
| F | 52.56 | 0.899 |
| p | 3.24e-305 | 0.624 |

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

Table 4: Balance tests. Column 1 regresses first-application approval on applicant characteristics; column 2 regresses examiner leniency on the same characteristics. The bottom rows report joint F-tests and their p-values. Standard errors are in parentheses.

may be rejected by every examiner. Although individual compliers cannot be identified, I can describe the group in several ways.

Following Maestas et al. (2013), under strict monotonicity the share of compliers can be estimated from Fig. 6 or as $\alpha \cdot (\bar{z_j} - \underline{z_j})$, where $\bar{z_j}$ and $\underline{z_j}$ are the highest and lowest values of the instrument. Under average monotonicity, this expression instead gives a lower bound on the complier share (Norris et al., 2021). Compliers account for roughly 23% of applicants, so the LATE is relevant for a substantial share of the applicant population.[16]

The estimates nevertheless remain local to the complier population. Without additional assumptions, they are not informative about treatment effects for applicants whom every examiner would approve or every examiner would reject.

Following Bhuller et al. (2020), I use complier weights to compare OLS and IV estimates for comparable populations (Table 6). These weights are constructed by a comparison of first-stage coefficients in different subsamples.

Although individual compliers are unobserved, Frandsen et al. (2023) show how to estimate their mean characteristics by simple regression. I report these estimates in Table 8. Compliers are less likely than the average applicant to have disabilities severe enough to qualify for adult Disability Insurance.

#### 4.2.4 Marginal Treatment Effects

A way of inspecting heterogeneous treatment effects and understanding the external validity of the IV estimates is to estimate Marginal Treatment Effects (MTEs) (Heckman and Vytlacil, 2007; Carneiro et al., 2010). The approach has attractive features: it takes into account the information available in the instrument in more detail and this can be used to estimate different parameters, even the Average Treatment Effect under strict assumptions.

The MTE approach has some drawbacks in my setting. First, the estimation and interpretation as usual requires *strict monotonicity* (Frandsen et al., 2023), instead of the average monotonicity I invoke in my setting. This assumption imposes rather implausible assumptions on the examiner behaviour (Mogstad and Torgovitsky, 2024). While it is possible that the MTEs can be regardless estimated and interpreted in a different way under violations of strict monotonicity (Sigstad, 2024), the literature on this is currently not settled.

Secondly, the MTE does not provide a clear way to estimate the effects on the large group of always-takers in my setting, especially the interesting group of individuals with low labor market prospects. The unobserved resistance does not directly map on any observable feature that could be used in understanding the mechanisms or targeting, and the results on the group that are most likely to be approved to the benefit are noisy and sensitive to the parametric assumptions made, as the instrument does not span the entire interval.

With these caveats in mind, I report MTE estimates in the Appendix as suggestive evidence. The drawbacks of the approach in my setting also motivate another way of understanding the effect of the Youth Rehabilitation Allowance on groups beyond the marginal applicants: comparison of observably similar rejected and approved applicants.

[16] The formula is $\hat{\alpha} \cdot (\bar{z_j} - \underline{z_j})$, evaluated at the first-stage coefficient $\hat{\alpha} = 0.803$ and at the highest and lowest values the instrument takes in the estimation sample.

### 4.3 Selection-on-observables approach

As mentioned in Section 4.1, a possible assumption is ***the conditional ignorability assumption***:

$$\big(Y_{it}(1),\, Y_{it}(0)\big) \perp Approved_i \mid X_i, \qquad 0 < P(Approved_i = 1 \mid X_i) < 1. \tag{6}$$

The first part says that among applicants with the same observed characteristics, approval is unrelated to what would have happened to them under either decision. The second part, overlap, says that no combination of characteristics is either always approved or always rejected. This assumption would mean that an adequate selection-on-observables strategy would yield unbiased estimates of treatment parameters.

A common perception since the seminal paper of LaLonde (1986) has been that when evaluating public policy programs, covariate-adjustment methods are hence unreliable, as individuals select into programs based on unobservable variables. Literature has pointed out that selection-on-observables methods have a hard time replicating the experimental benchmarks (LaLonde, 1986; Smith and Todd, 2005). However, with the increased use of rich administrative data and modern estimators, the case for selection-on-observables approaches is not as hopeless as once thought (Dehejia and Wahba, 2002; Imbens and Xu, 2025). By ensuring overlap, meaning that individuals with similar probabilities of treatment are compared, and using more detailed background data, experimental benchmarks can often be recovered. This is not a guarantee that such a strategy could work in my setting, but I argue that multiple details make this approach attractive.

My data allow using large amounts of background information to compare plausibly similar individuals and then verifying these estimates using placebo outcomes and balance tests measured prior to the treatment. My setting is an example of this: I have a natural control group of rejected applicants, plausibly more similar to the approved applicants than to the general population. I observe detailed data on the individuals, including much of the data available on the applications and a long history of healthcare use, allowing me to approximate the information observed by the examiners. As an additional benefit, I can compare my estimates (weighted to the complier population) to the IV estimates, providing an additional layer of validation.

The stability of the coefficients to additional covariates can also be assessed in the spirit of Altonji et al. (2005) and Oster (2019). I also observe many outcomes, like the healthcare use and parental labor market ones, prior to treatment. The treatment cannot have an effect on these outcomes, and any viable method should find a null effect here (Imbens and Rubin, 2015). These checks provide a solid basis for use of selection-on-observables methods. Still, the results rely on strong assumptions, which should be acknowledged.

Against the weight of the conditional ignorability assumption, the benefits of assuming conditional ignorability should be considered. There are two main benefits. While the IV estimates are sufficiently precise to establish economically important effects on adult employment and NEET status, they are less informative for some other outcomes like health and crime and cannot distinguish plausible effects among the highest-risk applicants, for whom examiner leniency generates much less approval variation.

This problem is not unique to my paper but shared with many papers using an (ex-

aminer) IV design. Secondly, the IV compliers are a selected group of the applicants, and extrapolating the effects outside of these compliers requires assumptions that are also not innocent. The problems are related in that the IV design has especially low power on some interesting subgroups that are under-represented in the complier population, for instance, the individuals with the most severe disabilities. For this group, the IV estimate on employment is $-0.07$ (SE $= 0.17$, F-statistic $= 24$, using similar choices as in Table 6). At 80% power and a 5% two-sided test, the minimum detectable effect is $\approx 2.8 \times 0.17 = 0.48$, more than three times the untreated employment rate of 15% in this group. Even an effect that eliminated employment entirely would be too small for the IV to detect. Usual approaches to extrapolating the LATE from IV designs without full support to the whole sample, like parametric MTE, rely on strong assumptions about the functional form as well as strong monotonicity. In addition, these approaches will also suffer from the power issues inherent with the IV. This suggests that if we wish to understand if the program can help anyone, other approaches should be tried.

More precisely, as the main selection-on-observables approach, I use Double/Debiased Machine Learning (DDML) (Chernozhukov et al., 2018; Ahrens et al., 2024). Relative to conventional selection-on-observables estimators, double machine learning is particularly well suited to settings like mine with a large number of pre-treatment covariates. It permits flexible, regularized estimation of the treatment and outcome models using different machine learning methods while using orthogonalization and cross-fitting to reduce overfitting and regularization bias in the treatment-effect estimate. I use an interacted model with logistic regression and gradient-boosted trees in my main estimation. In practice, different estimators give similar estimates when the most important predictors of treatment are included. Standard errors come from the influence function of the DDML estimator and are valid under the conditions in Chernozhukov et al. (2018).

Covariates used in the main estimation are (in total 356, and their interactions):

- **Applicant information:** Gender, Age, indicator for having 9th grade GPA, 9th grade GPA, applicant region, applicant lives in an urban area
- **Application information:** Year and month of application, diagnosis code group $\times$ year of application
- **Parental/family characteristics:** Parent ages, Parent education (3 categories), Parents employed, Parent's wage earnings, Parents in Disability Insurance
- **Healthcare use:** Indicators for any visit and number of visits for the ICD10 main categories + a more precise categorization for mental health disorders, antidepressant prescriptions filled, stimulant prescriptions filled, total drug expenses, and total hospital days

While I do not have access to the written application text data, the covariates well capture the selection process, as they represent much of the information plausibly available on the applications and used by the examiners. As argued by Angrist (1998) and Lindquist and Santavirta (2014), if we understand the selection criteria and have data on the variables that determine the selection, we can get a good handle on the selection bias

in this setting. The most important correlates of approval are the disability categories and previous educational performance. Within disability categories, individuals can yet have differently severe health problems. The measures of healthcare use plausibly capture much of this severity. Finally, parental background can proxy for the severity of the difficulties that applicants have or their skill in drafting the application. In practice, most parental characteristics have little correlation with approval to the program.

It is possible that some combination of, e.g., SES and disability category would affect approval in a particular way. The Double Machine Learning approach is a way to take this into account flexibly without overfitting: both the propensity score and the treatment effect are estimated using cross-folds. When necessary, I cluster the estimation of these at the individual level, so e.g. an individual's earlier observation is not used in the estimation of their propensity score.

### 4.3.1 Validity checks

Overlap means that the control and treatment groups have similar support in the covariate distributions. Ensuring overlap is important for any identification strategy relying on controlling observables. The most informative covariate is the propensity score, which is a combination of the covariates. Fig. 8 displays the propensity score distributions for the approved and rejected applicants. We can see that without any trimming, the propensity score distributions overlap pretty well. I drop the top and bottom 1% of the propensity scores. A similar graph, but for the high DI risk subgroup can be found in the Appendix Fig. A.17.

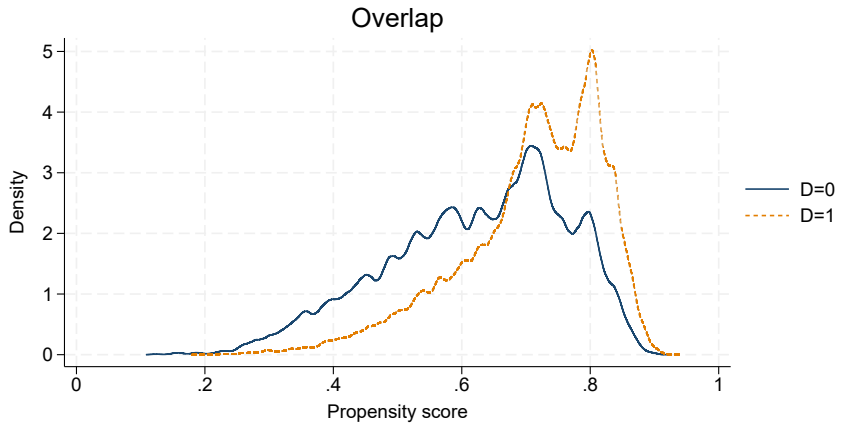



Figure 8: Overlap figure for the covariate adjustment method. The figure displays the distribution of the propensity scores estimated by the DDML procedure, using all covariates. 'D=1' refers to the distribution of the treated (approved) individuals, while 'D=0' refers to the distribution of the control (not approved) observations. Note that this propensity score is estimated for the main sample of individuals observed during years 6-12 from first application, for other samples, the figure would look slightly different. A similar figure for the low labor market prospects/high DI risk subsample is available in Fig. A.17.

Second, we can check whether, when using the chosen strategy, the treatment has an

effect on placebo outcomes (measured before the treatment) or immutable characteristics. For this exercise, I remove the chosen covariates (and any derived from them or very correlated with, like employment and earnings) from the covariate set. I then regress the variables individually on the treatment. Fig. 9 portrays the results from this exercise. While the OLS estimates without controls show that the approved applicants fare worse than the rejected in most characteristics, most of the coefficients from the DDML procedure are not statistically significant and are small also in magnitude. All of the coefficients displayed here and more are included in the estimation of the effects, so even if there were imbalances here it would not necessarily be damning for the design, as they are still controlled for in the results.

I perform a similar exercise separately for the high DI risk subgroup and find similarly mostly small imbalances (Fig. A.18). While some coefficients are significantly different from zero, there is no particular direction for these biases and even the covariates with any imbalance are included in the covariate set used for estimation.

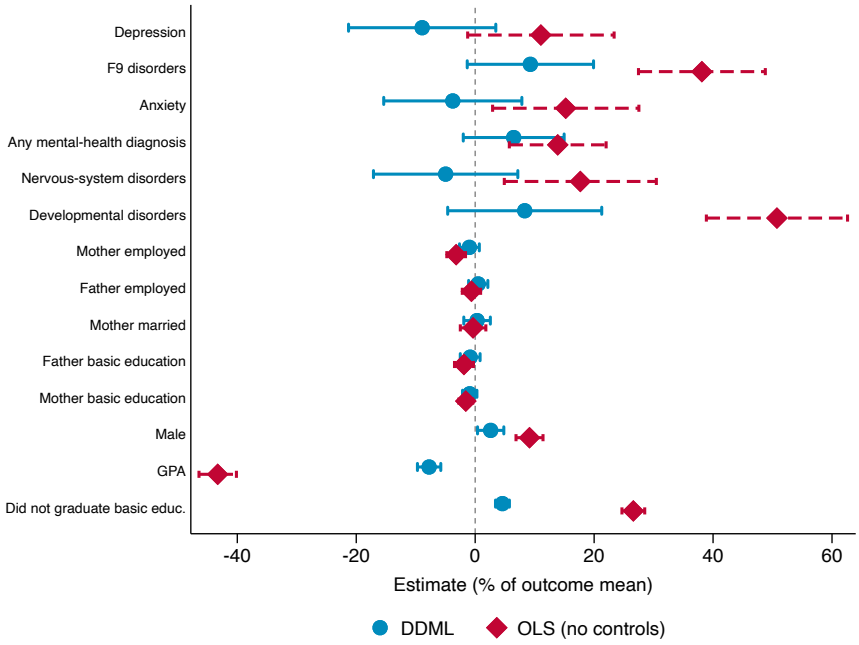



Figure 9: Balance test for the covariate adjustment design. For each coefficient, I run the DDML procedure with the variable listed here, measured at time -2, as the outcome. I leave out from each regression the closely correlated variables and the ones directly derived from this variable. The red markers display the raw associations between approval and a characteristic.

The validity checks presented here support the conditional ignorability assumption. In the results section, I present additional validity assessments. Stability of the results when including different sets of covariates is intuitively an indication of the extent of selection bias in a setting (Altonji et al., 2005). More formally, I also calculate the $\delta$ coefficients proposed by Oster (2019), setting the maximum $R^2$ to 1.3 times the $R^2$ of the controlled regression [17]. The Oster $\delta$ shows how much more important selection on unobservables would have to be compared to the selection on observables to make the estimates equal to

[17] Examples of the use of this logic in published papers include Nunn and Wantchekon (2011); Lindquist and Santavirta (2014); Mian et al. (2023); Dustmann et al. (2023) and many more.

0. A result $\delta > 1$ is often considered a robust one: the selection on unobservables would have to be stronger than selection on observables. I recover values between 2.6 and 6 for my key outcomes (Employment and NEET status). For the especially interesting subgroup of low labor market prospect individuals, the selection on observables biases the estimates away from zero. Unless the selection-on-unobservables has an opposite sign, the true effect would therefore be even more positive (for e.g. employment) than the estimates reported.

A further advantage in my setting is that under the assumptions A1-A4 I have a benchmark I can use to assess the performance of the covariate adjustment in removing selection bias: the IV estimates. A complication is that the IV estimates represent a treatment effect on the instrument compliers only. A simple comparison of the coefficients is therefore not informative of selection bias in the presence of heterogeneous treatment effects. I solve this issue by calculating complier weights using the approach used by Bhuller et al. (2020) and weighting my DDML sample with these weights [18]. The results obtained this way are very close to the IV results for the main outcomes.

The fact that adjusting for covariates can remove selection among the compliers is not a guarantee that the same covariates remove selection in other parts of the sample. For this validity check to be informative about the selection in other subgroups, the covariates that close the gap between approved and rejected applicants in the middle of the labor market prospect distribution should also close it at the bottom of it: that selection runs through the same observed variables for the severely disabled as for the marginal applicant, in a similar manner to Aizer et al. (2024). Together with the other validity checks, though, it provides good reasons to find the conditional ignorability assumption a plausible one.

# 5 Results

YRA approval reduces labor-market participation after benefit eligibility ends, despite having little effect on educational attainment. I first report short-run effects on education, work, and benefit receipt, which help interpret the longer-run results. I then examine health and healthcare use, test the sensitivity of the IV estimates, describe the compliers, and present results based on covariate adjustment.

## 5.1 IV results

### 5.1.1 Short-run impacts

Among marginal applicants, YRA approval generates little additional schooling but appears to displace earnings during benefit eligibility. Table 5 presents cumulative outcomes through age 21, when the education or rehabilitation condition ends. These measures summarize outcomes over the benefit period.

[18] In practice, this works by splitting the sample and comparing the first stage in these samples. The larger the first stage, the larger the share of compliers in this split of the sample. This restriction does not identify individual compliers; it selects the observable groups in which the first stage indicates the highest concentration of compliers. In the DDML procedure I restrict to the top 20 % of the sample with the highest complier likelihood. The methodology does allow for continuous weights, I instead approximate this for the purposes of my validation exercise. The results are not sensitive to the exact definition.

YRA receipt is conditional on attendance in secondary education or another rehabilitative activity. Most marginal applicants would attend secondary education even without YRA, leaving little scope for the benefit to raise enrollment. The IV estimates confirm this: years spent as a student barely increase (Table 5). The confidence intervals rule out gains as large as those found for conditional cash transfers in developing countries (Schultz, 2004; Glewwe and Kassouf, 2012); the estimates are closer to those for the UK Education Maintenance Allowance (Britton et al., 2025).

YRA approval raises secondary education completion by around 2 percentage points by age 21, although the estimate is imprecise. As rejected applicants catch up, the completion gap narrows by age 25. Recipients are no more likely to enter higher education (Fig. A.1).

As policymakers intended, YRA approval also reduces Disability Insurance (DI) receipt. This effect is mechanical because YRA replaces DI. Its magnitude indicates how often approved applicants would otherwise apply for and be awarded DI. The decline is small among compliers. Displacement may be greater among applicants with the weakest labor-market prospects and the highest probability of receiving DI in the absence of YRA, but this group is underrepresented among compliers.

| | (1) Cum. years schooling | (2) Secondary degree | (3) Cum. years on DI | (4) Cum. wages |
|---|---|---|---|---|
| *Panel A. Main estimates* | | | | |
| OLS | -0.024 | -0.076*** | -0.248*** | -2890.494*** |
| | (0.017) | (0.006) | (0.014) | (138.981) |
| IV | 0.005 | 0.022 | -0.149 | -2312.331* |
| | (0.162) | (0.058) | (0.127) | (1227.004) |
| AR $p$-value | 0.974 | 0.706 | 0.242 | 0.060 |
| First-stage $F$ | 386 | 386 | 386 | 386 |
| *Panel B. IV with additional controls* | | | | |
| IV | 0.046 | 0.041 | -0.189 | -1843.815 |
| | (0.157) | (0.056) | (0.118) | (1207.896) |
| AR $p$-value | 0.771 | 0.462 | 0.111 | 0.127 |
| First-stage $F$ | 388 | 388 | 388 | 388 |
| *Untreated complier mean* | *4.11* | *0.395* | *0.367* | *6475* |
| *Sample mean* | *4.08* | *0.382* | *0.430* | *5326* |
| N | 30420 | 30409 | 30420 | 30420 |

Table 5: Main IV results on cumulative education and labor-market outcomes by age 21 (the end of benefit eligibility). "Cum. years of schooling" refers to years spent as a student after age 15. The sample size is the number of individuals observed through age 21. Standard errors are clustered at the individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

YRA approval reduces employment and cumulative earnings during eligibility. Table 5 shows a decline in cumulative wage earnings by age 21, and Fig. 10 shows a decline in employment. At least some applicants would therefore have worked without YRA. The decline in cumulative earnings by age 21 amounts to 36% of the untreated complier mean. Early labor-market experience can be valuable, especially for young people with weak labor-market attachment (e.g. Heller, 2014). The earnings decline likely reflects reductions

in summer or part-time work, since few rejected applicants work full time.

Instrumenting current YRA receipt rather than first-application approval yields a similar earnings decline, but the estimates become less precise as recipients leave the program and the first stage weakens (Fig. A.9).

Although the IV estimates are imprecise, they consistently indicate less work during eligibility. Because the benefit is temporary, these short-run responses help interpret its long-run effects.

### 5.1.2 Long-run impacts

After eligibility ends, YRA approval reduces employment by 11.2 percentage points and raises NEET status by 11.7 percentage points. Table 6 presents the main labor-market results. I measure adult outcomes six to twelve years after application, roughly ages 21–28, when recipients are no longer subject to the program's education or rehabilitation condition. The regressions pool annual observations over this period.

This window captures adult labor-market outcomes in this sample. Although many Finns complete higher education between ages 25 and 30, relatively few applicants in my sample ever attend higher education. They are also on the margins of labor-market participation, so their earnings profiles are likely to be flatter than those of an average worker.

These effects amount to 28% and 26% of the untreated complier means, respectively. Both estimates are statistically significant. Adding extensive controls in panel B leaves the estimates nearly unchanged. The estimates remain statistically significant in most robustness specifications, with magnitudes close to the main estimates.

Annual earnings fall by €1,800, or 19% of the untreated complier mean. Applicants whose main activity is employment earn €20,028 per year on average, so the observed decline is smaller than the €2,243 implied by the employment change alone. This difference suggests that applicants induced not to work earn less than the average employed applicant.

There is no measurable change in student status. The nearly equal decline in employment and rise in NEET status already imply little change in other activities. The point estimates for student status are close to zero, and the confidence intervals are relatively tight. Continuing education beyond secondary school is rare in this population, so applicants move mainly between employment and being out of employment, education, or training.

The OLS estimates are similar to, or slightly more negative than, the IV estimates. This pattern can reflect either selection into treatment or differences between the compliers and the full applicant sample. Table 6 shows that untreated compliers are more likely to be employed and have higher income than the full sample. I discuss these differences further in Section 5.1.5.

The employment and NEET effects persist after eligibility ends and are larger than during YRA receipt. Figure 10 presents the yearly IV estimates of the main labor-market outcomes. Effects on earnings follow a similar pattern (Panel C).

Disposable income rises during YRA receipt but is nearly unchanged after eligibility ends because other transfers replace lost earnings. Panel D shows the initial increase in

disposable income. This increase is smaller than the estimated increase in YRA payments because rejected applicants can receive other benefits. Among applicants observed through age 27, the estimated increase in benefit receipt exceeds the estimated earnings loss (Section 6.2).

A more precise decomposition by benefit type is shown in Fig. A.3. Unemployment insurance and social-assistance payments increase the most, and both increases are statistically significant. This reflects the decline in labor-market participation. In contrast, there is no detectable change in pension or DI receipt. This null is also informative about the information and beliefs channel, which I return to in the mechanisms section.

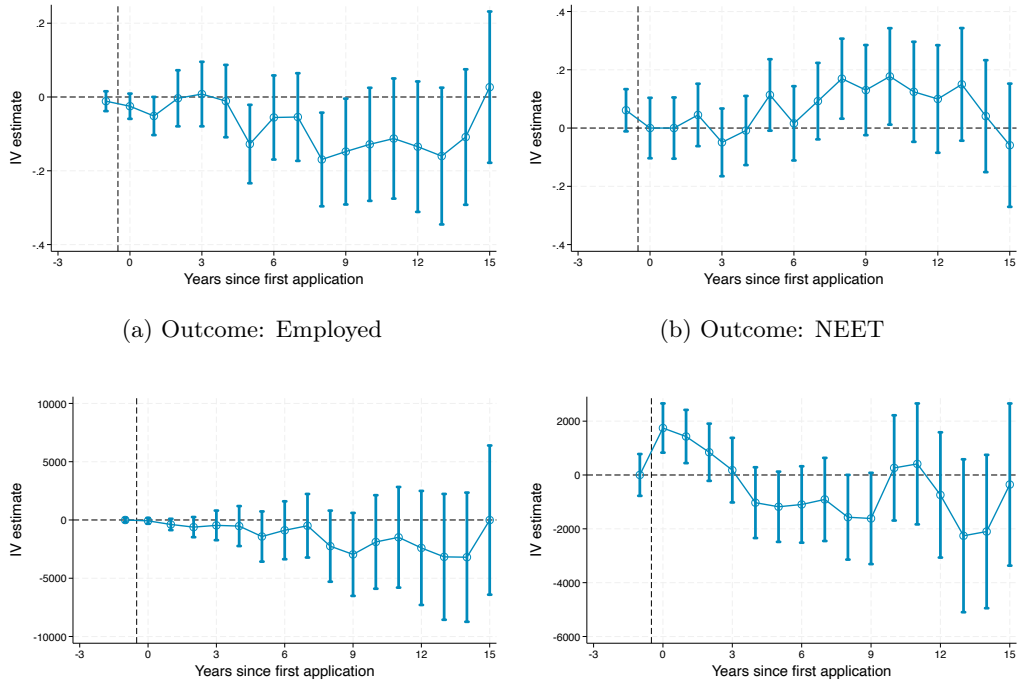
Applicants are on the margins of labor-market participation and may therefore respond strongly to YRA. The confidence intervals are wide but rule out large positive effects. The program therefore does not achieve its stated goal of increasing recipients' labor-market attachment.

| | (1) NEET | (2) Employed | (3) Empl. income (euros) | (4) Student |
|---|---|---|---|---|
| *Panel A. Main estimates* | | | | |
| OLS | 0.115*** | -0.100*** | -3,083*** | -0.015*** |
| | (0.003) | (0.003) | (77) | (0.002) |
| IV | 0.117** | -0.112** | -1,756 | -0.006 |
| | (0.054) | (0.050) | (1,331) | (0.032) |
| AR $p$-value | 0.031 | 0.026 | 0.189 | 0.842 |
| First-stage $F$ | 315 | 315 | 315 | 315 |
| *Panel B. IV with additional controls* | | | | |
| IV | 0.100* | -0.098** | -1,455 | -0.003 |
| | (0.051) | (0.049) | (1,298) | (0.032) |
| AR $p$-value | 0.052 | 0.043 | 0.263 | 0.927 |
| First-stage $F$ | 319 | 319 | 319 | 319 |
| *Untreated complier mean* | *0.454* | *0.399* | *9,177* | *0.147* |
| *Sample mean* | *0.567* | *0.295* | *6,802* | *0.138* |
| Observations | 102,595 | 102,595 | 102,595 | 102,595 |
| Individuals | 23,596 | 23,596 | 23,596 | 23,596 |

Table 6: Main results on labor-market outcomes using EB-IV. Outcomes are measured during years 6–12 after the first application, which roughly corresponds to ages 21–28. Labor-market outcomes are available through 2022. Both OLS and IV include region × year × language fixed effects. IV with additional controls also includes disability category, indicators for depression, anxiety, F9, and other healthcare visits, urban residence, parental employment, parent ages, and ninth-grade GPA. The sample size refers to individual-year observations. Standard errors are clustered at the individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

### 5.1.3 Health and healthcare use

YRA approval does not detectably change hospital days, overnight mental-health admissions, or healthcare use during benefit eligibility (Table 7). Administrative healthcare data reflect both underlying health and care-seeking. I use the severity of the outcomes to distinguish between these dimensions. Hospitalizations and overnight mental-health ad-




(a) Outcome: Employed

(b) Outcome: NEET

(c) Outcome: Wage earnings

(d) Outcome: Disposable income

Figure 10: Yearly IV estimates of the effects of YRA approval. Each point comes from a separate EB-IV regression for the indicated event year. The outcomes are not recorded before age 16, so the pre-treatment estimates are pulled toward zero. Estimates are not scaled, so an estimate of 0.1 for a binary outcome means an increase of 10 percentage points.

missions capture events that are serious enough that people are likely to seek care, so I treat them mainly as measures of underlying health. Intensive margin measures of visits and prescriptions also respond to resources and incentives to seek care. Healthcare co-payments are low or zero, especially for individuals under 18, but even small co-payments can reduce healthcare use (Haaga et al., 2024).

YRA may affect health and healthcare use in different ways. Additional income may reduce stress or increase treatment. Reduced work may lower strain, but it may also lead to a more passive lifestyle.

Columns 1 and 2 of Table 7 report effects on serious health outcomes over the years in which the benefit is most commonly received. I find no detectable effects on hospital days or overnight mental-health visits. The estimates are somewhat imprecise, but they provide no evidence of improvements in these measures of serious adverse health events.

I also find no evidence that the benefit substantially changes healthcare use. The IV coefficients are negative but not statistically significant, and small effects cannot be ruled out. This is consistent with the hypothesis that the individuals' healthcare use is not substantially constrained by money.

Table A.1 presents health outcomes after the benefit ends. The estimates indicate a decline in mental-health visits, although this effect is not statistically significant. Overall, I find no evidence that the YRA affects health or healthcare use, though the confidence intervals include medium-sized effects. The estimates based on the conditional-ignorability approach in Section 5.2 are more precise.

| | (1) Hosp. days | (2) Overnight MH visits | (3) MH visits | (4) Any antidepr. |
|---|---|---|---|---|
| OLS | 0.339*** | 0.020*** | 0.029*** | -0.015*** |
| | (0.064) | (0.004) | (0.003) | (0.002) |
| IV | 0.326 | 0.055 | -0.028 | -0.033 |
| | (0.904) | (0.062) | (0.041) | (0.040) |
| AR p-value (IV) | 0.719 | 0.374 | 0.493 | 0.407 |
| First stage F (IV) | 394 | 394 | 394 | 394 |
| IV, controls | 0.257 | 0.051 | -0.029 | -0.030 |
| | (0.884) | (0.062) | (0.038) | (0.039) |
| AR p-value (IV, contr.) | 0.771 | 0.411 | 0.456 | 0.443 |
| First stage F (IV, contr.) | 394 | 394 | 394 | 394 |
| Untreated complier mean | 1.85 | .0797 | .368 | .244 |
| Sample mean | 1.94 | .118 | .411 | .24 |
| N | 152014 | 152014 | 152014 | 152014 |
| N indiv. | 30534 | 30534 | 30534 | 30534 |

Table 7: Main results on health outcomes using EB-IV. Outcomes are measured years 0-5 after the first application, which corresponds to the time of benefit eligibility. Health outcomes are available until 2023. Both OLS and IV use the same, necessary controls. IV with additional controls uses controls for the disability category, indicators for any depression/anxiety/F9/other healthcare visits, urban area, parental employment, parent ages, and 9th grade GPA. Sample size refers to individual × year observations. Standard errors clustered at the individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

#### 5.1.4 Robustness of the IV estimates

I assess the sensitivity of the IV estimates to alternative estimators, instrument definitions, controls, and sample restrictions.

***Empirical Bayes*:** Using the conventional leave-one-out examiner-leniency instrument produces similar estimates, so Empirical Bayes shrinkage does not drive the results (Fig. A.13).

***Additional controls*:** The main tables already include a large set of additional controls. In the appendix, I also residualize the instrument by disorder category and add year × month fixed effects. The results remain similar. This reduces the concern that applicants may sort based on the timing or contents of their applications.

***UJIVE*:** I also report estimates from the Unbiased Jackknife IV estimator (UJIVE), following Kolesar (2013); Goldsmith-Pinkham et al. (2026). UJIVE performs well with many covariates. Because it requires constructing examiner leniency within the estimation sample, its first stage is noisier and its second-stage confidence intervals are wider. The estimates remain close to the baseline (Fig. A.13).

***Binary instrument*:** Mogstad and Torgovitsky (2024) argue that monotonicity is more plausible for a binary instrument than for a continuous measure of leniency. I therefore restrict the sample to applicants assigned to examiners in the top or bottom

25% of the leniency distribution and use assignment to the top group as the instrument. I repeat the exercise using the top and bottom 10%. Both specifications yield estimates close to the baseline (Fig. A.13).

***Examiner case-count threshold*:** I vary the minimum number of cases an examiner must have to enter the estimation sample. The estimates remain stable across these thresholds (Fig. A.14).

***Limit to a balanced sample*:** In Table A.11 I show the long-run estimates remain similar when limiting the sample to only the balanced sample.

***Dropping one region at a time*:** The estimates remain stable when I exclude each region in turn, so no single region drives the results (Fig. A.15).

### 5.1.5 Compliers, MTE, and the limits of the IV

The IV estimates apply to applicants whose approval depends on examiner leniency. Table 8 compares compliers with the full applicant sample and with approved applicants. I estimate the complier share following Bhuller et al. (2020). Compliers are more likely to be male and to have a developmental disorder, but less likely to have a mood disorder. Their diagnoses are less likely to lead to adult DI, and their parents have weaker labor-market attachment.

Treatment effects may vary with baseline labor-market prospects, but the local IV estimates are too imprecise to trace heterogeneity along this margin. Estimating these effects requires a different approach. One possible approach is the Marginal Treatment Effects (MTE) (Section 4.2.4). I estimate different versions of the MTE and report the results in Section A.3. All specifications are consistent with the treatment being beneficial for those with the lowest unobserved resistance, but the magnitudes vary widely for different specification choices. These estimates should be taken as suggestive also because of the precision: the confidence intervals in the low-resistance end span the entire plausible interval of the binary outcome (not included for clarity). The estimated ATE is around -0.07 using the normal distribution assumption for the error terms, which ends up being close to the ATE estimated using the selection-on-observables approach. The sensitivity and lack of precision confirm the inadequacy of the IV alone to answer the question about the effects on those with the lowest labor market prospects.

I therefore complement the IV with selection on observables. Under conditional ignorability and overlap, covariate adjustment yields more precise estimates for the broader applicant population and for groups underrepresented among compliers. Section 4.3 describes the approach.

## 5.2 Results under Conditional Ignorability

The covariate-adjusted estimates vary with baseline labor-market prospects. On average, YRA approval reduces employment and increases NEET status. Among applicants with low labor-market prospects, however, it raises employment and reduces NEET status. I present the main results and then show that my selection-on-observables estimates track closely the IV estimates on the key outcomes.

| | Complier share (%) | Sample share (%) | Share among accepted (%) |
|---|---|---|---|
| Male | 59.83 | 53.00 | 54.52 |
| Mood disorder (F3) | 7.28 | 13.86 | 12.48 |
| Anxiety disorder (F4) | 8.79 | 8.95 | 8.02 |
| Developmental disorder (F8) | 36.42 | 25.59 | 27.02 |
| ADHD/Conduct disorder (F9) | 12.48 | 14.12 | 16.63 |
| Other mental-health disorder | 6.00 | 6.12 | 4.11 |
| Chromosomal/intellectual disorder | 15.22 | 19.01 | 21.72 |
| Other physical disorder | 13.81 | 12.35 | 10.02 |
| Urban | 76.33 | 79.66 | 78.67 |
| Mother employed | 60.56 | 70.39 | 69.69 |
| Father employed | 66.57 | 72.29 | 72.10 |
| High DI risk | 12.46 | 18.91 | 21.39 |
| Low DI risk | 87.54 | 81.09 | 78.61 |

Table 8: Complier characteristics. $D_i$ indicates YRA approval, and $Y_i$ denotes the characteristic listed in each row. Following Bhuller et al. (2020), I estimate the complier mean of $Y_i$ from an IV regression of $Y_i(D_i - 1)$ on $D_i - 1$, using examiner leniency as the instrument. The table also reports the complier share implied by the first stage.

### 5.2.1 Main results

On average, YRA approval worsens long-run labor-market outcomes, although the effects are smaller than the corresponding complier estimates. Table 9 displays the main long-run results for three groups. In row 1, I report the estimates for the aggregate applicant sample, which I refer to as ATE. Approval lowers employment by 4.7 percentage points and raises NEET status by 4.6 percentage points. These effects are smaller than both the IV estimates and the DDML estimates for compliers. The covariate-adjusted estimates are also less negative than the OLS estimates in Table 6, consistent with negative selection into approval on observed characteristics. The estimates for long-run student status are close to zero and precisely estimated in all three groups.

By contrast, among applicants with low labor-market prospects, approval raises employment by 4.7 percentage points and reduces NEET status by 4.9 percentage points. The employment gain is about 29 percent of the group mean. Figure A.7 shows the same pattern across risk deciles. Employment effects are negative through most of the distribution but turn positive in the highest-risk deciles, while NEET effects move in the opposite direction. The highest-risk deciles also show the largest increases in years of education and the largest reductions in years on DI by age 21.

Among applicants with low labor-market prospects, selection on observables appears to attenuate the estimated effects. The negative Oster $\delta$ values indicate that adding controls moves the estimates away from zero. If selection on unobservables operates in the same direction, the reported effects are biased toward zero.

The earnings effects also differ across groups. The yearly estimates in Section A.1.3 show lower earnings for the full sample but higher earnings among applicants with low labor-market prospects. For applicants with low labor-market prospects, the increase is about €800 per year, or roughly 28 percent of mean earnings.

| | (1) Employed | (2) NEET | (3) Empl. income (euros) | (4) Student |
|---|---|---|---|---|
| ***ATE*** | -0.047*** | 0.046*** | -1632.184*** | 0.001 |
| | (0.003) | (0.003) | (77.075) | (0.002) |
| Oster delta(0) | 3.1 | 2.6 | 3.7 | -.043 |
| Outcome mean | .29 | .57 | 6802 | .14 |
| N | 102589 | 102589 | 102589 | 102589 |
| ***Moderate labor market prospects*** | -0.074*** | 0.073*** | -2306.439*** | -0.000 |
| | (0.004) | (0.004) | (95.259) | (0.003) |
| Oster delta(0) | 5.2 | 5.3 | 6 | 2.7 |
| Outcome mean | .33 | .51 | 7951 | .16 |
| N | 79592 | 79592 | 79592 | 79592 |
| ***Low labor market prospects*** | 0.047*** | -0.049*** | 679.445*** | 0.003 |
| | (0.005) | (0.007) | (113.602) | (0.004) |
| Oster delta(0) | -41 | -28 | -12 | -9.5 |
| Outcome mean | .16 | .77 | 2826 | .072 |
| N | 22997 | 22997 | 22997 | 22997 |
| Controls | All | All | All | All |

Table 9: Covariate-adjusted estimates obtained using double/debiased machine learning (DDML) (Chernozhukov et al., 2018). Outcomes are measured in years 6–12 after the first application, a period that roughly corresponds to ages 21–28. Estimation and SEs clustered at the individual level.

Short-run effects also differ sharply by applicants' labor-market prospects (Table 10). Cumulative earnings fall among applicants with moderate labor-market prospects, while the estimate for applicants with low labor-market prospects is close to zero. Among applicants with low labor-market prospects, YRA approval also increases time in education by 0.73 years and raises secondary-school completion by 14.3 percentage points. These results imply that many applicants in this group would otherwise have left education.

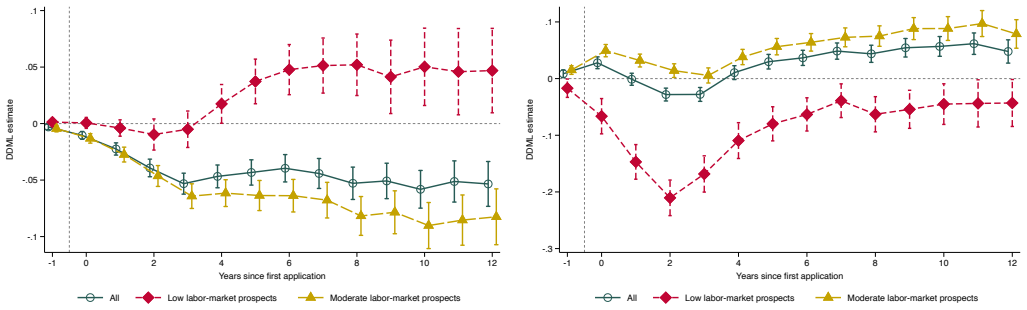
### 5.2.2 Validation

Two main checks support the conditional-ignorability design: the covariate-adjusted estimates closely match the IV estimates for compliers, and the estimates change little across covariate sets.

Table 11 compares the IV and complier-weighted DDML estimates for the main labor-market outcomes. The point estimates are numerically very close to each other. Fig. A.19 presents year-by-year estimates with the two designs, which show very similar patterns. While I have limited power to detect very small differences, the similarity between the estimates supports the assumption that most of the relevant selection into treatment can be captured by the observables.

The IV comparison is informative only for the complier-weighted population. It does not directly support the estimates for applicants with low labor-market prospects, who are underrepresented among compliers. For this group, the case for the design rests on propensity-score overlap and stability across covariate sets.

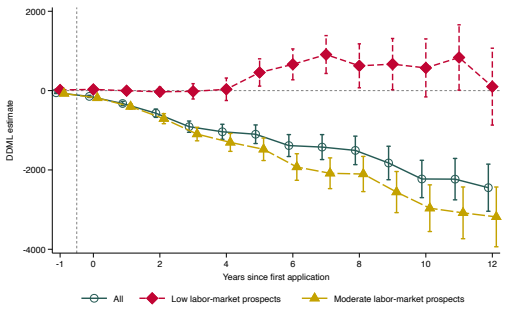
The estimates also change little across covariate sets. Only omitting application con-



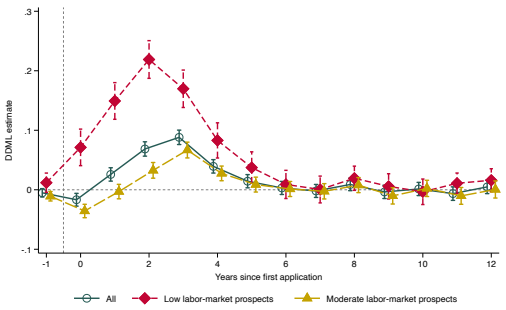

(a) Outcome: Employed




(b) Outcome: NEET




(c) Outcome: Wage earnings




(d) Outcome: Student

Figure 11: Main results using covariate adjustment. Each point is a DDML estimate for the corresponding event year. Employment, NEET status, wage earnings, and student status are not recorded before age 16, so the figure does not provide a consistent set of pre-application estimates. Coefficients are reported in the outcomes' original units; for a binary outcome, a coefficient of 0.1 represents a 10-percentage-point increase. "All" refers to the entire overlap sample. "Low labor-market prospects" refers to applicants in the top fifth of predicted DI risk, and "moderate labor-market prospects" refers to the remaining applicants.

| | (1) Cum. years schooling | (2) Secondary degree | (3) Cum. years on DI | (4) Cum. wages |
|---|---|---|---|---|
| ***ATE*** | 0.215*** | -0.006 | -0.513*** | -3101*** |
| | (0.019) | (0.006) | (0.019) | (219) |
| Oster $\delta(0)$ | -3.5 | 3.9 | -5.7 | 6.2 |
| Sample mean | 4.3 | .5 | .64 | 9097 |
| N | 30315 | 28948 | 30315 | 30315 |
| ***Moderate labor market prospects*** | 0.093*** | -0.040*** | -0.085*** | -3979*** |
| | (0.020) | (0.007) | (0.017) | (261) |
| Oster $\delta(0)$ | -2.7 | 19 | -2.9 | 11 |
| Sample mean | 4.5 | .53 | .38 | 10383 |
| N | 24604 | 23551 | 24604 | 24604 |
| ***Low labor market prospects*** | 0.729*** | 0.143*** | -2.320*** | 349 |
| | (0.056) | (0.015) | (0.073) | (364) |
| Oster $\delta(0)$ | 21 | 17 | 12 | -4.7 |
| Sample mean | 3.5 | .39 | 1.7 | 3542 |
| Controls | All | All | All | All |
| N | 5711 | 5397 | 5711 | 5711 |
| Age at obs. | 21 | 21 | 21 | 21 |

Table 10: Short-run estimates obtained from adjusting for covariates using DDML. 'Cum. years of schooling' refers to years spent as a student after age 15. Estimation and SEs clustered at the individual level.

trols changes them, which is unsurprising because these variables capture information available to examiners. I also report the Oster $\delta$ coefficients. For employment and NEET status, selection on unobservables would need to be 3.5 and 4.6 times as strong as selection on observables, respectively, to reduce the estimates to zero.

### 5.2.3 Other outcomes

YRA approval has no clear effects on crime or overnight mental-health visits (Fig. A.6). The selection-on-observables approach provides more precise estimates for these outcomes. There is a clear gradient of responses by the DI index in disability insurance and probability of being married, in the direction of the main outcomes.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| ***Employed*** | -0.112** | -0.097*** | -0.089*** | -0.104*** | -0.100*** | -0.105*** |
| | (0.050) | (0.007) | (0.007) | (0.007) | (0.007) | (0.007) |
| Oster delta(0) | | 3.5 | | | | |
| Outcome mean | .29 | .29 | .29 | .29 | .29 | .29 |
| ***NEET*** | 0.117** | 0.105*** | 0.091*** | 0.101*** | 0.105*** | 0.102*** |
| | (0.054) | (0.007) | (0.006) | (0.006) | (0.006) | (0.007) |
| Oster delta(0) | | 4.6 | | | | |
| Outcome mean | .57 | .57 | .57 | .57 | .57 | .57 |
| ***Student*** | -0.006 | 0.001 | 0.001 | 0.004 | -0.001 | 0.001 |
| | (0.032) | (0.005) | (0.005) | (0.005) | (0.005) | (0.005) |
| Oster delta(0) | | 1.4 | | | | |
| Outcome mean | .14 | .14 | .14 | .14 | .14 | .14 |
| Controls | IV controls | All | No parent | No applicant | No application | No healthcare |
| N | 102592 | 21784 | 21784 | 21784 | 21784 | 21784 |

Table 11: Comparison of the IV estimates and covariate-adjusted DDML estimates on the complier-weighted sample 6–12 years after the first application (Chernozhukov et al., 2018). The covariate sets are listed in Section 4.3. The Oster $\delta(0)$ measures the strength of selection on unobservables needed to reduce the estimate to zero. For instance, $\delta(0) = 2$ means that selection on unobservables would need to be twice as strong as selection on observables. I calculate $\delta(0)$ using an OLS regression with all controls and set $R^2_{max} = 1.3\tilde{R}^2$. The measure is not informative for student status, for which the estimate is already close to zero, but I report it for completeness. Estimation and SEs clustered at the individual level.

# 6 Discussion, mechanisms, and welfare effects

## 6.1 Mechanisms

The benefit has negative effects on the marginal applicants, but the evidence suggests positive effects for those with the lowest prospects. In this section I aim to understand why. It is harder to explain the negative effects, and for this reason I start with theories that could explain this.

The effects on the labor market are still visible a decade after the payments have stopped. This rules out the usual explanation in the adult disability insurance literature, where the negative effects follow from the income and substitution effects of a benefit currently being paid (Bound, 1991; Maestas et al., 2013; Autor et al., 2019). The YRA is paid mostly for four years and ends around age 20. The explanations for the effects observed in previous sections must therefore be something that changed during the benefit recipiency.

The heterogeneity in Section 5.2 suggests a possible direction. Every accepted applicant receives the same transfer, and the screening and the conditions are the same. The applicants differ in the counterfactuals. For the compliers of the IV design, who have moderate labor market prospects, the benefit reduced work: cumulative wage earnings by age 21 fall by 36 % of the untreated complier mean, while not shifting participation in education by much (Table 5). These individuals were not displaced from the Disability

Insurance, as intended by the policymakers.[19]

For the applicants with the lowest labor market prospects, the benefit replaced disability insurance, earnings during the benefit respond little, and the long-run effects on the labor market are positive (Table 9, Section A.1.3). It is also suggested as a potential mechanism in previous studies of temporary cash transfers that find negative effects (Barr et al., 2021; Price and Song, 2025; Britton et al., 2025). Consistent with this mechanism, Filippucci (2025) finds that active labor-market measures offset the negative effects of the cash transfer by changing recipients' activity. I will formalize this in the form of a standard human capital theory in the style of e.g. Becker (1964); Mincer (1974) and in a similar fashion as in the literature on conditional cash transfers in developing countries (Skoufias and Parker, 2001).

Previous literature on cash transfers also offers possible other explanations. One is the formation of 'Welfare cultures', so changes in beliefs, information or stigma related to welfare use (Moffitt, 1983; Lindbeck et al., 2003; Dahl et al., 2014). A second one is that cash can have negative effects on health (Dobkin and Puller, 2007; Evans and Moore, 2011; Black et al., 2024), possibly through consumption of harmful goods. Below, I will also consider the evidence for these theories.

### 6.1.1 Employment: Lost human capital among the compliers

The YRA can change what applicants do between ages 16 and 20. An applicant can attend education, work, receive disability insurance, or spend time in another activity. Approval combines a cash transfer with a requirement to attend education or rehabilitation. It may therefore increase education, but it may also replace work or disability insurance. These activities will accumulate different human capital and would result in ambiguous long-run effects.

I summarize this idea with a simple human-capital model. At the end of compulsory school, applicant $i$ has an initial stock of academic human capital, proxied by ninth-grade GPA, $G_i$. Let $d \in \{0, 1\}$ denote YRA approval, as in a usual potential outcomes framework. Adult market-relevant human capital as a function of the YRA is

$$H_i^A(d) = h(G_i) + \theta_S S_i(d) + g(X_i(d)), \qquad \theta_S \geq 0, \quad g' > 0, \quad g'' < 0. \tag{7}$$

where $S_i$ is subsequent time in education, and $X_i$ is productive early labor-market exposure. The coefficient $\theta_S$ denotes the effect of schooling on the human-capital index. The function $g(X_i)$ captures the standard Mincerian idea that workers acquire skills through experience at a declining rate (Mincer, 1958, 1974; Dustmann and Meghir, 2005).

The paper's main adult outcome is employment rather than a wage rate.[20] I therefore add a participation decision to the human-capital relationship. Applicant $i$ is employed as an adult under treatment state $d$ when the offered wage exceeds the minimum offer needed

[19] Interestingly, in informal discussions I had with the social workers who are in touch with the applicants, this also came up. Many social workers did not think the applicants were even near the margin to qualify for disability insurance.

[20] I do not observe hourly/monthly wages or hours worked, and in this population the selection into employment and therefore into being observed would regardless be severe.

to work:

$$E_i^A(d) = \mathbf{1}\left\{\log w_i^*(d) = \alpha + \lambda H_i^A(d) + u_i \geq R_i\right\}, \qquad \lambda > 0. \tag{8}$$

The reservation index $R_i$ summarizes health, the cost of working, and available benefits in adulthood. [21] $u_i$ is a wage offer shock. Equation (8) links the standard Mincerian wage mechanism to the employment outcome observed here: schooling and early work raise the wage offer, and a higher offer increases the probability that work clears the participation threshold.

The YRA affects this decision by changing the human-capital inputs accumulated before adulthood. Its effect on applicant $i$'s human capital is

$$\Delta H_i \equiv H_i^A(1) - H_i^A(0) = \theta_S \Delta S_i + \Delta g_i. \tag{9}$$

[22] Here $\Delta S_i = S_i(1) - S_i(0)$, and $\Delta g_i = g(X_i(1)) - g(X_i(0))$ denote the corresponding changes in the human-capital inputs.

Because a higher wage offer raises employment, a first-order approximation maps the average change in human capital into the average change in employment:

$$\Delta p \approx \underbrace{a\theta_S \Delta \bar{S}}_{\text{education channel}} + \underbrace{a\Delta \bar{g}}_{\text{early-work channel}}. \tag{10}$$

Here $\Delta p$ is the average change in the adult employment probability, $\Delta \bar{S}$ and $\Delta \bar{g}$ are the average changes in schooling and early-work human capital, and $a \geq 0$ scales changes in human capital into changes in employment.[23]

The model gives a simple prediction of the treatment-effect patterns. If approval produces little additional education but reduces early work, then adult employment should fall. If it moves applicants from disability insurance into education without displacing work, the schooling component is positive and the work component is small, so adult employment should rise. If approval increases education and reduces work, the net effect will depend on the size of the changes in the human capital stocks and their prices in the labor market. Because studying and working can occur in the same year, changes in the two inputs need not offset each other one-for-one.

### Empirical implementation

While the sign of the predicted effects seems broadly consistent with the empirical results, the magnitude is yet to be examined. I assess the predictions in a simple empirical exercise. With all the simplifications attached, this should be more interpreted as a back-of-the-envelop calculation than an attempt to model the labor market in a realistic way.

I first estimate the effects of YRA approval on the inputs in equation (7): years recorded

[21] Thus, $R$ could be extended to depend on receiving the benefit as well, in the form of either habituation or decline in health that would result in increased disutility of work, but I keep it fixed here for simplicity.

[22] Ninth-grade GPA does not appear as a treatment-induced change because it is measured before the application. It can still determine the initial stock of human capital.

[23] Formally, $\Delta p = \mathbb{E}[p_i(1) - p_i(0)]$, $\Delta \bar{S} = \mathbb{E}[\Delta S_i]$, and $\Delta \bar{g} = \mathbb{E}[\Delta g_i]$. If $m_i$ is the slope of applicant $i$'s employment probability at the baseline threshold, then $a = \lambda \mathbb{E}[m_i]$. The approximation requires this probability to be differentiable and $\mathrm{Cov}(m_i, \Delta H_i) = 0$.

as a student and early labor-market exposure. These are measured as cumulative stocks during ages 16-20, so they do not overlap the period of observing adult employment. I measure labor market attachment as cumulative earnings. As Ashworth et al. (2021) argue, it is important to measure realized experience instead of potential experience. As hours worked are not available, months spent working is not well recorded for short spells, and the wages of young individuals with little qualifications are likely to be more homogeneous than experienced workers, I argue this is both the best approximation in my setting as well as a reasonable one. I report these estimates for the IV compliers and for the lower- and higher-DI-risk groups used in the heterogeneity analysis.

Next, I estimate predictive employment gradients for schooling and early work. I use a quadratic in early labor-market exposure and estimate the following regression among initially rejected applicants:

$$E_i^A = \beta_0 + f(G_i) + \beta_S S_i + \beta_X X_i + \beta_{X^2} X_i^2 + C_i'\gamma + \varepsilon_i, \tag{11}$$

where $C_i$ contains pre-application healthcare use and socioeconomic background variables [24]. The coefficients measure how each human-capital input predicts adult employment in this population, conditional on initial academic human capital and the other inputs [25]. Without instruments for the different terms, the estimates from this equation are likely biased toward an unknown direction due to some confounders that cannot be controlled here. Thus, they should be taken as suggestive.

For each empirical group $k$ (e.g., applicants with low labor-market prospects), define the treatment-induced changes $\Delta\bar{S}_k = \mathbb{E}[S_i(1) - S_i(0) \mid i \in k]$, $\Delta\bar{X}_k = \mathbb{E}[X_i(1) - X_i(0) \mid i \in k]$, and $\Delta\overline{X^2}_k = \mathbb{E}[X_i(1)^2 - X_i(0)^2 \mid i \in k]$. I combine the estimated counterparts of these changes with the empirical gradients:

$$\widehat{\Delta p}_k^{\,\text{proj}} \equiv \widehat{\beta}_S \widehat{\Delta\bar{S}}_k + \widehat{\beta}_X \widehat{\Delta\bar{X}}_k + \widehat{\beta}_{X^2} \widehat{\Delta\overline{X^2}}_k. \tag{12}$$

Equation (12) is a descriptive empirical analogue of the channel decomposition in equation (10). The $\beta$ coefficients are the predictive regression coefficients from equation (11), not the model slope $a$. The projection translates the treatment-induced changes in schooling and early work into a predicted change in the adult employment rate. I compare this prediction with the estimated treatment effect on adult employment for each group. The most useful test is whether the projection reproduces both sides of the heterogeneity result: a negative predicted effect for the IV compliers and other higher-prospect applicants, whose early work falls with little educational gain, and a positive predicted effect for the applicants with the lowest prospects, whose educational attainment rises without a comparable loss of work. Due to simplicity of the approach and the assumptions imposed in the estimation of the returns, a perfect match of the magnitudes can hardly be expected.

Are the estimates from Eq. (11) reasonable? Some previous studies find that employment while in school predicts improved subsequent labor-market outcomes, although

[24] I include indicators for any visit for each ICD10 Chapter A-Z as well as parent education, employment, region, and child gender, all measured at age 15.

[25] I call these coefficients below returns, consistent with the previous literature, but they only reflect causal parameters under strict assumptions.

| | Effect of approval on inputs, by age 21 | | | | Implied effect on adult employment (pp) | | | |
|---|---|---|---|---|---|---|---|---|
| | $\Delta S$ (years) | $\Delta X$ (EUR 1,000) | $\bar{X}_0$ (EUR 1,000) | $\Delta(X^2)$ (EUR 1,000)$^2$ | Education $\hat{\beta}_S \Delta S$ | Early work $\hat{\beta}_X \Delta X + \hat{\beta}_{X^2} \Delta(X^2)$ | Implied total | Actual $\Delta E$ |
| Examiner IV (compliers) | 0.005 | −2.312* | 6.48 | −24.6 | 0.0 | −7.0 | −7.0 | −11.2** |
| | (0.162) | (1.227) | | | | | | (5.0) |
| DDML, higher-prospect applicants | 0.093*** | −3.979*** | 10.38 | −66.8 | 0.1 | −10.8 | −10.8 | −7.4*** |
| | (0.020) | (0.261) | | | | | | (0.4) |
| DDML, lowest-prospect applicants | 0.729*** | 0.349 | 3.54 | 2.6 | 0.7 | 1.1 | 1.9 | 4.7*** |
| | (0.056) | (0.364) | | | | | | (0.5) |
| *Employment gradient used in the calculation (pp per unit of input)* | | | | | | | | |
| $\hat{\beta}_S$ (per year as student) | 1.00 | | | | | | | |
| $\hat{\beta}_X$ (per EUR 1,000) | 3.58 | | | | | | | |
| $\hat{\beta}_{X^2}$ (per (EUR 1,000)$^2$) | −0.051 | | | | | | | |

Table 12: Treatment effects on the human-capital inputs of Eq. (7) and the adult employment effect they imply through Eq. (12). $S$ is years recorded as a student after age 15, and $X$ is cumulative wage income through age 20, measured in thousands of euros. Columns 2–5 report the estimated effects on the inputs: the examiner IV row comes from Table 5 and the two DDML rows from Table 10. Columns 6–8 multiply those effects by the gradients estimated from Eq. (11) among initially rejected applicants; column 9 is their sum and column 10 the estimated effect on adult employment in years 6–11 (Table 6 and Table 9). Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

the estimated returns are sensitive to how selection into youth employment is addressed (Ruhm, 1997; Hotz et al., 2002). Credible direct estimates on return to this type of experience/attachment are not common. The closest evidence for this setting on returns to work experience relevant comes from studies of early job matches, summer-job programs, and subsidized employment, where the returns are sometimes disappointing but vary considerably across settings (Hensvik et al., 2023; Knutsson and Tyrefors, 2024; Heller, 2014; Gelber et al., 2016; Le Barbanchon et al., 2020). [26] (Bruhn et al., 2025) use quasi-random variation in early occupational experience from a natural (non-subsidized employment) setting and show large and persistent effects of initial exposure to a profession.

A change in adult health, work costs, or benefit options would operate through $R_i$ rather than $H_i^A$. If approval changes this threshold by $\Delta R_i$, the employment index changes by $\lambda \Delta H_i - \Delta R_i$. I examine those channels separately using the adult health and benefit outcomes. The exercise here asks how much of the employment pattern is consistent with the changes in education and early work documented during the benefit period.

I also conduct a descriptive mediation analysis, where I include the accumulated earnings by age 20 in the equation where the outcome is adult employment. Table A.5 presents the results. The IV estimates change somewhat when the cumulative wage earnings are included, but the difference is not large. The DDML estimates for the whole sample and for the complier-weighted population change substantially, which would be consistent with a large role for the mediator. These numbers should be interpreted carefully and as descriptive: the mediator is not randomized, so conditioning on a post-treatment variable does not generally identify causal mediation, and in the IV specification the adjusted treatment coefficient need not retain the original complier-LATE interpretation.

[26] Unlike most of these interventions, the YRA does not create work experience in special-protected or sheltered contexts but displaces attachment formed in the ordinary labor market. This type of connections could plausibly be more valuable than the connections induced by subsidized employment programs, which might be not profitable for both parties without government intervention.

### 6.1.2 Information, beliefs or other forms of welfare dependency

Another possibility is that receiving the YRA changes how recipients interact with the social insurance system. It may teach them how to apply for related benefits or reduce the stigma of claiming them (Goffman, 1963; Moffitt, 1983). It could also change attitudes toward work and self-sufficiency or medicalization of "normal" life situations (like poor focus in school or negative emotions stemming from the current environment). While this type of story is common in policy debates, there is little credible empirical evidence isolating this type of channels, making it harder to formulate a testable hypothesis about how it would operate and for which groups it would be especially prevalent.

If the YRA teaches recipients how to navigate health-conditioned benefits or reduces the stigma of claiming them, adult disability insurance is the natural outcome to examine. Adult DI resembles the YRA in its health conditions and application requirements, although its assessment is separate and more demanding. I observe no increase in DI recipiency after the benefit ends in the IV analysis (Fig. A.3). Using the selection-on-observables approach, a small increase is detected for the overall group, but not enough to explain the employment effects Fig. A.6[27]. Looking at another health-based benefit, the more short-term sickness benefit, I also find null effects (Fig. A.3). Unemployment insurance and social assistance, the general benefits available to anyone out of work, increase instead, which follows from the decline in employment rather than from anything specific to the YRA.

Based on the previous literature, we would expect the opposite pattern. Dahl et al. (2014) find that a parent receiving disability insurance has a large effect on their children eventually receiving it as well, an effect hypothesized to work through information about the application process and beliefs about success in the applications, and one that operates through the same benefit and even the same diagnosis codes. For the informational channel there is also evidence from other settings (Finkelstein and Notowidigdo, 2019; Anders and Rafkin, 2024).[28]

The administrative outcomes leave two limitations. The IV confidence intervals include meaningful increases in DI recipiency, and there could be an increase in DI applications, which I do not observe in my data. Beyond that, these channels are hard to test without detailed survey data on information, attitudes, and beliefs, which I do not have. The existing qualitative work does not support this kind of changes[29], but these hypotheses remain untestable.

[27] And much smaller than in Dahl et al. (2014) where the treatment is parent being approved to a benefit (there DI), not the focal child earlier in life, like in my setting.

[28] A mechanism like this should not be understood to necessarily be a bad outcome: the welfare impact would depend on many things, including how the channel affects targeting of the social insurance benefits. In many informational experiments, the recipients induced to take up a benefit are more likely to be eligible than the average recipient, or the ones with the highest need face the highest barriers (Finkelstein and Notowidigdo, 2019; Castell et al., 2025).

[29] Evidence from Basic Income experiments generally does not suggest large changes in attitudes (Blomberg et al., 2021; Broockman et al., 2024), and previous qualitative research on YRA recipients does not document attitudes of this kind (Salminen, 2019).

### 6.1.3 Changes in health and consumption of harmful goods

The recipients could also spend the cash on harmful or addictive goods. Stories of this kind are documented by Evans and Moore (2011) and Chorniy et al. (2025), and suggested as a reason for the negative health effects of disability insurance by Black et al. (2024). My data allow me to test this directly. Using crime data, I test whether the accepted are more likely to be convicted of or reported to the police for drug-related crimes. Using healthcare data, I test whether they are more likely to be diagnosed with a substance abuse disorder in specialized or primary care; these data do not include stays in drug detox or rehabilitation institutions, but a diagnosis from healthcare typically precedes such a stay. In either data source and with either identification strategy, I find no increase in substance abuse (Fig. A.2). If anything, there is a decrease in the first years of benefit recipiency.

A possibility in a similar vein is that the benefit allows the recipients to get used to a level of consumption that exceeds their post-benefit ability to fund this consumption. They would then accumulate debt after the benefit ends, which could lead to financial distress and garnishment of wages, and inhibit the later labor market outcomes. I find no evidence of increased wage garnishment in Fig. A.2. If anything, there is reduction during the benefit receipt, although not statistically significant.

## 6.2 Welfare effects

The MVPF framework of Hendren and Sprung-Keyser (2020) is an approach to accounting for the benefits and costs of a public policy. The MVPF is the recipients' willingness to pay per euro of net cost to the government. In my setting I can observe, among other things, the use of healthcare and criminal activity, allowing for a detailed accounting of the expenses. I also observe all benefits paid to the individuals.

$$\text{MVPF} = \frac{\text{WTP}}{\text{Net government cost}} = \frac{\sum_i \text{WTP}_i}{M - FE}, \tag{13}$$

where $M$ is the mechanical cost of the transfer, holding behaviour fixed, and $FE$ is the fiscal externality, the change in the government budget caused by the behavioural responses to the policy:

$$FE = \underbrace{\Delta\tau}_{\text{taxes}} - \underbrace{\Delta B}_{\text{other transfers}} - \underbrace{\Delta H}_{\text{healthcare}} - \underbrace{\Delta CJ}_{\text{criminal justice costs}} - \underbrace{\Delta E}_{\text{Costs of providing education}}. \tag{14}$$

Under this sign convention, behavioural responses that save the government money, such as higher tax revenue or lower spending on healthcare and the criminal justice system, reduce the net cost and raise the MVPF.

The largest category of government expenses is the social insurance transfers paid to the individuals. In the MVPF, this amount features on both sides of the ratio: recipients value a euro of transfer at approximately a euro, so the transfer enters the numerator at its cost and the mechanical cost $M$ in the denominator, and everything I estimate is a deviation from that benchmark. I discount all amounts by 3% and present them in 2023 euros.

I exclude earnings during benefit recipiency from the calculation because I assume these losses reflect individuals choosing to work less due to the income effect of the benefit. I instead include earnings losses after the benefit ends in willingness to pay, since, in my judgment, they reflect at least partly lower productivity. While I prefer this interpretation, the long-run earnings response could instead reflect a choice to work less. Under this interpretation, the accompanying increase in other benefit receipts is part of the same behavioral response: neither the reduction in earnings nor the increase in other benefits should enter willingness to pay, although both continue to affect the fiscal cost of the program. Under this alternative, willingness to pay consists only of the YRA payments. As shown in the last row of Table 13, this choice has a substantial effect on the MVPF.

I include only the costs of hospital and specialized healthcare as well as the drug costs to the government in the cost estimates. I do not include the changes in primary care visits for two main reasons: (1) the data are available only from 2011 onward and (2) the costs of this care are not straightforwardly calculated from the data. Second, whether to put changes in health on the WTP side of the equation. I do not do so, since it would require mapping the healthcare data into quality-adjusted life years or similar measures, and it is also unclear whether there are changes in health overall, so the impact is likely to be small. Also excluded are the administrative costs, since they are difficult to estimate, and schooling costs, since these are approximately zero. Finally, the calculations assume zero effects beyond age 27, an age chosen because the later ages rely on estimates that only represent a few cohorts, which might not be representative of the whole sample.

With my preferred choices, the benefit has a marginal value of public funds of 0.8 (Table 13). A pure transfer that induced no behavioral response would have an MVPF of one, so the YRA is not an efficient way of moving resources to this group: to justify it, we would need the recipients to have a 1.2 times larger welfare weight than the general population.

Instead, on the individuals with the lowest prospects, the government costs are negative. This means that the policy pays for itself and the MVPF, as outlined by Hendren and Sprung-Keyser (2020), is defined to be infinite. The majority of this comes from the increase in tax revenue from these individuals. While there is also a large change in other benefits paid, this enters both sides of the MVPF formula. This number is estimated using the DDML procedure, and it inherits the strong identification assumptions highlighted in earlier sections.

In general, these numbers should be taken with some caveats. There are substantial choices involved in the procedure, which I explain in Section A.5.4. These choices largely shape the final MVPF number. However, the results of this exercise line up with the main analysis in suggesting that the transfer seems to provide good value for those with the lowest prospects, but not for the marginal applicant.

# 7 Conclusion

In this paper I provide evidence that a temporary cash transfer intended to support adolescents with health problems can inhibit their later labor market success when provided to individuals who would have remained in education even without the benefit. The purpose

| | Euros per recipient | |
|---|---|---|
| | IV (compliers) | Low prospects |
| *Panel A. Willingness to pay* | | |
| YRA payments | 16559 | 19894 |
| Other benefits | 3991 | -21771 |
| Reduction in earnings | -8731 | 1756 |
| Crime victim costs | 2365 | 953 |
| **Total** | 14184 | 832 |
| | | |
| *Panel B. Net cost to the government* | | |
| YRA payments | 16559 | 19894 |
| Other benefits | 3991 | -21771 |
| Reduction in tax revenue | 1765 | -1359 |
| Healthcare costs | -3922 | 1746 |
| Criminal justice costs | -694 | 752 |
| **Total** | 17700 | -738 |
| **MVPF** | **0.80** | **-1.13** ($\infty$) |
| **MVPF, without earnings loss** | **1.07** | **-28.25** ($\infty$) |

Table 13: MVPF calculation, using the sample of individuals observed until age 27. IV and DDML estimates. Healthcare costs include hospital/specialized healthcare visits and drug expenses. The row 'without earnings loss' reports the numbers when considering the post-treatment effects on labor supply as a choice rather than something imposed by the benefit. Values are not discounted.

of the benefit was to incentivize individuals to attend education instead of relying on passive disability benefits in adolescence, which would improve their labor market outcomes in adulthood. I evaluate whether this purpose was fulfilled using two identification strategies with different assumptions to understand the heterogeneous effects of the program. Combining the applications with comprehensive administrative data allows me to estimate effects on labor market outcomes and other dimensions of life.

For marginal applicants, I use the quasi-randomly allocated application examiners' approval rates as instrumental variables to identify the effects of receiving the cash transfer on adult outcomes. The main finding is that the program did not achieve its goals in this population: receiving the benefit reduced employment and increased the welfare benefits received by the applicants.

These estimates apply to the IV compliers: applicants whose approval depends on the assigned examiner. Individuals with the lowest labor market prospects are underrepresented among these compliers. To evaluate the effects on this subgroup, I compare accepted and rejected applicants, using observables to control for selection. An array of validity checks supports the underlying assumptions. Most importantly, I show that among the compliers, the IV and selection-on-observables approaches yield similar results.

These results show that approval for the benefit improved the labor market outcomes of individuals with the lowest labor market prospects. This result becomes understandable when I turn to the short-run responses by the labor market prospect group. For individuals with low labor market prospects, the benefit moves individuals from welfare benefits, such as disability insurance, to education. It does not displace much full-time or part-time work. Instead, among applicants with moderate labor market prospects, approval for the program did not meaningfully increase attendance in education, which was already high,

and substantially reduced earned income while receiving the benefit. While not conclusive, the clear differences in the counterfactual activities of applicants in the two groups offer an intuitive explanation for the heterogeneous effects.

For the marginal applicants, the MVPF of the program is below 1, while for the low prospect sample the benefit pays for itself. The heterogeneity in results suggests potential for improving welfare by simple changes in targeting, even if the paper does not weigh in on the optimal targeting procedure. My low prospect sample is defined by variables observable to the policymakers and the most important predictor is the diagnosis in the application. Limiting the transfer based on these characteristics would therefore be achievable.

The results and the methodological approach of this paper suggest that it would often be useful to look beyond simple LATEs in economics research. While this is not a new insight by any means, this paper shows how these different identification strategies can complement each other. With improvements in data and new methods, the use of selection-on-observables models can offer opportunities for additional insights (see also e.g. Chetty et al., 2026), as long as the strict assumptions are kept firmly in mind.

There are plenty of open questions related to the labor market outcomes of young individuals with health problems. There are two possible readings of the positive effects on the group with low prospects. One, more optimistic, reading is that there seems to be room for improving their labor market outcomes with public policy. A second, more pessimistic, reading is that around 80% of them are still not employed even after the considerable investment made in the form of the Youth Rehabilitation Allowance. Overall, more economics research would be useful to understand the barriers and opportunities young individuals with health problems face in the labor market.

An increasing share of adolescents in high-income countries are diagnosed with health problems. Individuals with these problems have a higher chance of ending up outside the labor market and becoming a burden on public finances rather than an asset. My results suggest that when planning programs to help these individuals, careful consideration should be given to what kind of human capital accumulation is incentivized and what the prospects of the recipients are. The bad news from this paper is that a single solution is unlikely to fit all disadvantaged individuals. The good news is that evidence suggests that public policy can help individuals with severe disabilities to successfully transition into adulthood. Designing these policies will be a key challenge in bringing welfare states into the 21st century.

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# Appendix to

*Cash Transfers, Health, and the Transition to Adulthood*

Ossi Tahvonen

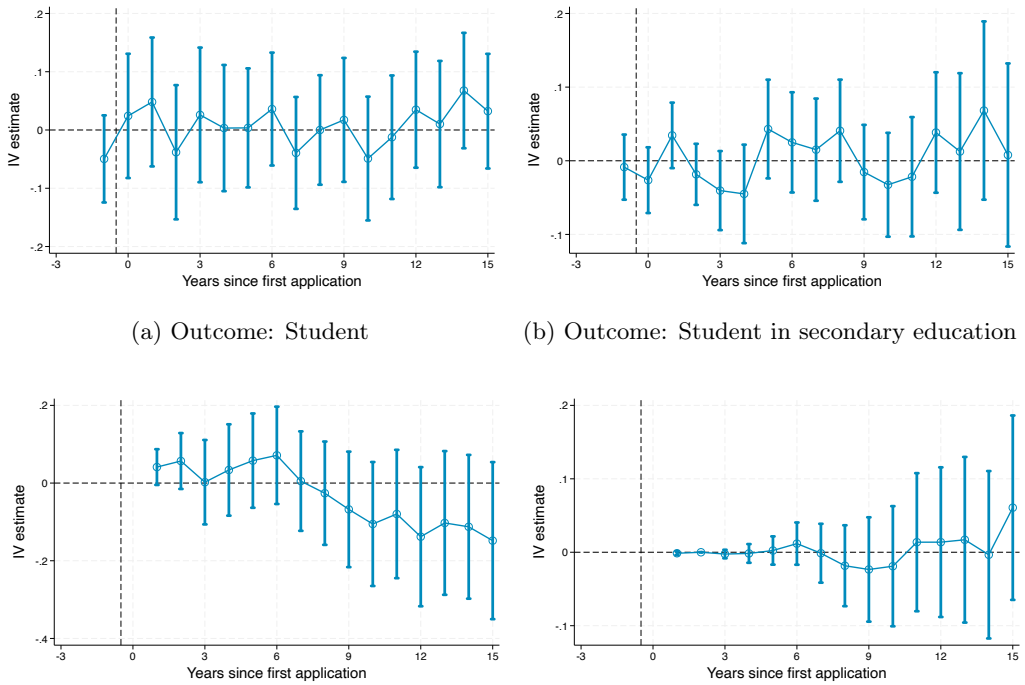
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## A.1 Additional Results

### A.1.1 Education, health, and high-risk-group outcomes



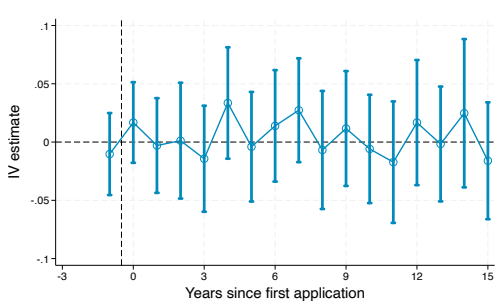

(a) Outcome: Student

(b) Outcome: Student in secondary education

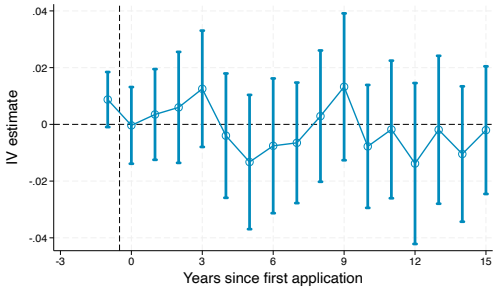
(c) Outcome: Secondary education degree

(d) Outcome: Higher education degree

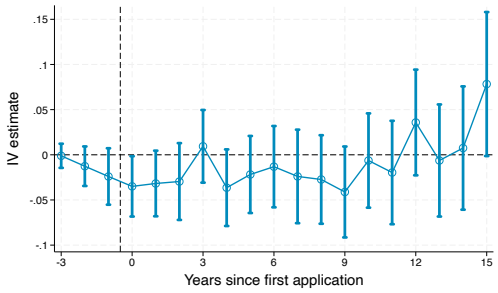
Figure A.1: Long-run (EB) IV estimates on educational outcomes. Each estimate represents an IV-estimate from a separate regression using time-specific samples. Outcomes apart from health are not necessarily informative for individuals younger than 16 years old, which moves pre-treatment estimates toward 0. Estimates are not scaled, so in the case of binary outcomes an estimate of 0.1 means 10p.p. increase.



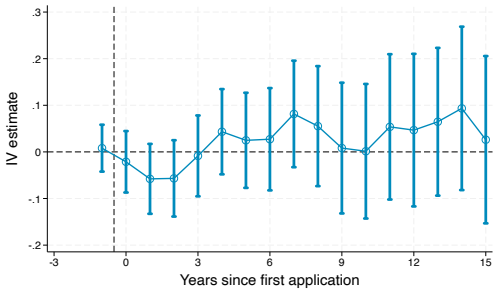

(a) Outcome: Any criminal conviction




(b) Outcome: Any drug related criminal conviction




(c) Any addiction diagnosis




(d) Outcome: Financial trouble (garnishment)

Figure A.2: Long-run (EB) IV estimates on crime, drug-related crimes, addiction diagnoses, and financial trouble. Each estimate represents an IV-estimate from a separate regression using time-specific samples. Outcomes apart from health are not necessarily informative for individuals younger than 16 years old, which moves pre-treatment estimates toward 0. Estimates are not scaled, so in the case of binary outcomes an estimate of 0.1 means 10p.p. increase.

| | (1) Hosp. days | (2) Overnight MH vis. | (3) Any MH. | (4) Any antidepr. |
|---|---|---|---|---|
| OLS | 0.092 | 0.011** | 0.004 | -0.010*** |
| | (0.065) | (0.005) | (0.003) | (0.003) |
| IV | 0.024 | 0.005 | -0.070 | -0.029 |
| | (1.105) | (0.064) | (0.044) | (0.046) |
| AR p-value (IV) | 0.983 | 0.941 | 0.112 | 0.521 |
| First stage F (IV) | 343 | 343 | 343 | 343 |
| IV, controls | -0.087 | -0.003 | -0.077* | -0.030 |
| | (1.104) | (0.064) | (0.043) | (0.045) |
| AR p-value (IV, contr.) | 0.937 | 0.965 | 0.068 | 0.512 |
| First stage F (IV, contr.) | 346 | 346 | 346 | 346 |
| Untreated complier mean | 1.84 | .127 | .346 | .261 |
| Sample mean | 1.56 | .106 | .341 | .239 |
| N | 117649 | 117649 | 117649 | 117649 |
| N indiv. | 26617 | 26617 | 26617 | 26617 |

Table A.1: Main results on health outcomes using EB-IV. Outcomes are measured years 6-12 after the first application, which roughly corresponds to ages 21-28. Health outcomes are available until 2023. Both OLS and IV use the same, necessary controls. IV with additional controls uses controls for the disability category, indicators for any depression/anxiety/F9/other healthcare visits, urban area, parental employment, parent ages, and 9th grade GPA. Sample size refers to individual $\times$ year observations. Standard errors clustered at the individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

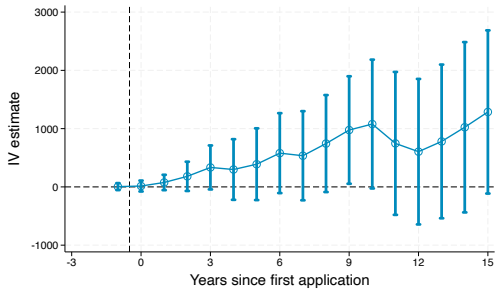
| | (1) Cum. Years of school | (2) Sec. educ. degree | (3) Cum. years in DI | (4) Cum. wages |
|---|---|---|---|---|
| OLS | 0.673*** | 0.059*** | -2.042*** | -185.257 |
| | (0.049) | (0.014) | (0.048) | (226.305) |
| IV | -0.089 | -0.027 | -2.031*** | -664.134 |
| | (0.618) | (0.174) | (0.574) | (2545.893) |
| AR p-value (IV) | 0.885 | 0.876 | 0.002 | 0.794 |
| First stage F (IV) | 39.1 | 39.1 | 39.1 | 39.1 |
| IV, controls | -0.054 | -0.002 | -2.019*** | -474.953 |
| | (0.614) | (0.172) | (0.572) | (2526.122) |
| AR p-value (IV, contr.) | 0.929 | 0.993 | 0.002 | 0.851 |
| First stage F (IV, contr.) | 39.5 | 39.5 | 39.5 | 39.5 |
| Untreated complier mean | 2.7 | .204 | 2.53 | 2789 |
| Sample mean | 3.33 | .25 | 1.24 | 2054 |
| N | 5724 | 5720 | 5724 | 5724 |
| N indiv. | 5724 | 5720 | 5724 | 5724 |

Table A.2: Main results on short-run labor market outcomes using EB-IV for the high DI risk sample. Both OLS and IV use the same, necessary controls. IV with additional controls uses controls for the disability category, indicators for any depression/anxiety/F9/other healthcare visits, urban area, parental employment, parent ages, and 9th grade GPA. Standard errors clustered at the individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

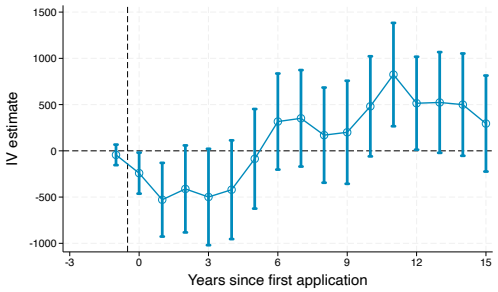
| | (1) NEET | (2) Employed | (3) Wage inc. | (4) Student |
|---|---|---|---|---|
| OLS | -0.034*** | 0.031*** | 93.450 | 0.003 |
| | (0.007) | (0.006) | (122.774) | (0.004) |
| IV | 0.051 | -0.094 | -2839.161 | 0.040 |
| | (0.182) | (0.160) | (3416.189) | (0.094) |
| AR p-value (IV) | 0.781 | 0.556 | 0.402 | 0.669 |
| First stage F (IV) | 24.5 | 24.5 | 24.5 | 24.5 |
| IV, controls | 0.018 | -0.066 | -2395.388 | 0.045 |
| | (0.178) | (0.156) | (3335.072) | (0.093) |
| AR p-value (IV, contr.) | 0.920 | 0.671 | 0.470 | 0.624 |
| First stage F (IV, contr.) | 25.1 | 25.1 | 25.1 | 25.1 |
| Untreated complier mean | .763 | .211 | 5914 | .0248 |
| Sample mean | .77 | .157 | 2826 | .0725 |
| N | 22997 | 22997 | 22997 | 22997 |
| N indiv. | 4892 | 4892 | 4892 | 4892 |

Table A.3: Main results on long-run labor market outcomes using EB-IV for the high DI risk sample. Outcomes are measured years 6-12 after the first application, which roughly corresponds to ages 21-28. Both OLS and IV use the same, necessary controls. IV with additional controls uses controls for the disability category, indicators for any depression/anxiety/F9/other healthcare visits, urban area, parental employment, parent ages, and 9th grade GPA. Sample size refers to individual $\times$ year observations. Standard errors clustered at the individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

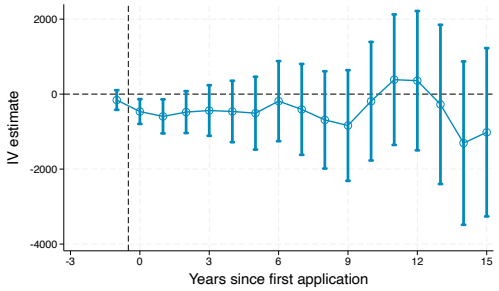
### A.1.2 Transfer receipt and public costs



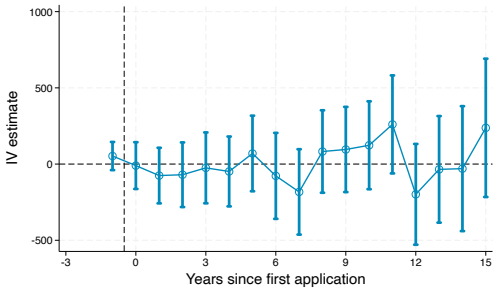

(a) Outcome: Unemployment benefits (euros)




(b) Outcome: Social Assistance (euros)

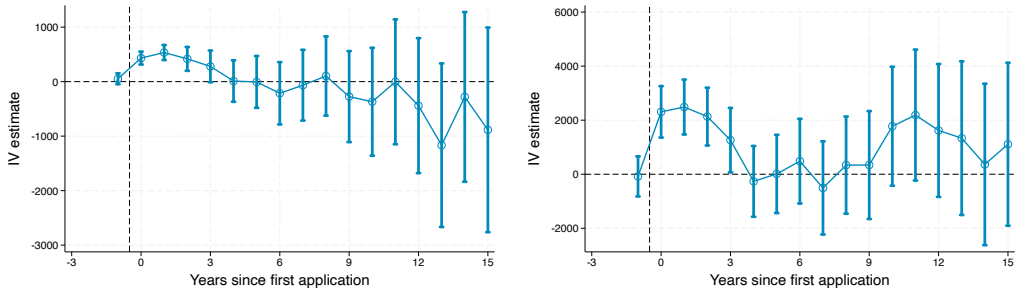



(c) Outcome: Pensions paid (euros) (disability insurance)




(d) Outcome: Illness benefits (euros)

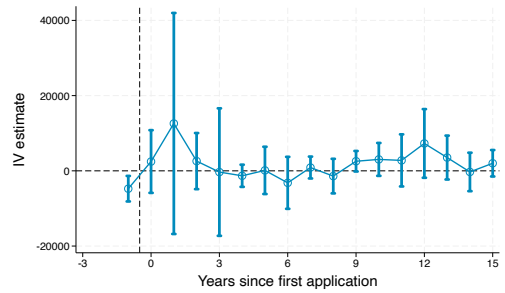
Figure A.3: Long-run (EB) IV estimates on benefits paid to individuals. Each estimate represents an IV-estimate from a separate regression using time-specific samples. Outcomes apart from health are not necessarily informative for individuals younger than 16 years old, which moves pre-treatment estimates toward 0.



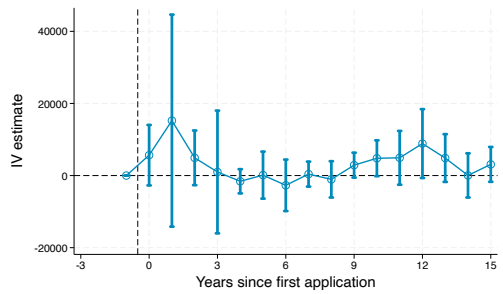

(a) Outcome: Paid taxes (euros)




(b) Outcome: Total transfers (euros)



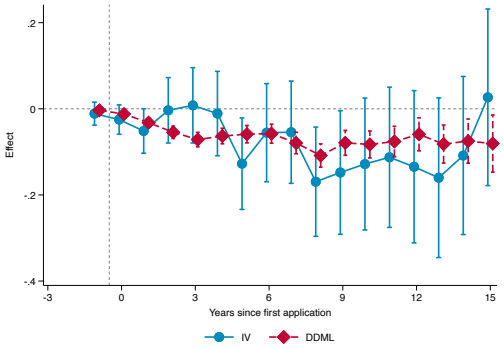

(c) Outcome: Hospital and specialized care costs (euros)



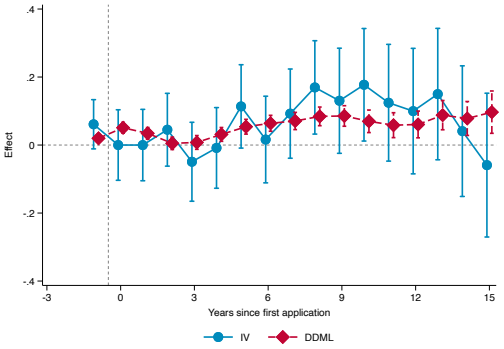

(d) Outcome: All public health expenses (euros)

Figure A.4: Long-run (EB) IV estimates on cumulative costs to the public sector. Each estimate represents an IV-estimate from a separate regression using time-specific samples.

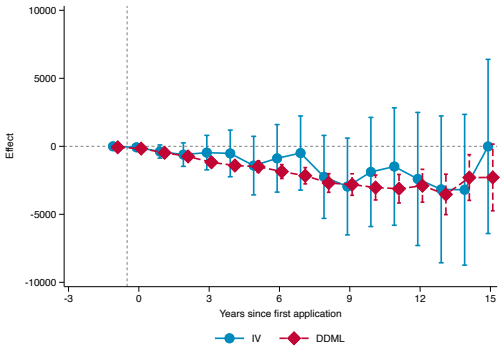
### A.1.3 Covariate-adjusted estimates and treatment-effect heterogeneity



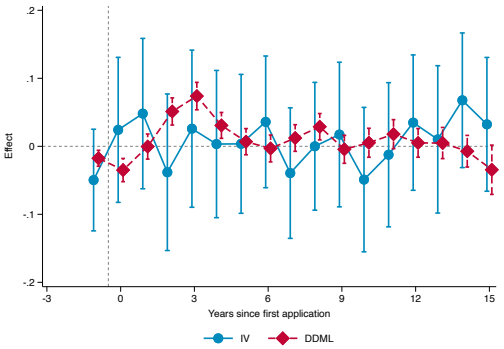

(a) Outcome: Employed



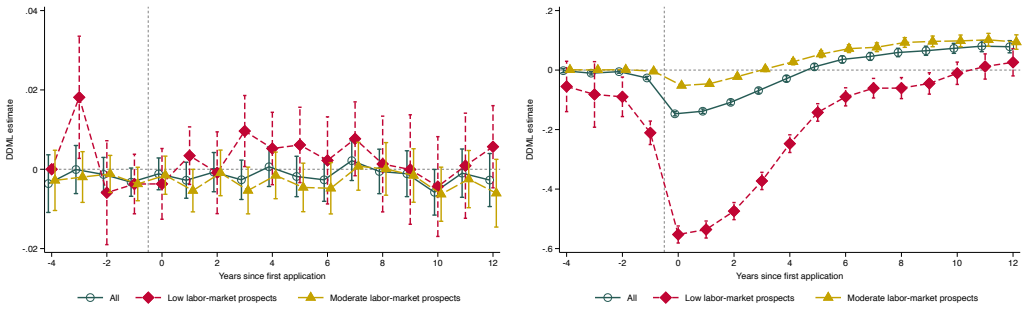

(b) Outcome: NEET




(c) Outcome: Wage earnings



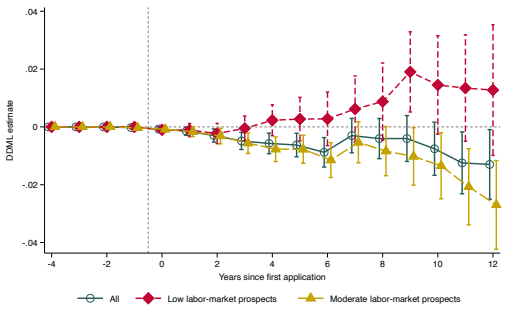

(d) Outcome: Student



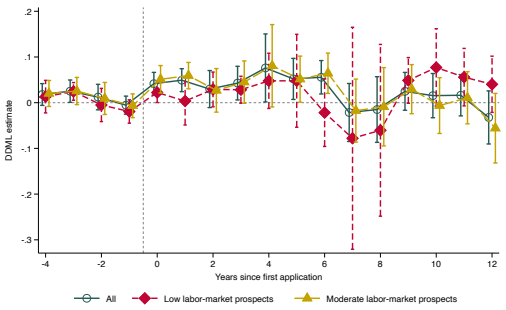

(a) Outcome: Any criminal conviction




(b) Outcome: Disability Insurance



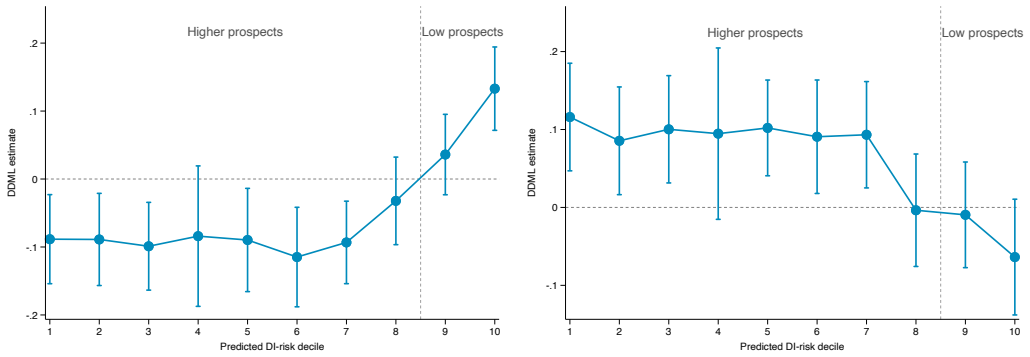

(c) Outcome: Married




(d) Outcome: Overnight MH visit

Figure A.6: Additional results using the covariate adjustment approach. Each estimate represents a DDML estimate from a separate estimation using time-specific samples. Marriage is not possible for individuals younger than 18 years old, which is why the pre-treatment outcomes are not informative. Estimates are not scaled, so in the case of binary outcomes, an estimate of 0.1 means 10p.p. increase. 'All' refers to the entire sample with overlap, 'Low Labor Market prospects' to the applicants with the lowest labor market prospects and 'Moderate Labor Market prospects' to the rest.

*Adult outcomes, pooled over years 6–12 after first application*

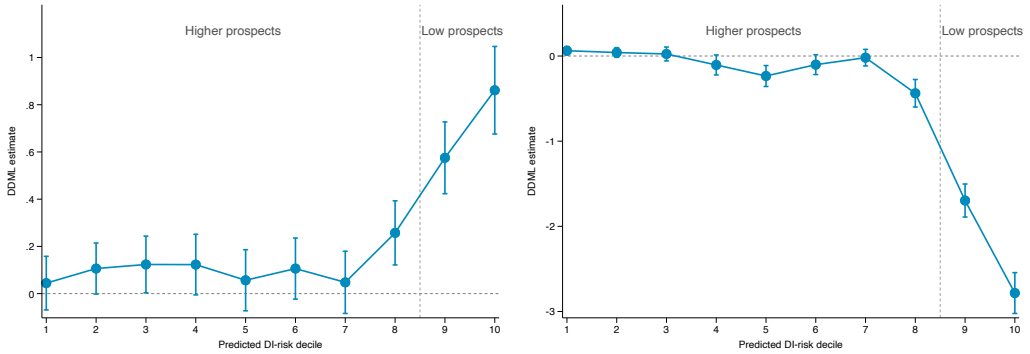



(a) Outcome: Employed




(b) Outcome: NEET

*Cumulative outcomes by age 21*



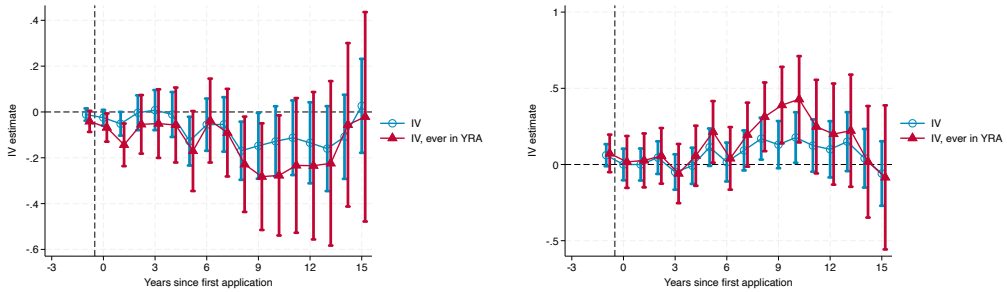

(c) Outcome: Years in education




(d) Outcome: Years on DI

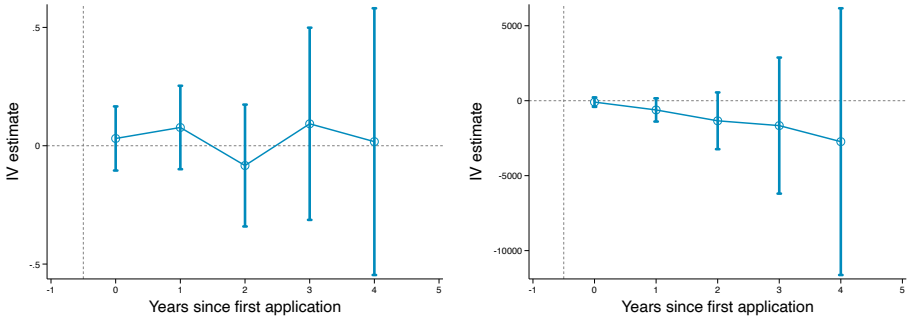
Figure A.7: DDML estimates by predicted Disability Insurance (DI) risk. The index predicts DI receipt two years after application using pre-treatment covariates and is estimated out of fold among rejected applicants. Applicants are grouped into deciles ordered from higher to lower labor-market prospects. The dashed vertical line separates the top fifth of predicted DI risk (deciles 9–10), which defines the low-labor-market-prospect group in the main analysis, from the remaining applicants. Each point is a separate DDML estimate for the corresponding decile using the specification described in Section 4.3. Vertical bars show 95% confidence intervals. Estimates are not scaled, so an estimate of 0.1 for a binary outcome denotes a 10 percentage-point effect. Standard errors are clustered at the individual level.

### A.1.4 Alternative treatment definitions, reduced forms, and mechanisms



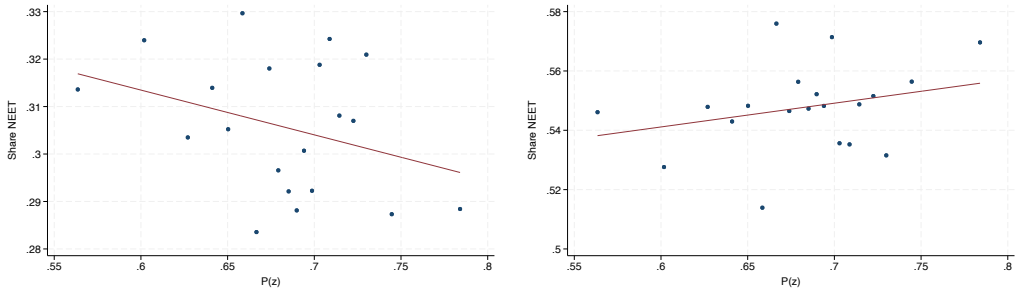

(a) Outcome: Employed (b) Outcome: NEET

Figure A.8: Results using 'first application accepted' as the endogenous variable and a comparison to using 'Ever Receive any YRA benefits' as the endogenous variable. 95 % confidence intervals included




(a) Outcome: Student (b) Outcome: Wage earnings

Figure A.9: IV results using 'Currently in the YRA program' as the endogenous variable. 95 % confidence intervals included.




(a) Outcome: Employed (b) Outcome: NEET

Figure A.10: Reduced-form graphs. The x-axis is the EB instrument and y axis the outcome, measured at time horizon 6-12.

| | (1) NEET | (2) Employed | (3) Wage inc. |
|---|---|---|---|
| Reduced form | 0.0955** | -0.0911** | -1430.3 |
| | (0.0444) | (0.0410) | (1092.3) |
| Observations | 102592 | 102592 | 102592 |

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

Table A.4: Reduced form version of the main results (as in Table 6).

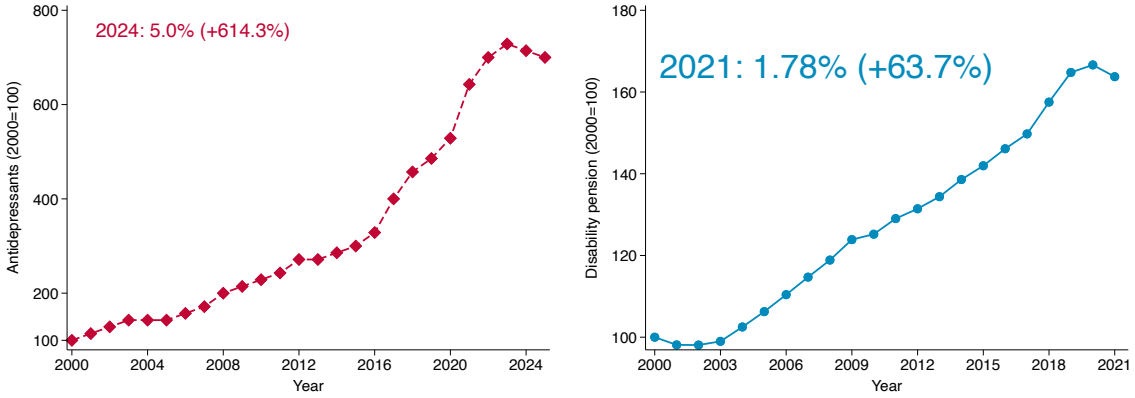
| | (1) IV | (2) IV, mediation | (3) DDML | (4) DDML, mediation | (5) DDML (compliers) | (6) DDML (compliers), mediation |
|---|---|---|---|---|---|---|
| First applic. accepted | -0.112** | -0.0936** | -0.0461*** | -0.00618** | -0.0930*** | -0.0224*** |
| | (0.0500) | (0.0424) | (0.00303) | (0.00274) | (0.00676) | (0.00615) |
| Outcome mean | 0.36 | 0.36 | 0.29 | 0.29 | 0.36 | 0.36 |
| Controls | IV controls | IV controls + cum. earnings | All | All + cum. earnings | All | All + cum. earnings |
| Observations | 102592 | 102592 | 102595 | 102595 | 102595 | 102595 |

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A.5: Mediation analysis. Different specifications with and without controlling for accumulated wage earnings at age 20. The outcome is employment 6-12 years from the application. Since the mediator is not randomized and there is no instrument for it available, the results should be taken as suggestive. Standard errors clustered at the individual level.

### A.1.5 Additional descriptive evidence




(a) Antidepressant-reimbursement recipients, ages 13–17.

(b) Disability-pension recipients, ages 16–29.

Figure A.11: Yearly disability pension receipt among individuals aged 16–29 (panel b) and antidepressant reimbursement receipt among individuals aged 13–17 (panel a) in Finland. Each series is indexed to 100 in 2000. The colored annotation reports the raw 2024 population share, with the percentage change from 2000 to 2024 in parentheses. Sources: Finnish Centre for Pensions (ETK), Statistics Finland, and THL Sotkanet based on Kela reimbursement data.

| | (1) Gen. pop. | (2) Approved | (3) Rejected |
|---|---|---|---|
| Student | 0.976 | 0.997 | 0.998 |
| Employed | 0.024 | 0.003 | 0.002 |
| Distinct admissions | 1.02 | 3.85 | 3.85 |
| Healthcare costs | 559 | 2,424 | 2,164 |
| Anxiety diagnosis | 0.053 | 0.019 | 0.020 |
| Depression diagnosis | 0.052 | 0.015 | 0.017 |
| Single parent HH | 0.211 | 0.246 | 0.253 |
| Urban | 0.807 | 0.747 | 0.762 |
| Mother age | 44.04 | 44.51 | 44.80 |
| Father age | 46.45 | 47.60 | 47.69 |
| Mother DI | 0.027 | 0.102 | 0.099 |
| Father DI | 0.040 | 0.091 | 0.088 |
| Mother married | 0.697 | 0.577 | 0.598 |
| Father married | 0.691 | 0.563 | 0.577 |
| Mother employed | 0.814 | 0.652 | 0.691 |
| Father employed | 0.831 | 0.707 | 0.728 |
| Mother taxable income | 4,467 | 26 | 167 |
| Father taxable income | 6,174 | 40 | 290 |
| Accepted (1st appl.) | 0 | 1 | 0 |
| Ever in YRA | 0 | 0.996 | 0.242 |
| Total YRA benefits | 0 | 25,857 | 5,846 |
| Observations | 1,476,350 | 4,471 | 1,275 |

Table A.6: Summary statistics for the applicants, by approval status, for the low labor market prospect group.
Notes: All variables are measured in the year before the first application. Euro amounts are in 2023 prices. Gen. pop. is the sample of all other individuals aged 15.

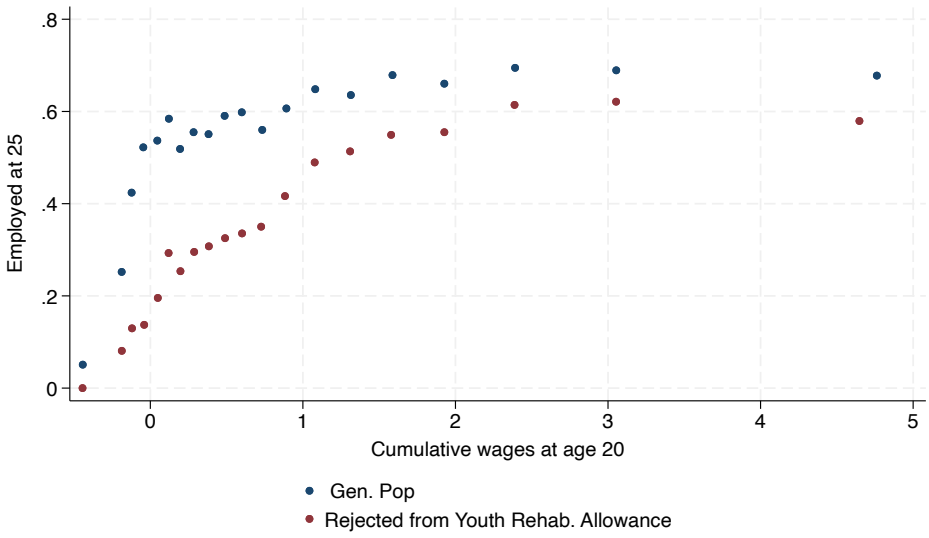



Figure A.12: Descriptive figure of the association between accumulated work earnings (x-axis) and the probability of employment (y-axis) between years 6–12 after the application to YRA. It displays 20 even-sized bins. 'Gen. Pop.' presents to a group with a similar cohort profile as the applicants for comparison.

| Code | Diagnosis group | Approval rate | Share in DI after 5y | $N$ |
|---|---|---|---|---|
| C7 | Eye, brain/CNS, endocrine and unspecified cancers | 79.6% | 20.4% | 54 |
| C9 | Leukemia, myeloma and other blood cancers | – | 13.3% | 45 |
| D3 | Benign neoplasms of urinary, nervous and endocrine organs | 75.0% | 25.0% | 28 |
| D6 | Aplastic and other anemias | 45.8% | – | 24 |
| E1 | Diabetes mellitus and related glucose disorders | 26.0% | 4.0% | 150 |
| E2 | Other endocrine-gland disorders | 45.0% | – | 20 |
| E7 | Inherited/metabolic disorders of amino acids, carbohydrates and lipids | 65.6% | 52.5% | 61 |
| E8 | Other metabolic disorders | 74.3% | – | 35 |
| F0 | Organic mental disorders, including dementia | 40.5% | 13.5% | 37 |
| F1 | Mental and behavioral disorders due to substance use | 52.9% | – | 34 |
| F2 | Schizophrenia, schizotypal and delusional disorders | 75.3% | 40.1% | 887 |
| F3 | Mood (affective) disorders | 61.4% | 11.0% | 4,234 |
| F4 | Anxiety, stress-related and somatoform disorders | 60.9% | 9.2% | 2,739 |
| F5 | Eating, sleep and other physiologically linked behavioral syndromes | 54.8% | 11.1% | 389 |
| F6 | Personality and adult behavioral disorders | 61.0% | 19.5% | 77 |
| F7 | Intellectual disabilities | 80.1% | 55.8% | 4,322 |
| F8 | Disorders of psychological development | 72.0% | 17.7% | 7,823 |
| F9 | Childhood/adolescent-onset behavioral and emotional disorders | 66.7% | 7.2% | 4,041 |
| G1 | Systemic atrophies affecting the central nervous system | 67.4% | 30.4% | 46 |
| G3 | Alzheimer disease and other degenerative nervous-system disorders | 75.0% | 28.6% | 28 |
| G4 | Epilepsy, migraine, TIA, sleep and other episodic neurologic disorders | 59.0% | 18.1% | 537 |
| G6 | Polyneuropathies and other peripheral nervous-system disorders | 63.0% | – | 54 |
| G7 | Neuromuscular-junction and muscle diseases | 77.9% | 36.2% | 149 |
| G8 | Cerebral palsy and other paralytic syndromes | 77.9% | 32.9% | 721 |
| G9 | Autonomic and other nervous-system disorders | 65.7% | 25.7% | 70 |
| H2 | Iris, ciliary-body and lens disorders | – | – | 20 |
| H3 | Choroidal and retinal disorders | 91.4% | 21.6% | 116 |
| H4 | Glaucoma, vitreous and optic-nerve disorders | 86.7% | 11.7% | 60 |
| H5 | Eye-movement/refraction disorders and visual impairment | 78.7% | 13.8% | 94 |
| H9 | Hearing loss and other ear disorders | 79.6% | 5.2% | 574 |
| I4 | Cardiomyopathy, conduction disorders and cardiac arrhythmias | 42.4% | – | 33 |
| I6 | Cerebrovascular disease (stroke and related disorders) | 63.9% | – | 36 |
| J4 | Chronic lower respiratory diseases (including COPD and asthma) | 21.5% | – | 93 |
| K5 | Inflammatory bowel disease and other intestinal disorders | 38.1% | – | 84 |
| L2 | Dermatitis and eczema | 12.9% | – | 31 |
| M0 | Infectious arthropathies | 47.5% | 2.5% | 510 |
| M3 | Systemic connective-tissue disorders | 53.2% | – | 47 |
| M4 | Spinal deformities and spondylopathies | 37.5% | – | 40 |
| M5 | Intervertebral-disc and other back disorders | 19.4% | – | 31 |
| M8 | Osteoporosis and other bone-density/structure disorders | 38.1% | – | 21 |
| P1 | Birth trauma | 31.2% | – | 32 |
| Q0 | Congenital malformations of the nervous system | 79.8% | 29.2% | 168 |
| Q1 | Congenital malformations of eye, ear, face and neck | 86.5% | 14.6% | 96 |
| Q2 | Congenital malformations of the circulatory system | 56.3% | 19.4% | 103 |
| Q6 | Congenital malformations of urinary tract and selected musculoskeletal sites | 64.8% | 18.5% | 54 |
| Q7 | Other congenital musculoskeletal malformations | 67.8% | 15.0% | 180 |
| Q8 | Other congenital malformations | 77.5% | 39.0% | 377 |
| Q9 | Chromosomal abnormalities | 65.4% | 73.8% | 503 |
| R4 | Cognitive, consciousness, perception and behavioral symptoms/signs | 40.3% | 11.3% | 62 |
| R6 | Edema, developmental and other general symptoms/signs | 79.2% | 29.2% | 24 |
| S0 | Injuries of the head | 55.8% | – | 43 |
| T9 | Sequelae of injuries, poisoning and other external causes | 42.9% | 16.3% | 49 |
| Z9 | Health status, follow-up and other factors influencing care | 39.3% | – | 28 |
| – | Other (diagnosis groups with $N < 20$) | 47.9% | 2.4% | 451 |

Table A.7: Diagnoses on the applications. *Share approved* refers to the share of applications initially approved. *Share in DI after 5y* refers to the share of both rejected and accepted applicants who receive Disability Insurance/Pension at age 25. Due to data privacy, any shares that would correspond to less than 6 individuals have been changed to "–" in accordance with the Statistics Finland and Findata disclosure rules.

## A.2 Additional validity and robustness checks

### A.2.1 Monotonicity

This appendix describes the monotonicity tests summarized in Section 4.2.2 and reports their results.

The traditional test of monotonicity is to estimate the first stage for different subsamples. Monotonicity implies that the positive relationship between the instrument and the acceptance probability should hold in each sample (Maestas et al., 2013). Table A.8 reports the first-stage regression across subsamples.

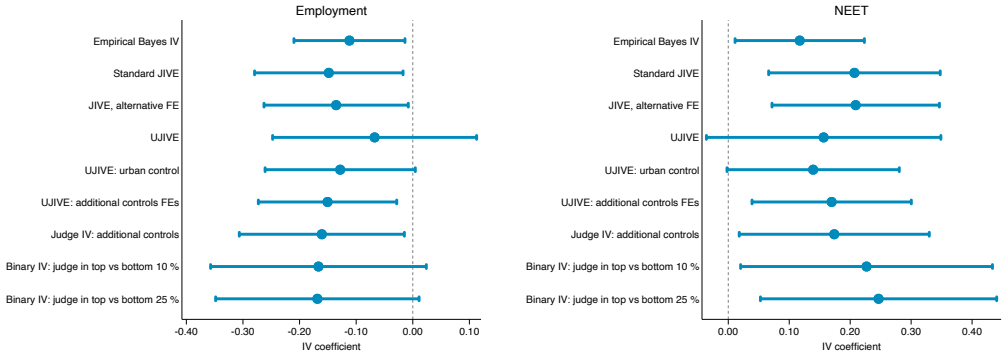
| Subsample | First-stage coefficient | Standard error | $N$ |
|---|---|---|---|
| Male | 0.785 | 0.058 | 16185 |
| Female | 0.819 | 0.065 | 14354 |
| Urban | 0.820 | 0.049 | 24353 |
| Rural | 0.689 | 0.094 | 6202 |
| Mood disorder | 0.887 | 0.121 | 4214 |
| Anxiety disorder | 0.964 | 0.153 | 2714 |
| Physical condition | 0.722 | 0.098 | 5964 |
| Attention-deficit disorder | 0.757 | 0.157 | 2375 |
| Mother: higher education | 0.758 | 0.080 | 9579 |
| Mother: no higher education | 0.826 | 0.052 | 20077 |
| Decision year after 2007 | 0.791 | 0.053 | 23827 |
| Finnish-speaking | 0.820 | 0.044 | 29217 |

Table A.8: First-stage estimates by subgroup. This is an indirect test of the monotonicity assumption: the first stage should be positive in all covariate groups.

Bhuller et al. (2020) propose a second test. An implication of the assumption is that a leniency measure estimated in one sample should be informative of the first stage in another sample. If the converse held, examiners who are lenient toward one group would be strict toward another, which would violate the assumption. I implement this test for groups based on the illness categories and gender in Table A.9. The first-stage coefficient is positive and significant in all splits, which supports the assumption.

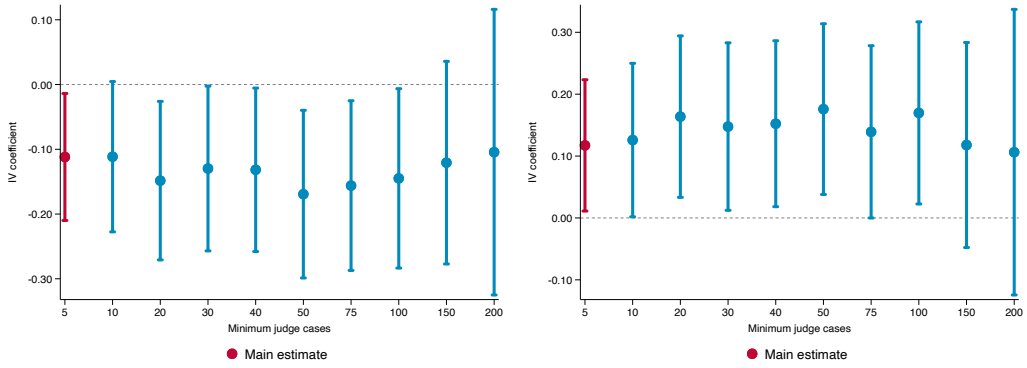
As Mogstad and Torgovitsky (2024) point out, different constructions of the instrument might make the assumption of monotonicity more plausible. For instance, it is more likely that an examiner who is among the most lenient will be more lenient than an examiner who is among the strictest (Sigstad, 2024). I report results using this binary instrument in Fig. A.13. While the compliers using this instrument are plausibly different than with the continuous measure, the results are very similar to the main results.

### A.2.2 Robustness




(a) Employment (b) NEET

Figure A.13: Robustness to different IV estimation choices for the main outcomes. 95 % confidence intervals included.




(a) Employment (b) NEET

Figure A.14: Robustness to different restrictions on the number of cases each examiner has to have handled. The default with the Empirical Bayes shrunk IVs indicated in red. Outcomes measured at 6–11 years from the first application. Standard errors clustered at the individual level. 95 % confidence intervals included.

| | First application approved | N | F |
|---|---|---|---|
| Female [Male] | 0.412*** | 14301 | 83.05 |
| | (0.0453) | | |
| Male [Female] | 0.365*** | 16080 | 88.72 |
| | (0.0388) | | |
| Int./chrom. dis. [Others] | 0.117*** | 26549 | 48.93 |
| | (0.0168) | | |
| Mood/anx. [Others] | 0.181*** | 22722 | 57.81 |
| | (0.0238) | | |
| Dev. dis. [Others] | 0.296*** | 20562 | 71.52 |
| | (0.0351) | | |
| Physical [Others] | 0.229*** | 18165 | 40.24 |
| | (0.0361) | | |
| High acc. rate [Low] | 0.456*** | 9141 | 60.40 |
| | (0.0587) | | |
| Low acc. rate [High] | 0.280*** | 21188 | 86.99 |
| | (0.0300) | | |

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A.9: Monotonicity test using the split-sample method (Bhuller et al., 2020). In the regressions, I estimate the first stage for some groups, defining the instrument using the complement of that group. Here for instance, "Male [female]" denotes a regression where I define the instrument using males and estimate the first stage using this instrument for the complement sample of female applicants.

| | (1) | (2) | (3) |
|---|---|---|---|
| | First application approv. | First application approv. | First application approv. |
| EB IV | 0.844*** | 0.803*** | 0.794*** |
| | (0.0434) | (0.0433) | (0.0425) |
| N | 30623 | 30623 | 30623 |
| F-Statistic | 378.8 | 344.4 | 348.7 |
| Controls | No | Resid. FE | Res. + Additional |

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

Table A.10: First stage estimates with different specifications. *Resid. FE* refers to the necessary controls: the region-year-language FEs used in the construction of the IV. Additional controls include parent characteristics, diagnosis code group indicators, and number of healthcare visits before the application. The small difference between columns 2 and 3 is consistent with the instrument being independent of the background variables.

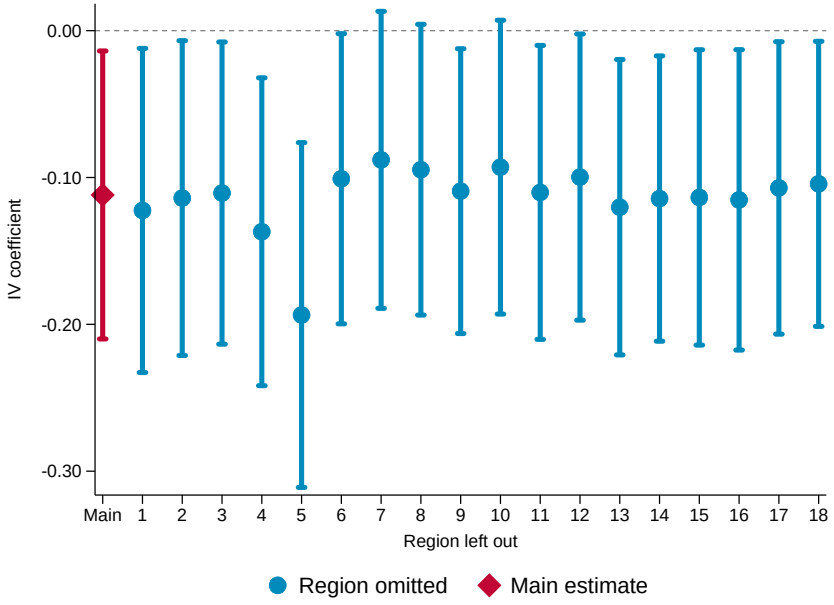
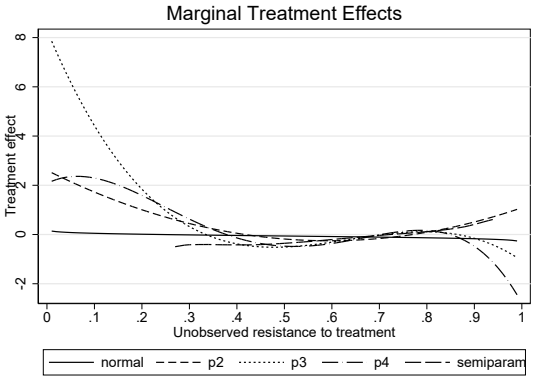



Figure A.15: Robustness of the employment estimate to excluding one region at a time. Blue circles show the leave-one-region-out EB IV estimates, while the red diamond shows the main full-sample estimate. Vertical bars indicate 95% confidence intervals.

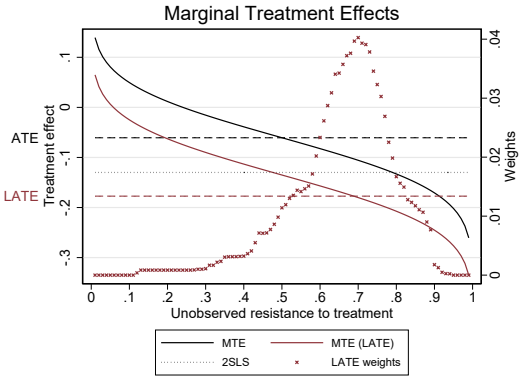
| | (1) NEET | (2) Employed | (3) Wage inc. | (4) Student |
|---|---|---|---|---|
| OLS | 0.121*** | -0.105*** | -3217.951*** | -0.016*** |
| | (0.004) | (0.003) | (86.227) | (0.003) |
| IV | 0.160** | -0.133** | -1853.206 | -0.029 |
| | (0.066) | (0.061) | (1616.619) | (0.039) |
| AR p-value (IV) | 0.017 | 0.032 | 0.255 | 0.461 |
| First stage F (IV) | 206 | 206 | 206 | 206 |
| IV, controls | 0.148** | -0.124** | -1736.468 | -0.025 |
| | (0.063) | (0.060) | (1583.215) | (0.038) |
| AR p-value (IV, contr.) | 0.019 | 0.038 | 0.274 | 0.520 |
| First stage F (IV, contr.) | 208 | 208 | 208 | 208 |
| Untreated complier mean | .443 | .407 | 9309 | .15 |
| Sample mean | .585 | .283 | 6416 | .131 |
| N | 77901 | 77901 | 77901 | 77901 |
| N indiv. | 13323 | 13323 | 13323 | 13323 |

Table A.11: The main long-run IV estimates using a balanced sample. The sample is limited to individuals observed each year from years 6 through 11 after the first application. IV with additional controls uses controls for the disability category, indicators for any depression/anxiety/F9/other healthcare visits, urban area, parental employment, parent ages, and 9th grade GPA. Sample size refers to individual × year observations. Standard errors clustered at the individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

## A.3 Marginal Treatment Effects




(a) MTE curves, using different parametric forms. 'pX' denotes polynomial degree X specification, normal denotes using normal distribution for the error terms.




(b) MTE using normal distribution for the unobserved error term.

Figure A.16: Outcome: employed, averaged over year 6-12 after application. LIV approach. Using the stata package `mtefe` (Andresen, 2018). Confidence intervals are not displayed due to computational as well as informational reasons: for most specifications, they span most of the possible interval for the binary outcome.

## A.4 Derivations of the human capital model

Starting from the human capital equation: $H_i^A(d) = h(G_i) + \theta_S S_i(d) + g(X_i(d))$

we define that the change in human capital caused by the treatment is:

$\Delta H_i \equiv H_i^A(1) - H_i^A(0) = \theta_S \Delta S_i + \Delta g_i$.

Enter the definition of employment:

$E_i^A(d) = \mathbf{1}\{\log w_i^*(d) = \alpha + \lambda H_i^A(d) + u_i \geq R_i$. The treatment can cause changes in employment (and the parameter of interest is these changes in some population)

$\Delta E_i^A = \mathbf{1}\{\log w_i^*(1) \geq R_i\} - \mathbf{1}\{\log w_i^*(0) \geq R_i\}$.

Let $Z_i \equiv \log w_i(0) - R_i$. Conditional on a given treatment-induced change in human capital, $\Delta H_i = h$, the change in employment is

$$\Delta E_i = \mathbf{1}\{Z_i + \lambda h \geq 0\} - \mathbf{1}\{Z_i \geq 0\}.$$

Taking expectations conditional on $\Delta H_i = h$ gives

$$\begin{aligned}
\mathbb{E}[\Delta E_i \mid \Delta H_i = h] &= \mathbb{E}\left[\mathbf{1}\{Z_i + \lambda h \geq 0\} - \mathbf{1}\{Z_i \geq 0\} \mid \Delta H_i = h\right] \\
&= P(Z_i + \lambda h \geq 0 \mid \Delta H_i = h) - P(Z_i \geq 0 \mid \Delta H_i = h) \\
&= P(Z_i \geq -\lambda h \mid \Delta H_i = h) - P(Z_i \geq 0 \mid \Delta H_i = h).
\end{aligned}$$

Let $F_{Z|\Delta H=h}(z) \equiv P(Z_i \leq z \mid \Delta H_i = h)$ denote the conditional CDF of $Z_i$. Assuming the conditional distribution of $Z_i$ is continuous,

$$\begin{aligned}
\mathbb{E}[\Delta E_i \mid \Delta H_i = h] &= \left[1 - F_{Z|\Delta H=h}(-\lambda h)\right] - \left[1 - F_{Z|\Delta H=h}(0)\right] \\
&= F_{Z|\Delta H=h}(0) - F_{Z|\Delta H=h}(-\lambda h).
\end{aligned}$$

Taking a first-order Taylor approximation of the second term around zero:

$$F_{Z|\Delta H=h}(-\lambda h) \approx F_{Z|\Delta H=h}(0) - \lambda h f_{Z|\Delta H=h}(0),$$

where $f_{Z|\Delta H=h}$ is the corresponding conditional density function. Now the two instances of the term $F_{Z|\Delta H=h}(0)$ cancel out. Hence,

$$\mathrm{E}[\Delta E_i \mid \Delta H_i = h] \approx \lambda h f_{Z|\Delta H=h}(0).$$

Averaging over the distribution of $\Delta H_i$ and applying the law of iterated expectations,

$$\begin{aligned}
\Delta p \equiv \mathrm{E}[\Delta E_i] &= \mathrm{E}[\mathrm{E}[\Delta E_i \mid \Delta H_i]] \\
&\approx \lambda \, \mathrm{E}\left[\Delta H_i f_{Z|\Delta H=\Delta H_i}(0)\right].
\end{aligned}$$

Defining $m_i \equiv f_{Z|\Delta H=\Delta H_i}(0)$ (density/slope at the employment margin) gives

$$\Delta p \approx \lambda \, \mathrm{E}[m_i \Delta H_i].$$

Then, using

$$\Delta H_i = \theta_S \Delta S_i + \Delta g_i,$$

we obtain

$$\begin{aligned}\Delta p &\approx \lambda \,\mathrm{E}[m_i \Delta H_i] \\ &= \lambda \,\mathrm{E}[m_i(\theta_S \Delta S_i + \Delta g_i)] \\ &= \lambda \theta_S \,\mathrm{E}[m_i \Delta S_i] + \lambda \,\mathrm{E}[m_i \Delta g_i].\end{aligned}$$

Next, I assume that $m_i$ is independent of $\Delta H_i$. This assumption simplifies the formula. In practical terms it means that the size of the treatment-induced change in human capital is unrelated to how close an individual is to the employment margin.

$$\mathrm{E}[m_i \Delta H_i] = \mathrm{E}[m_i]\mathrm{E}[\Delta H_i].$$

Define

$$a \equiv \lambda \,\mathrm{E}[m_i] \geq 0.$$

Lets denote for exposition that

$$\Delta \bar{S} \equiv \mathrm{E}[\Delta S_i], \qquad \Delta \bar{g} \equiv \mathrm{E}[\Delta g_i],$$

Since

$$\mathrm{E}[\Delta H_i] = \theta_S \,\mathrm{E}[\Delta S_i] + \mathrm{E}[\Delta g_i],$$

we obtain

$$\boxed{\Delta p \approx a \theta_S \Delta \bar{S} + a \Delta \bar{g}}.$$

## A.5 Methodological Details

### A.5.1 Selection-on-observables approach

I implement the covariate-adjustment design using the interactive DDML estimator described in Section 4.3. For each outcome, the procedure estimates two nuisance functions: the conditional outcome regression, $E[Y \mid X, D]$, and the propensity score, $E[D \mid X]$, where $D$ indicates approval. The outcome regression short-stacks predictions from OLS and gradient-boosted trees, while the propensity-score model short-stacks predictions from logistic regression and gradient-boosted trees. The DDML estimate is then formed from the cross-fitted nuisance predictions.

I use five cross-fitting folds. Fold assignment is clustered by applicant, so all observations for the same individual remain in the same fold. This prevents another observation for an applicant from entering the training sample used to predict that applicant's outcome or treatment status. Inference is also clustered at the applicant level. I estimate the nuisance functions separately within each analysis sample, including the full applicant sample and the labor-market-prospect subgroups, rather than imposing nuisance relationships estimated in one population on another.

The specification combines transparent linear learners with a gradient-boosted tree learner that can capture nonlinearities and interactions. Cross-fitting and the orthogonal DDML score reduce the first-order sensitivity of the treatment-effect estimate to errors in these nuisance models.

For the coefficient-stability diagnostic reported with the DDML estimates, I first estimate an OLS model containing the same covariate set and cluster its standard errors by applicant. I calculate the Oster $\delta$ for a zero treatment effect using $R_{\max} = \min\{1, 1.3R^2\}$, where $R^2$ is from the controlled regression. While the OLS-based specification does not perfectly correspond the DDML specification, a benefit is that the Oster $\delta$ is a familiar statistic that can be compared with other settings, unlike the approach of (Chernozhukov et al., 2026) designed for these cases.

### A.5.2 Overlap and placebo diagnostics for the high-DI-risk subgroup

The main text reports the corresponding diagnostics for the full applicant sample. Figure A.17 shows the estimated propensity-score distributions for approved and rejected applicants in the high-DI-risk subgroup. Figure A.18 applies the DDML procedure to pretreatment outcomes and fixed characteristics that approval cannot affect. Together, the figures assess whether approved and rejected high-risk applicants have common support and whether the adjustment procedure removes the large raw differences between them along observed pretreatment dimensions.

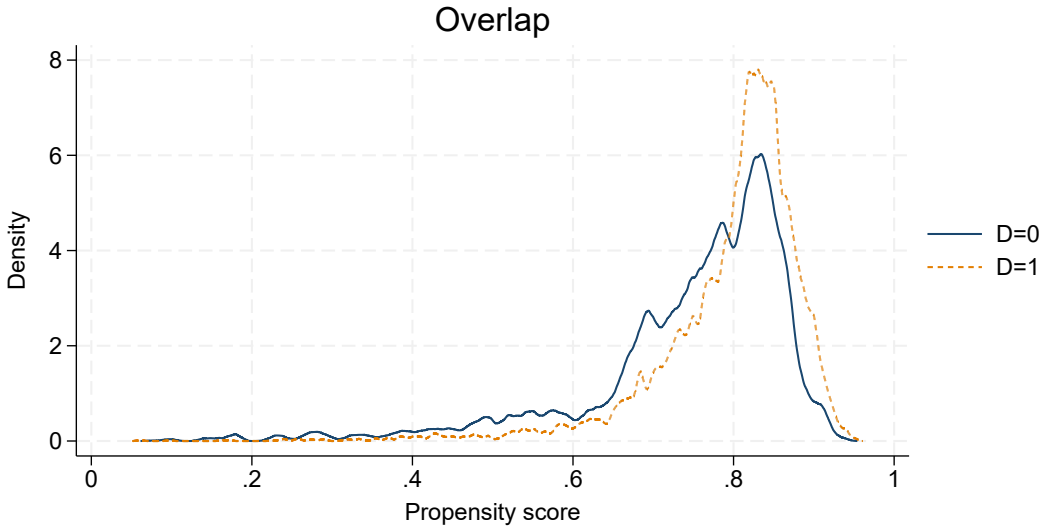



Figure A.17: Overlap for the covariate-adjustment design in the high-DI-risk subgroup. The figure shows the distributions of the propensity scores estimated by the DDML procedure using the full covariate set. $D = 1$ denotes approved applicants and $D = 0$ rejected applicants. The propensity score is estimated for the sample observed in years 6–11 after the first application; its distribution may therefore differ in other analysis samples. Figure 8 reports the corresponding figure for the full applicant sample.

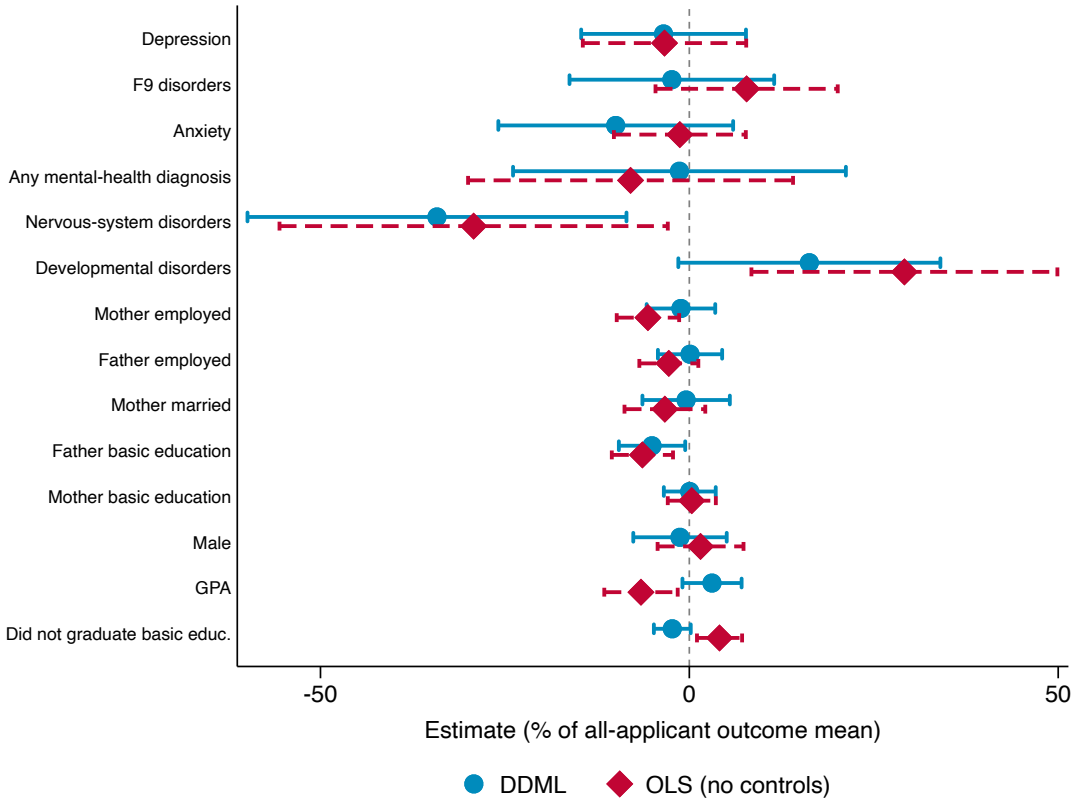
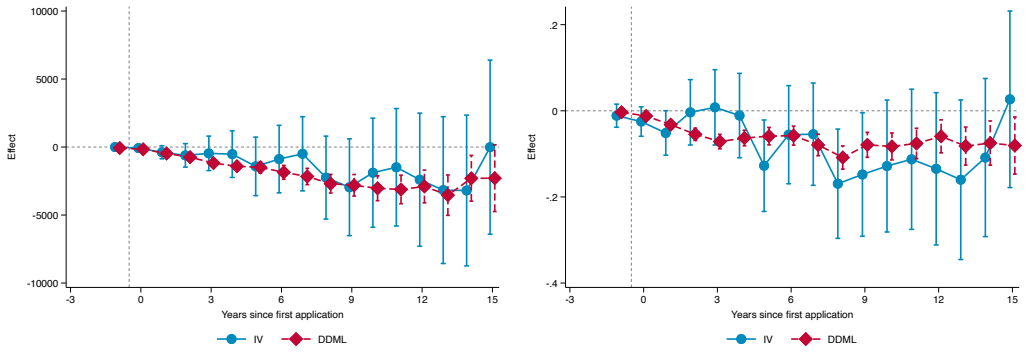



Figure A.18: Placebo and balance tests for the covariate-adjustment design in the high-DI-risk subgroup. Each DDML estimate uses the characteristic listed in the figure, measured before the application, as the outcome. Closely related variables and variables constructed from the outcome are omitted from that regression's covariate set. Figure 9 reports the corresponding figure for the full applicant sample.

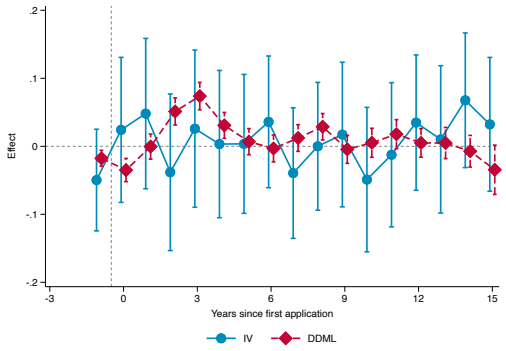
### A.5.3 Additional DDML-IV comparisons



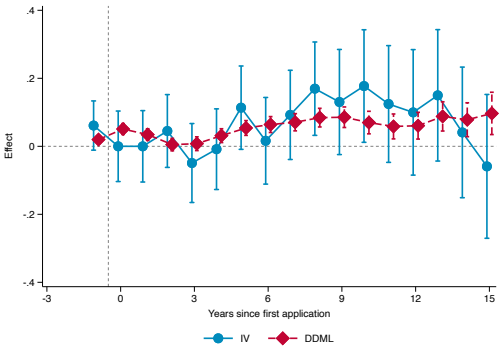

(a) Outcome: Employment income




(b) Outcome: Employed




(c) Outcome: Student




(d) Outcome: NEET

Figure A.19: Complier-weighted DDML and IV estimates.

### A.5.4 Cost-effectiveness and MVPF calculations

This section presents some details of the calculation of costs and benefits omitted from the main body of the paper. These provide the arguments for the choices I have made. Robustness to all alternative choices is hard to ascertain, as the number of different decisions that could be made is rather large.

The estimates are produced with similar methods to the rest of the paper, looking at only the sample that can be followed until age 27. I calculate cumulative amounts, using only post-application data. The IV specification is the one in the IV method section, while the DDML choices and sample definition are explained in the corresponding sections.

Most of these numbers include considerable uncertainty and should be taken as suggestive. Since administrative data sometimes include outliers that have very large costs associated with them, I winsorize all costs at the 95th percentile.

***Taxes*:** I use the taxes paid -amount from the FOLK TULO module, adjusted for inflation. This includes the taxes paid on the Youth Rehabilitation Allowance itself.

***Other transfers*** I similarly use the transfers received -amount from the FOLK TULO module, adjusted for inflation. This includes all transfers.

***Criminal Justice costs*** I use an estimate of 495 euros per police investigation (Hinkkanen, 2014; Valtioneuvosto, 2026). This is reported as cost per offence. In the data, one offence can have multiple reports associated with it. To get closer to the real costs, I cap the number of offences to 10 per month. While somewhat arbitrary, this helps to remove the possibility that single crimes entered into the system in erratic fashion would drive the results. In addition, for convictions I use a cost of 2000 euros as estimated in (Valtioneuvosto, 2024). This cost does not include costs of providing punishment, like costs of prison time. As young individuals are not commonly convicted to a closed prison but instead receive suspended sentences, there are very few that receive prison time in my sample. I do not attempt to estimate the costs of providing punishment.

***Victimization costs*** I value the harm to victims at 2,000 euros per property offence and 10,000 euros per violent or sexual offence. These are round numbers based on the unit costs of crime for England and Wales in Heeks et al. (2018). I use only their costs as a consequence of crime (property stolen or damaged, physical and emotional harm, lost output, and health and victim services) and leave out their police and criminal justice costs, which are already counted under criminal justice costs above. In 2015/16 prices, these consequence costs range from £580 (theft from a vehicle) to £4,670 (theft of a vehicle) for property offences, and from £3,750 (violence without injury) to £31,450 (rape) for violent and sexual offences. Unlike the other items, these costs fall on the victims rather than the government, so they enter willingness to pay and not the net cost.

***Costs of providing education*** It would be misleading to use the average educational costs since many of the costs of education are fixed (buildings etc.). I instead use as an estimate of the marginal cost the average cost of materials and supplies in

general upper secondary schools, which is estimated to be 467 euros per student-year (Opetushallitus, 2025). I do not value the private costs of education in this calculation.

***Healthcare costs*** The healthcare costs are based on the healthcare records with information on the provider costs. This information is not available for all of the records. For instance, prior to year 2012 this information is not recorded, and after this it is also not recorded in some administrative units. I make some steps to arrive at a final number. I calculate the healthcare costs in the following way:
1) Cap the visits to 1 visit per person per day per diagnosis to avoid transfers across administrative units counting as multiple visits.
2) Based on the records with costs indicated, I calculate the average costs per length of visit with a particular main diagnosis in ICD-10 classification.
3) Impute the costs based on the main diagnosis and length of stay for all the visits.
4) Validate that the imputation yields sensible costs for a sample of visits where the costs are known.

These costs are not exactly the same as the calculated costs using the NordDRG weights. My data does not include this information, and it would not even be calculable for all of the years in the data.

I do not include the costs of providing primary care in my estimates. These data are only available from the year 2011 onward, and the data do not include costs nor is there a clear template from the official sources to estimate this with the available information on the nature and length of each visit. The point estimates suggest small effects, so this omission is unlikely to turn around the conclusions of the MVPF calculation.

***Medical drug costs*** The Kela Drug expenses data include the public sector costs of the drugs for each purchase, which I use directly.

### A.5.5 Empirical Bayes shrinkage and simulation evidence

The empirical Bayes shrinkage I implement in this paper is relatively straightforward: following literature on eg. teacher Value-Added literature (Chetty et al., 2014). I use linear shrinkage with normal prior, shrinking all examiner-specific instruments toward the *grand mean*, that is, the mean of all judges. As the basis estimates, I take the standard leave-out residualized acceptance rates:

$$\hat{\theta}_{ij} = \frac{1}{N_j - 1} \times \left( \sum_{k=1}^{N_j} (Approved^*_{jk}) - Approved^*_i \right) \tag{15}$$

where $Accepted^*$ are the residuals from a regression with Acceptance in LHS and fixed effects in RHS. Then we shrink these ($\hat{\theta}_{ij}$'s) using:

$$\theta^*_{ij} = \left( \frac{\sigma^2_\theta}{\sigma^2_\theta + s^2_j} \right) \hat{\theta}_{ij} + \left( \frac{s^2_j}{\sigma^2_\theta + s^2_j} \right) \mu_\theta \tag{16}$$

I show using simulations that IV using empirical Bayes shrinkage appears to be unbiased and the standard deviation of the estimates is smaller than with the standard leave-out procedure. In these simulations I use 200 judges, each handling between 10 and 50 cases. Treatment is determined by judge leniency, randomly drawn from uniform distribution, confounder and a random component. A continuous outcome variable is determined by a random component, confounder, and possible treatment. I estimate treatment effects using OLS, 2SLS (not JIVE/leave-one-out), oracle type 2SLS using true leniency (serves as a comparison), standard leave-one-out, and my EB procedure. I repeat this procedure 1000 times to get a distribution of the estimates.

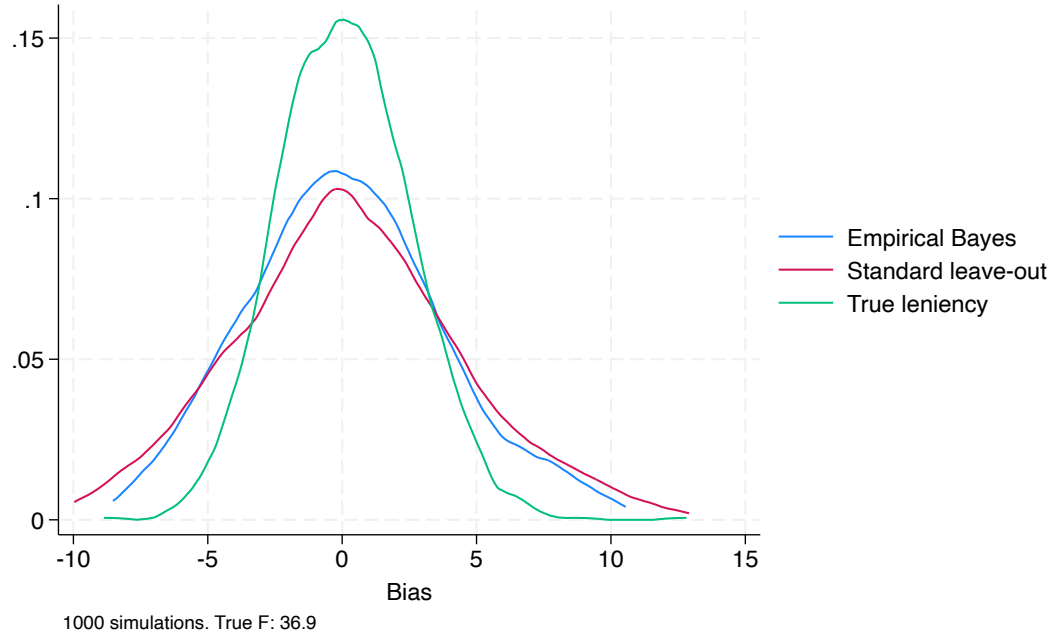



Figure A.20: Distribution of the biases from the second-stage estimates in 1000 simulations using different estimators.

As can be seen in Fig. A.20, the (in practice unattainable) true leniency performs best. Shrinking the estimated leniencies toward the mean of all judges does improve from the standard leave-one-out procedure: the estimates are more centered around 0. In Fig. A.21 it can be seen that when the instrument(s) is/are weaker, the performance of EB in comparison to standard leave-one-out procedure is improved even more.

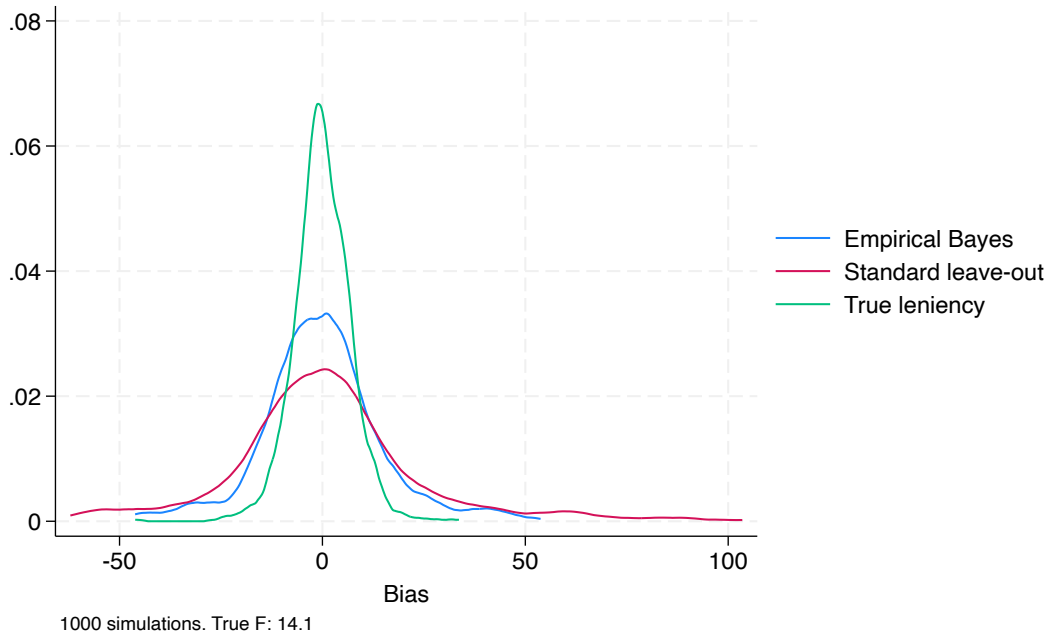



Figure A.21: Distribution of the biases in the second-stage estimates. 1000 simulations using different estimators. With lower "true F" (F-statistic if the true leniency is known), the EB-IV substantially improves from the standard leave-out/jackknife estimator.

In practice, after shrinking, I use the shrunk instrument just as any other instrument, with no special steps in inference to account for the fact that the EB-IV is estimated. This follows the previous literature on Empirical Bayes. In my simulations I find that the SEs provided by the JIVE procedure are on the conservative side, while the SEs from the EB-IV have roughly correct coverage, though this might not be true with all configurations of instrument strength and other factors.

**Empirical Bayes shrinkage in examiner designs.** Empirical Bayes shrinkage has previously been used to reduce noise in examiner-specific propensities. Humphries et al. (2025) use Beta-Binomial empirical Bayes shrinkage of leave-one-out judge propensities as a robustness check, shrinking estimates toward either the overall mean or a district-year-specific mean. My implementation uses the same basic idea, but applies a linear empirical Bayes shrinkage rule to the residualized leave-one-out approval rates, with the amount of shrinkage determined by the sampling variance of each examiner's estimated leniency.

More recently, Frandsen et al. (2026) study this idea formally. They show that shrinking noisy leave-one-out first-stage propensities can increase the precision of examiner-IV estimates without additional identifying assumptions or changing the underlying causal estimand and derive the shrinkage rule that minimizes the asymptotic variance of the IV

estimator. Their optimal rule additionally depends on first- and second-stage error moments, whereas I use a simpler approach. The simulations in this section examine the finite-sample performance of this implementation.